

# Competing for Its Own Sake: Experimental Evidence on the Welfare Effects of Competition

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## Abstract

Economists often view competition as a means to motivate effort, improve efficiency, and therefore enhance welfare. However, this instrumental perspective may overlook that competition itself can directly influence individuals' welfare. This paper investigates how competition affects utility derived solely from the act of competing, independent of material outcomes. I conduct a series of experiments which show that competition affects utility through two opposing channels: a belief channel, in which competition lowers expectations of success and reduces utility; and a preference channel, through which individuals derive enjoyment directly from competing. The preference channel dominates, resulting in a net positive impact on utility. I further show that these welfare effects influence future choices: competition induces attribution bias, leading individuals to misattribute the enjoyment of competition to the underlying task and increase their willingness to engage in it again, even in the absence of competition. These effects also extend to social interactions, reducing post-competition zero-sum thinking and fostering altruism.

**Keywords:** Competition, Reference-dependent Utility, Attribution Bias, Zero-sum

**JEL Codes:** C91, D60, D91, I31

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# 1 Introduction

One of the central principles in economics is that markets are generally an effective way to organize economic activity, and that perfect competition maximizes social welfare (Mankiw, 2018). This classical perspective emphasizes the incentives that competition creates and the economic benefits it yields. Yet it overlooks a subtle aspect of welfare: the direct utility consequences of how people experience competition. Consumers may experience excitement when competing for limited-edition products during sales events, while job seekers may experience anxiety when competing for scarce positions in the labor market. Such experienced utility,<sup>1</sup> arising directly from the act of competing, can have a meaningful impact on overall welfare. Ignoring these effects may lead us to overestimate welfare when competition brings distress, or underestimate it when competition generates enjoyment.

This paper investigates how competition affects individuals' utility derived solely from the act of competing, independent of material outcomes.<sup>2</sup> I examine whether this impact is context dependent, specifically, when competition generates utility and when it yields disutility. I also explore the implications of these welfare effects: I conjecture that they may shape individuals' beliefs about the enjoyment of specific tasks and thus their future decisions about whether to undertake these tasks again, and may even affect their attitudes toward social and strategic interactions.

To address these questions, I develop a conceptual framework and conduct a series of pre-registered randomized controlled experiments on Prolific<sup>3</sup> with two treatment dimensions: (i) competition versus no competition, and (ii) reward structures framed as gains versus losses. The typical design in the experimental literature on competition employs a gain-framed reward structure in which participants compete for a prize: the winner receives the reward, while the loser receives nothing (Niederle and Vesterlund, 2007; Croson and Gneezy, 2009; Abbink et al., 2010; Fallucchi et al., 2020; Reuben et al., 2024). Although some designs incorporate proportional-prize structures in which second- and third-place participants receive smaller payoffs, the lowest-ranked participant still earns no reward (Harbring and Irlenbusch, 2003, 2011; Cason et al., 2010, 2020). I introduce a loss-framed design in which competition imposes explicit losses on losers: only the winner retains their endowment and remains at the status quo, while the loser forfeits part

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<sup>1</sup>Experienced utility refers to hedonic and affective experience, which can be derived from immediate reports of current subjective experience or from physiological indices (Kahneman et al., 1997).

<sup>2</sup>"Material outcomes" is used in a broad sense, including tangible rewards as well as recognition, status, or any other benefits realized through competition.

<sup>3</sup>Prolific is a widely used online platform for recruiting high-quality participants for social science research (Peer et al., 2022).

of their endowment. This design highlights a key feature of many real-world competitive environments: losing a competition entails real losses, not merely the absence of gains. For example, in the workplace, "winning" may mean keeping one's job, while losing could mean being laid off. I hypothesize that competition has a positive impact on utility when rewards are gain-framed but a negative impact when loss-framed, as individuals are more sensitive to potential losses than to equivalent gains (Tversky and Kahneman, 1979). This design creates four experimental treatments: gain-framed competition, gain-framed control, loss-framed competition, and loss-framed control.

To conduct the experiment, participants were randomly assigned to one of the four experimental treatments and asked to complete a timed quiz. The quiz consists of fifty puzzles similar to the Raven Progressive Matrices, a widely used measure of cognitive ability (Malamud and Pop-Eleches, 2011; Gill and Prowse, 2016; Alan et al., 2019; Drobner and Goerg, 2024; Willadsen et al., 2024). Participants have four minutes to complete the quiz. Each correct answer earns one point, each incorrect answer deducts one-quarter of a point, and skipped questions receive zero points.<sup>4</sup>

In the competition treatments, participants were randomly paired and competed for a monetary prize based on their quiz performance.<sup>5</sup> Within each pair, under the gain-framed reward structure, each participant received a \$2 show-up fee. The participant with the higher quiz score earned an additional \$4 reward, while the other received no additional payment. Under the loss-framed structure, each participant started with a \$6 show-up fee. The participant with the higher score kept the full \$6, while the other forfeited \$4. During the competition, participants observed their opponent's raw score in real time through a live score bar, which displayed the opponent's cumulative points earned for correct answers without deductions for incorrect ones. Because participants received no feedback on their own performance, this noisy feedback about the opponent's performance allowed them to gauge their relative standing in the competition but not to perfectly infer the final outcome.<sup>6</sup> Since participants cannot physically observe their rivals online, which often limits competitive engagement, the live score bar helped mitigate this limitation and foster a stronger sense of competition.

In the non-competition treatments, participants completed the quiz individually. Under

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<sup>4</sup>This scoring rule encourages participants to engage thoughtfully rather than guess randomly. The cognitive task is an IQ quiz. I chose this task for two reasons. First, it mirrors cognitively demanding real-world work, enhancing the external validity of the experimental findings. Second, IQ test performance is ego-relevant and naturally elicits utility responses, making it easier to identify competition's utility effects.

<sup>5</sup>The real-time online matching was efficient: the average waiting time for pairing was 19 seconds.

<sup>6</sup>Providing partial information about opponents approximates real-world situations, where people often know something but not everything about their rivals. Preventing participants from learning the final outcomes was crucial for this study's objective: isolating the utility effects of competition itself.

the gain-framed reward structure, each participant received a \$2 show-up fee and earned an additional \$4 reward if their quiz score exceeded a *predetermined threshold*;<sup>7</sup> otherwise, they received no additional reward. Under the loss-framed structure, each participant received a \$6 show-up fee and retained the full amount if their score surpassed the threshold; otherwise, they forfeited \$4. The threshold was set at the median score from the competition treatments, ensuring that participants across all treatments faced the same ex-ante success probability of one-half and the same expected monetary payoffs.<sup>8</sup> With monetary incentives held constant, the presence of competition and the framing of rewards were the only factors that varied across treatments.

After completing the quiz, participants filled out a survey evaluating their enjoyment of the task and their willingness to take a similar quiz again in the future.<sup>9</sup> They then answered questions assessing their zero-sum thinking and participated in decision-making tasks designed to elicit attitudes toward social interaction, including a dictator game and a public goods game. At the end of the experiment, and before learning the final outcomes,<sup>10</sup> participants were asked to indicate whether they believed they had won the competition or, in the non-competition treatments, whether they had exceeded the threshold.

The experiment shows that competition increases utility not only under the gain-framed reward structure but also, unexpectedly, under the loss-framed structure. While the treatment effect of competition does not differ significantly across frames, utility levels are consistently higher in the gain frame. To better understand this pattern, I disentangle the overall effect of competition on utility into two channels: a belief channel, through which competition reduces utility by lowering individuals' beliefs about their chances of winning, and a preference channel, through which individuals derive enjoyment directly from the act of competing. The positive preference channel outweighs the negative belief channel, resulting in a net positive effect of competition on utility. Interestingly, participants who believe they rank lower in the quiz experience greater enjoyment from competition than those who believe they rank higher. The utility gains from competition do not vary system-

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<sup>7</sup>Before the quiz, participants were informed that the threshold was a prior session's median score. I depart from the standard piece-rate design (Niederle and Vesterlund, 2007; Möbius et al., 2022) for non-competition treatments because a linear payment scheme would introduce confounding factors such as inequality aversion or risk aversion, which could contaminate the estimated effect of competition on utility.

<sup>8</sup>The gain- and loss-framed controls use the median scores from their respective competition treatments. Because participants are randomly paired in the competition treatments and the non-competition threshold is set at the median, the ex-ante success probability is one-half across all treatments.

<sup>9</sup>Participants in competition treatments were asked about taking a similar quiz without competition; those in non-competition treatments were asked about taking another similar quiz under the same conditions.

<sup>10</sup>In competition treatments, final outcomes included participants' quiz score, their opponent's score, and whether they won. In non-competition treatments, final outcomes included participants' quiz score and whether they exceeded the predetermined median threshold.

atically across demographic characteristics such as gender, age, education, or income.

These welfare effects have important implications. Intrinsically enjoyable competition increases individuals' preferences for the underlying task and their willingness to engage in it again, even in the absence of competition. This pattern suggests attribution bias in competitive settings:<sup>11</sup> individuals may misattribute the enjoyment derived from competition to the inherent qualities of the task itself, leading them to be more willing to undertake it again even after the competitive element has been removed. The positive welfare effects of competition also extend to social interactions. Participants who engaged in competition exhibited weaker zero-sum mindsets afterward and allocated more money to their partners in a dictator game. This finding contradicts the preregistered hypothesis that competition would foster stronger zero-sum thinking. One possible explanation is that the pleasurable experience of competing fostered more positive social attitudes, leading them to act more cooperatively and generously in subsequent social interactions.

This paper contributes to several strands of literature. First, it connects to the literature on experienced utility, which demonstrates how individuals' immediate affective responses to ongoing experiences shape their utility. This literature defines experienced utility and develops various methods for measuring subjective well-being (Kahneman et al., 1997, 1999; Kahneman and Krueger, 2006; Kahneman and Deaton, 2010; Benjamin et al., 2014). It also shows that individuals derive utility from the process of political participation (Frey et al., 2004; Frey and Stutzer, 2005) and from the intrinsic experience of work (Benz and Frey, 2008). I contribute to this literature by showing that competition, a central feature of economic interaction, generates its own experienced utility independent of final outcomes. This paper also relates to the behavioral welfare economics literature, particularly studies documenting the emotional value individuals attach to their own decisions and extending the traditional welfare framework to incorporate decision utility (Bartling et al., 2014; Bernheim et al., 2024). My findings suggest a related extension: individuals derive utility from the act of competing, and therefore complete welfare assessments in competitive environments should account for not only outcomes but also the utility generated by the competition itself.

Second, this work contributes to the literature on state-dependent preferences, particularly attribution bias. Prior work shows that individuals misattribute temporary states such as thirst, weather, and fatigue to the stable properties of goods or activities (Haggag et al., 2019, 2021; Bushong and Gagnon-Bartsch, 2023). I contribute to this literature by showing that attribution bias also arises in strategic settings: individuals engaged in com-

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<sup>11</sup>Attribution bias refers to the tendency to misattribute the influence of a temporary emotional state to a stable property of an activity (Haggag et al., 2019).

petition misattribute the enjoyment derived from competition to the underlying task and express greater willingness to undertake the task again in the future.

Finally, this paper speaks to the experimental literature on competition. A large body of work examines how competition affects effort, performance, tournament entry, and beliefs such as self-confidence, typically by comparing behavior under piece-rate and tournament payment schemes (Gneezy et al., 2003; Niederle and Vesterlund, 2007, 2011; Möbius et al., 2022; Hauge et al., 2023; Englmaier et al., 2024). I contribute to this literature by introducing a novel noncompetitive benchmark: a threshold-based incentive scheme that preserves the binary payoff structure of tournaments, thereby eliminating potential confounds arising from differences between linear and binary payment structures. This approach enables a cleaner estimate of the causal effect of competition.<sup>12</sup> Moreover, most laboratory studies of competition employ gain-framed tournaments, in which winners receive a prize and losers receive nothing (Carpenter et al., 2010; Cason et al., 2010; Dohmen and Falk, 2011; Buser et al., 2014; Krishna et al., 2025). Only a few examine loss-framed contests (e.g., Hong et al., 2015; Dato et al., 2018). I contribute to this literature by introducing a loss-framed competitive environment and providing experimental evidence on the welfare consequences of competition in such settings.

The remainder of the paper proceeds as follows. [Section 2](#) introduces a conceptual framework of the experienced utility from competition. [Section 3](#) describes the experimental design, procedures and data. [Section 4](#) provides an explicit definition of competition in the experimental context. [Section 5](#) presents the main results on how competition affects utility and decomposes the total effect into belief and preference channels. [Section 6](#) discusses the implications of the welfare effects of competition. [Section 7](#) concludes.

## 2 Theoretical Framework

This section introduces a theoretical framework for utility derived solely from the act of competing, independent of final outcomes. The framework builds on the classical winner-take-all tournament model developed by Lazear and Rosen (1981) and incorporates reference-dependent preferences from Kőszegi and Rabin (2006) to allow utility to vary across different contexts. [Section 2.1](#) presents the model setup, while [Section 2.2](#) outlines its testable predictions. The non-competitive setting can be viewed as a special case of this framework in which individuals compete against a fixed performance threshold rather than against another person.

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<sup>12</sup>If the objective were to compare the effects of competitive versus piece-rate payment schemes, this threshold design would be unnecessary.



## 2.1 Setup

Consider a winner-take-all competition with  $N \geq 2$  competitors, indexed by  $i \in \{1, \dots, N\}$ . Each competitor  $i$  possesses an ability parameter  $a_i > 0$  that is drawn independently from a common distribution  $F$  and then simultaneously chooses an effort level  $e_i \geq 0$ . Exerting effort entails a cost  $c : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  that is differentiable, strictly increasing, and convex, with  $c(0) = 0$ ,  $c'(\cdot) > 0$  and  $c''(\cdot) > 0$ . The monetary payoffs for the winner and the loser of the competition are denoted by  $(w^{win}, w^{lose}) \in \mathbb{R}^2$ .

Each competitor  $i$  has reference-dependent utility. The reference point  $r \in \mathbb{R}$  is exogenously given and represents a monetary endowment. Payoffs can be framed in two ways: as gains or as losses. In the gain frame, each competitor  $i$  receives an initial endowment  $b$  ( $r = b$ ). The winner receives a reward  $w^{win}$ , yielding a final payment of  $b + w^{win}$ , while the loser receives no reward, ending up with  $b$ . Under the loss frame, each competitor is initially endowed with  $b + w^{win}$  ( $r = b + w^{win}$ ). The winner keeps the full endowment  $b + w^{win}$ , while the loser forfeits  $w^{win}$ , ending up with  $b$ . Holding all else equal, expected payoffs are identical across both frames.<sup>13</sup>

Competitor  $i$ 's utility from the act of competing consists of two components: pecuniary and non-pecuniary. The pecuniary component<sup>14</sup> is the expected utility from monetary rewards net of the cost of effort.<sup>15</sup> It is expressed as the probability-weighted average of the monetary payoffs from winning and losing, minus the cost of effort. The non-pecuniary component, denoted by  $\psi_i$ , captures utility derived from competing per se, independent of monetary outcomes. It comprises two elements: immediate affective responses such as excitement or stress experienced during competition, and anticipatory utility from envisioning winning or losing.<sup>16</sup> Anticipated success may bring satisfaction from imagining outperforming others, a sense of superiority, or self-fulfillment, whereas anticipated failure may generate feelings of diminished status, disappointment, or self-doubt. Taken together, competitor  $i$ 's experienced utility from competing can be expressed as

$$P_i(e_i) \cdot v(w_i^{win}) + (1 - P_i(e_i)) \cdot v(w_i^{lose}) - c(e_i) + \psi_i$$

<sup>13</sup>In both frames, the winner receives  $b + w^{win}$  while the loser receives  $b$ . When winning probabilities are equal, the expected payoffs are identical across both frames.

<sup>14</sup>This study aims to isolate utility from the act of competing, separate from the material outcomes that result from competition. The monetary rewards considered here reflect the *anticipated* payoffs during the competitive process and therefore contribute to the experienced utility of competition.

<sup>15</sup>The cost of effort is included in the pecuniary component for simplicity of classification. That said, this classification is conceptually reasonable since effort is often driven by monetary incentives.

<sup>16</sup>Anticipatory utility has been documented in the literature, showing that individuals derive utility from anticipating future outcomes (Caplin and Leahy, 2001; Brunnermeier and Parker, 2005; Marzilli Ericson and Fuster, 2011).

The first three terms represent the pecuniary component of utility, while the fourth term captures the non-pecuniary component.<sup>17</sup> The perceived probability of winning is given by

$$P_i(e_i) = \Pr\left(\theta(a_i, e_i) > \max_{j \neq i} \theta(a_j, e_j)\right)$$

where  $\theta(a, e)$  is a performance function that is continuous and strictly increasing in both ability  $a$  and effort  $e$ . Competitor  $i$  wins if their performance exceeds that of all others; in the event of a tie, the winner is selected uniformly at random, with probability  $\frac{1}{N}$ .  $v(x)$  is a reference-dependent value function (Kőszegi and Rabin, 2006)

$$v(x) = \begin{cases} x & \text{if } x \geq 0, \\ \lambda x & \text{if } x < 0, \end{cases}$$

with a loss aversion parameter  $\lambda > 1$ .

In the non-competitive scenario, the comparison benchmark  $\max_{j \neq i} \theta(a_j, e_j)$  is replaced by a fixed threshold  $\theta^*$ , with all other elements of the model remaining unchanged. An individual succeeds if their performance exceeds  $\theta^*$ , so payoffs depend solely on their own ability and effort. The threshold  $\theta^*$  is set to match the objective probability of success in the corresponding competitive condition. For example, when  $N = 2$ , the threshold is set at the median performance of the competition treatments, ensuring a 50 percent ex-ante success probability. When  $N = 10$ , it is set at the 90th percentile of the performance distribution, so that the probability of surpassing the threshold matches the 10 percent winning probability in a winner-take-all tournament.<sup>18</sup>

More generally, let  $s \in \{C, NC\}$  denote the setting, where  $C$  represents competition and  $NC$  represents non-competition. Let  $f \in \{G, L\}$  denote the framing of the reward structure, where  $G$  corresponds to the gain frame and  $L$  to the loss frame. Individual  $i$ 's experienced utility in scenario  $(s, f)$ , denoted  $U_i^{s,f}(e_i)$ , can be written as

$$P_i^{s,f}(e_i) \cdot v(w_i^{win,f}) + (1 - P_i^{s,f}(e_i)) \cdot v(w_i^{lose,f}) - c(e_i) + \psi_i^{s,f}$$

The first three terms capture the pecuniary component of utility, denoted as  $U_{i,pec}^{s,f}(e_i)$ , while the fourth term captures the non-pecuniary component, denoted as  $U_{i,nonpec}^{s,f}$ . The perceived probability of winning,  $P_i^{s,f}(e_i)$ , depends on scenario  $(s, f)$ , reflecting that the

<sup>17</sup>Utility from the task alone is normalized to zero and thus not explicitly modeled in experienced utility.

<sup>18</sup>Winner-take-all is used in its strict sense: a contest with exactly one winner.



same effort level may translate into different perceived probabilities depending on the competitive setting and reward framing. The non-pecuniary utility  $\psi_i^{s,f}$  also varies across scenarios, as affective responses and anticipatory utility may differ between settings and frames. Monetary payoffs ( $w^{win}, w^{lose}$ ), by design, vary only with framing; this ensures that within each frame, competition is the only distinguishing factor, allowing us to cleanly isolate its impact. The cost of effort function,  $c(e_i)$ , is assumed identical across all scenarios for a given individual  $i$ . Any emotional factors that may affect the marginal cost of effort are captured in the non-pecuniary component  $U_{i,nonpec}^{s,f}$ .

Given this framework, I now examine how experienced utility differs between competitive and non-competitive settings. I introduce two empirically testable assumptions and derive predictions about the utility relationship in the following subsections. These assumptions are not axiomatic and will be validated through experimental evidence.

**Assumption 1** . *Fix a frame  $f \in \{G, L\}$ . For every effort level  $e \geq 0$ ,<sup>19</sup> the perceived probability of success is (weakly) lower in the competitive setting than in the non-competitive setting:*

$$P_i^{C,f}(e) \leq P_i^{NC,f}(e) \text{ for all } e \geq 0.$$

The intuition behind this assumption is as follows. In the competitive setting, winning requires performance to exceed  $\max_{j \neq i} \theta(a_j, e_j)$ , which is an endogenous and uncertain benchmark determined by opponents' abilities and effort choices. In the non-competitive setting, success is evaluated against a fixed and transparent performance threshold  $\theta^*$ , which is calibrated to match the corresponding objective winning probability in the competitive context. Although the objective probability of success is identical across the two settings by design, the potentially larger perceived uncertainty (Bruttel et al., 2023; Balafoutas and Sutter, 2019) in competition may depress individuals' beliefs about their chances of success,<sup>20</sup>. Therefore, participants' perceived probability of winning is smaller relative to scenarios with fixed standards. Formally, under a slightly stronger assumption, we can posit that for any effort level  $e$ ,  $P_i^{C,f}(e) \leq P_i^{NC,f}(e)$ .

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<sup>19</sup>This assumption may be strong. A more general model that relaxes these assumptions and derives the relationship between utilities based on the ability distribution is being developed in subsequent work.

<sup>20</sup>Bruttel et al. (2023) show that when the source of uncertainty is strategic rather than exogenous, beliefs are systematically reshaped; Balafoutas and Sutter (2019) show that uncertainty and ambiguity in tournament rules shift competitive behavior.

**Assumption 2** *The non-pecuniary utility under competition is weakly higher than under non-competition in the gain frame and weakly lower in the loss frame:*

$$\psi_i^{C,G} \geq \psi_i^{NC,G} \quad \text{and} \quad \psi_i^{C,L} \leq \psi_i^{NC,L} \quad \text{for all } i.$$

The intuition behind this assumption is as follows. Non-pecuniary utility consists of two components: affective responses to the act of competing and anticipatory utility derived from expectations of winning or losing. The first component arises only in the competitive setting, whereas the second exists in both competitive and non-competitive contexts. Under the gain frame, competition provides an opportunity to gain extra reward; if losing, individuals at least remain at the status quo. In such cases, emotional responses to competition are more likely to be positive. In terms of the anticipatory utility from winning or losing, it is essentially similar between competition and non-competition scenarios. While winning in competition involves a sense of superiority from outperforming others, which does not exist in the non-competitive setting, anticipatory losing in competition also involves a sense of lost status, which is absent in non-competition. On average, these effects may offset each other. Taking both affective responses and anticipatory utility into account, under the gain frame, we have  $\psi_i^{C,G} \geq \psi_i^{NC,G}$ . Under the loss frame, competition becomes a survival game: winning a competition simply allows individuals to stay at the status quo, while losing incurs actual losses relative to their initial position. Individuals naturally fear losses, and according to loss aversion, losses typically evoke stronger emotional reactions than equivalent gains (Tversky and Kahneman, 1979; Kahneman et al., 1991; Brown et al., 2024). In such cases, emotional responses to competition are more likely to be negative. As for the anticipatory utility from winning or losing, the same reasoning applies as under the gain frame, so it is similar across the two settings. Taking both affective responses and anticipatory utility into account, under the loss frame, we have  $\psi_i^{C,L} \leq \psi_i^{NC,L}$ .

## 2.2 Predictions

The theoretical framework generates two main predictions about how competition affects utility, guiding the empirical analysis. First, a belief channel: competition affects utility via its impact on beliefs about winning,  $P_i^{s,f}(e_i)$ . Second, a preference channel: individuals derive direct utility from the act of competing; this is captured by the non-pecuniary utility component  $\psi_i^{s,f}$ . These predictions are experimentally tested in Section 5. Proofs and auxiliary assumptions are in Appendix A.

**PREDICTION 1.** (*Belief channel*) Fix a frame  $f \in \{G, L\}$  and hold the preference channel constant. Under Assumptions 1, the competitive setting, relative to the non-competitive benchmark, yields lower experienced utility at the optimal effort level  $e^{f,*}$ :

$$U_i^{C,f}(e_C^{f,*}) \leq U_i^{NC,f}(e_{NC}^{f,*}).$$

The intuition for this prediction is as follows. When the preference channel, which captures the non-pecuniary component of utility, is held constant, the pecuniary component becomes the main determinant of experienced utility. Expected pecuniary utility is the probability-weighted expected monetary payoff net of effort costs. Since monetary prizes are fixed within each frame, any variation in expected pecuniary utility across settings must arise from differences in the perceived probability of winning or in effort costs. Because the cost function is assumed to remain unchanged across settings, expected pecuniary utility ultimately depends on beliefs about winning and on effort levels. Assumption 1 implies that the perceived probability of winning is lower under competition. At the same time, since monetary incentives, which are the main drivers of effort, are held constant across settings, effort levels are likely to remain similar in both environments. As a result, pecuniary utility is lower under competition, driven primarily by the reduced perceived probability of winning. With the preference channel held constant, overall experienced utility under competition is therefore lower than under non-competition.

**PREDICTION 2.** (*Preference channel*) Fix a frame  $f \in \{G, L\}$  and hold the belief channel constant. Under Assumption 2, the competitive setting, relative to the non-competitive benchmark, yields:

- i Gain frame: (weakly) higher experienced utility at the optimal effort level  $e^{G,*}$ , i.e.,

$$U_i^{C,G}(e_C^{G,*}) \geq U_i^{NC,G}(e_{NC}^{G,*})$$

- ii Loss frame: (weakly) lower experienced utility at the optimal effort level  $e^{L,*}$ , i.e.,

$$U_i^{C,L}(e_C^{L,*}) \leq U_i^{NC,L}(e_{NC}^{L,*})$$

The intuition for this prediction is as follows. When the belief channel, which captures the pecuniary component of utility, is held constant, the non-pecuniary component be-

comes the primary determinant of experienced utility. Under the gain frame, Assumption 2 implies that non-pecuniary utility under competition is higher than under non-competition, leading experienced utility under competition to exceed that under non-competition. Conversely, under the loss frame, Assumption 2 implies that non-pecuniary utility under competition is lower than under non-competition, resulting in lower experienced utility under competition. In other words, when monetary considerations do not affect utility, differences in experienced utility arise from affective responses to competition. Under the gain frame, competition elicits positive affect such as excitement, motivation, and engagement, which enhance experienced utility relative to non-competition. Under the loss frame, competition evokes negative feelings such as stress and anxiety, which reduce experienced utility relative to non-competition.

### 3 Experiment

The experiment is designed to investigate whether competition itself carries utility independent of final outcomes, whether this utility varies across different contexts, and the implications of these effects. To provide clean and reliable causal evidence, I conduct a series of experiments. Section 3.1 introduces the experimental design, Section 3.2 describes the experimental procedures, and Section 3.3 summarizes the data.

#### 3.1 Experimental Design

##### *A. Treatments*

The experiment employs a between-subjects design that varies two treatment dimensions: competition versus non-competition, and reward structure framed as gains versus losses. Participants are randomly assigned to one of four treatments: Gain Framing with Competition, Gain Framing without Competition, Loss Framing with Competition, and Loss Framing without Competition. All participants complete a timed IQ quiz under their assigned treatment, then respond to a post-task survey. Figure 1 presents a visual summary of the experimental design, with details described in what follows.

**Incentives.** In the Gain Framing with Competition treatment, participants are randomly paired and compete for a monetary prize based on their performance in an IQ quiz. Each participant receives a \$2 show-up fee. Within each pair, the participant with the higher

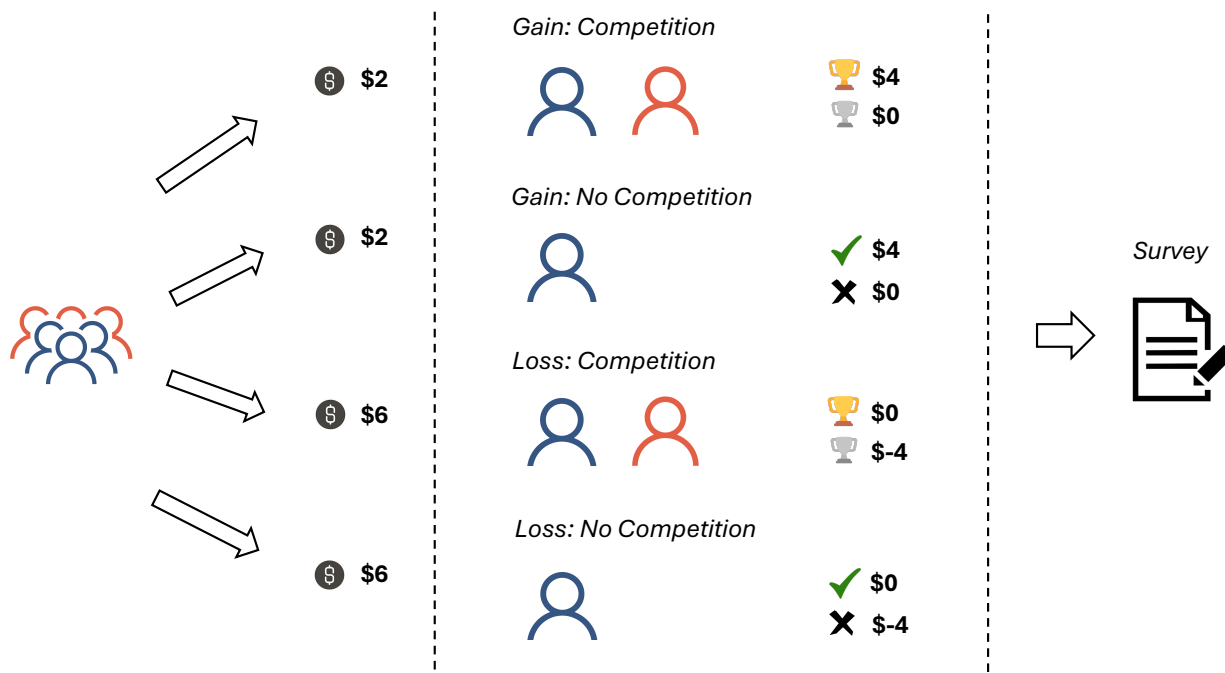


Figure 1: Visual Representation of Experimental Design

*Notes:* This figure illustrates the structure of the experimental design. Participants are randomly assigned to one of four treatments that vary competition (competition vs. no competition) and reward framing (gain vs. loss). In gain-framed treatments, participants receive a \$2 show-up fee and can earn \$4 more by winning in competition or exceeding a performance threshold in non-competition. In loss-framed treatments, participants receive a \$6 show-up fee and can lose \$4 by losing in competition or failing to exceed the threshold in non-competition. The threshold is set at the median score from the corresponding competition treatment within the same reward framing. All participants complete a post-task survey after the experimental task.

quiz score earns an additional \$4 reward, while the participant with the lower score receives no additional payment. In the Gain Framing without Competition treatment, each participant also receives a \$2 show-up fee but completes the quiz individually. Participants earn an additional \$4 reward if their quiz score exceeds a predetermined threshold; otherwise, they receive no reward. In the event of a tie, one participant was randomly selected to receive the reward. The performance threshold is set at the median score from the Gain Framing with Competition treatment, and participants are informed of this median benchmark before starting the quiz. This design ensures an identical ex-ante success probability across treatments: In the competition condition, the pairing structure generates a one-half probability of winning. In the non-competition condition, because the median represents the midpoint of the score distribution, half of all participants score above it and half below it, yielding a one-half success probability by design. With the same reward structure (\$2 base payment plus \$4 performance bonus) across both treatments, their expected payoffs are therefore identical.

In the Loss Framing with Competition treatment, the structure parallels the gain framing but shifts the reference point. Each participant starts with a \$6 show-up fee and is randomly paired with a competitor. Within each pair, the participant with the higher quiz score retains the full \$6, while the participant with the lower score forfeits \$4, keeping only \$2. In the Loss Framing without Competition treatment, participants complete the quiz individually. Those whose scores exceed a predetermined threshold keep the full \$6, while those falling short forfeit \$4, keeping only \$2. In the event of a tie, one participant is randomly selected to keep the \$6. The threshold is set at the median score from the Loss Framing with Competition treatment,<sup>21</sup> and participants are informed of this median benchmark before starting the quiz. This median threshold design ensures that participants in both loss-framed conditions face a one-half probability of avoiding the loss. With the same reward structure (\$6 base payment and a potential loss of \$4) across both loss-framed treatments, the expected payoffs are identical. In fact, expected monetary payoffs are the same across all four treatments.<sup>22</sup> With monetary incentives held equal, the presence of competition and the framing of the reward structure are the only factors that vary across treatment conditions, enabling clean identification of competition effects.

***IQ Quiz.*** The choice of an IQ quiz as the experimental task is motivated by two factors. First, performance on IQ tests is inherently ego-relevant, naturally eliciting utility responses from participants. This feature increases the likelihood of observing potential utility effects of competition, should they exist. Second, the cognitively demanding nature of the IQ quiz resembles tasks commonly encountered in real-world work environments, enhancing the external validity of the experimental findings. One potential concern with using an IQ quiz is that some participants might be particularly averse to IQ-related competition, leading to higher attrition in competition treatments relative to non-competition treatments and creating imbalance across conditions. Fortunately, attrition rates were similar across all treatments in practice, alleviating this concern.

The IQ quiz is designed as follows. It consists of 50 puzzles adapted from [Civelli and Deck \(2017\)](#), similar to Raven’s Progressive Matrices, a widely used measure of cognitive ability ([Drobner and Goerg, 2024](#); [Willadsen et al., 2024](#)). Each puzzle provides four answer choices, only one of which is correct; an example is shown in [Figure 2](#). All treatments

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<sup>21</sup>In practice, quiz scores in the Gain Framing with Competition treatment and the Loss Framing with Competition treatment are very close.

<sup>22</sup>In the gain-framed treatments, the expected payoff equals \$2 plus one-half of either \$4 or \$0, yielding an expected value of \$4. In the loss-framed treatments, the expected payoff equals \$6 plus one-half of either \$0 or -\$4, also yielding an expected value of \$4.

use the same set of puzzles. The scoring rule awards one point for each correct answer, deducts one-quarter point for each incorrect answer, and assigns zero points for skipped questions. Participants cannot return to questions they have answered or skipped. This is designed to incentivize thoughtful engagement: answering a question yields a higher expected payoff than skipping it, and the penalty for incorrect answers discourages random guessing. Participants have four minutes to complete as many puzzles as possible. This fixed time limit ensures that all participants spend the same amount of time on the task, eliminating potential confounds from variations in task duration.

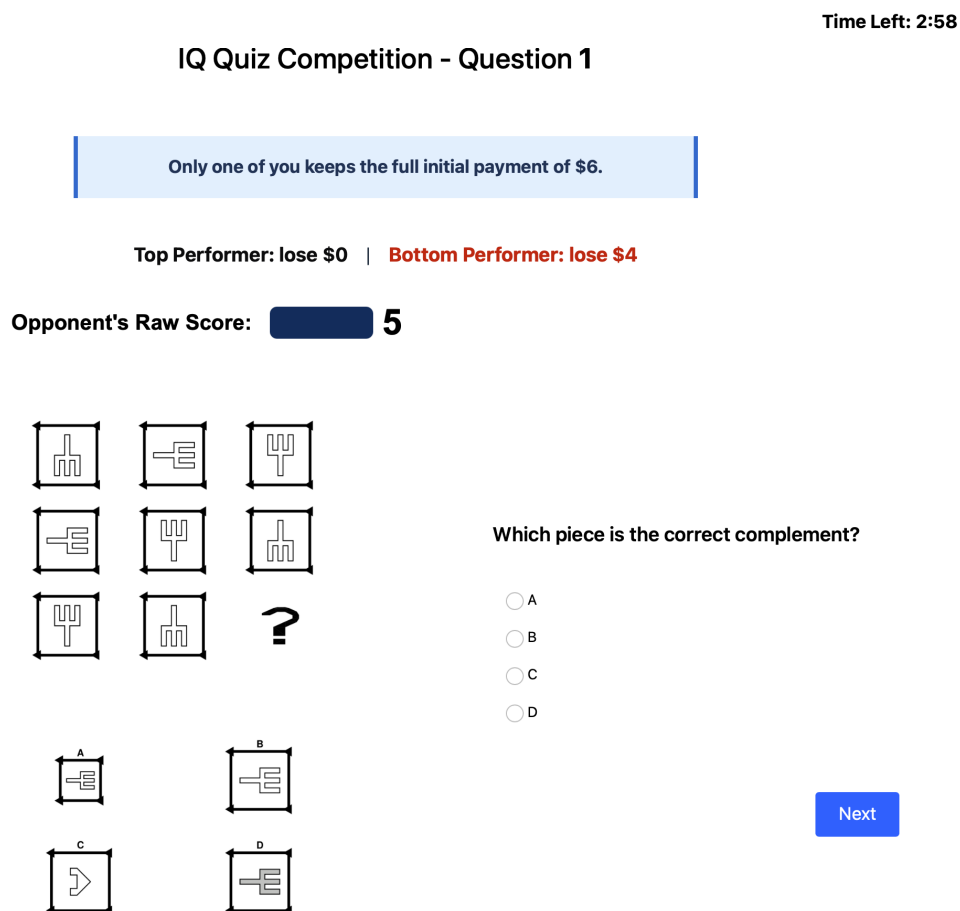


Figure 2: IQ Quiz Interface

*Notes:* This figure shows a screenshot of the IQ quiz in the loss-framed competition treatment. Participants are informed that only one of the two competitors retains the full initial payment of six dollars, while the other loses part of it based on relative performance. The top performer loses zero dollars and the bottom performer loses four dollars. Each question presents a matrix reasoning problem in which participants select the correct complement from four options labeled A, B, C, and D. The opponent's raw score is displayed to enhance the sense of competition, while the remaining time indicates the total time left to complete the four-minute experimental task.



**Online Matching.** In the competition treatments, participants in both framing conditions are randomly matched in real time with another participant. The maximum waiting time is five minutes, though the average was 19 seconds, making systematic treatment differences unlikely. Appendix Figure D.2 displays the matching screen at this stage. Once matched, participants begin the IQ quiz immediately. In the non-competitive treatments, participants proceed directly to the quiz without matching.

**Interface.** Figure 2 shows the IQ quiz interface for the loss-framing competition group. The interfaces for the other three groups are presented in Section C. Two design features of the task screen merit attention. The first is the *real-time opponent score bar*. In both competitive treatments, participants observe their opponent’s *raw* score in real time. The raw score represents the cumulative number of correct answers without deductions for incorrect ones. This design serves two purposes. First, its dynamic nature heightens the salience of competition and may evoke feelings such as excitement, nervousness, or stress, enabling detection of whether participants derive utility from competing. Online experiments often limit engagement because participants cannot physically observe rivals; the score bar mitigates this limitation by making competition more tangible. Second, because only *raw* scores are shown, participants learn about their opponents’ performance only partially and cannot infer the final competition outcome. This feature is crucial for isolating the effect of competition itself. The score bar appears only in the competitive treatments. The second design feature is treatment-specific interface wording to reinforce treatment salience. In the competitive conditions, the interface states “Only one of you earns the \$4 bonus” in the gain frame and “Only one of you keeps the full initial payment of \$6” in the loss frame, underscoring both rivalry and the zero-sum nature of the task. In the non-competitive conditions, the wording emphasizes individual performance: “Reach the target score to earn the \$4 bonus” in the gain frame and “Reach the target score to keep the initial payment of \$6” in the loss frame.

**IQ Quiz Performance Feedback.** Participants receive no feedback about their own performance during the quiz. At the end of the experiment, they are informed of their own quiz score, their opponent’s score, and whether they beat their opponent (in competition treatments) or met the threshold (in non-competition treatments). This design ensures isolating the utility derived from competition itself, independent of realized outcomes. Although in competitive conditions the display of the opponent’s real-time score may enable participants to form beliefs about outcomes, such beliefs are conjectural rather than precise, since only raw scores were observed.

## B. Outcome Measures

The outcome variables are measured through survey questions and incentivized decision-making tasks administered after the IQ quiz, as shown in [Figure 1](#). Utility, task preferences, and zero-sum thinking are elicited through the survey, while altruistic behavior is measured using a dictator game and a public goods game.

**Measuring Utility.** A self-reported measure of utility is commonly used to capture individuals' subjective well-being ([Kahneman and Krueger, 2006](#); [Kahneman et al., 1997](#); [Frey et al., 2004](#); [Haggag et al., 2019](#)). Following this approach, I use a survey question with a seven-point Likert scale to measure experienced utility: "On a scale from 1 (not at all enjoyable) to 7 (extremely enjoyable), how enjoyable was the IQ Quiz (or, in treatment groups, the IQ Quiz Competition)?" To capture specific emotional components of utility, I adapt the approach of [Bernheim et al. \(2024\)](#), who elicit the feelings associated with individuals' choices. Participants are asked: "Thinking back to how you felt during the IQ Quiz (or IQ Quiz Competition for the treatment groups), please indicate the extent to which you experienced the following emotions (excitement, satisfaction, stress, embarrassment, and anxiety) on a scale from 1 (not at all) to 7 (extremely)." The first two emotions are classified as positive and the latter three as negative. These measures are administered uniformly across all experimental conditions.<sup>23</sup>

**Measuring Task Preference.** Individuals are generally aware of state-dependent preferences. For example, food tastes better when one is hungry and vacations are more enjoyable when the weather is sunny. However, they may fail to account for the transient nature of such states and instead misattribute their influence to stable characteristics of the good or activity ([Haggag et al., 2019](#); [Bushong and Gagnon-Bartsch, 2023](#)). While the literature has documented such attribution biases in non-strategic contexts such as hunger and thirst, less is known about whether similar biases arise in strategic environments. This study provides an opportunity to examine whether competitive settings shape individuals' preferences for a task through such biases.

To measure task preferences, I adapt survey questions from [Haggag et al. \(2019\)](#). The primary measure asks respondents whether they would accept the task under varying hypothetical payments. They are presented with the following scenario:<sup>24</sup>"Imagine you have

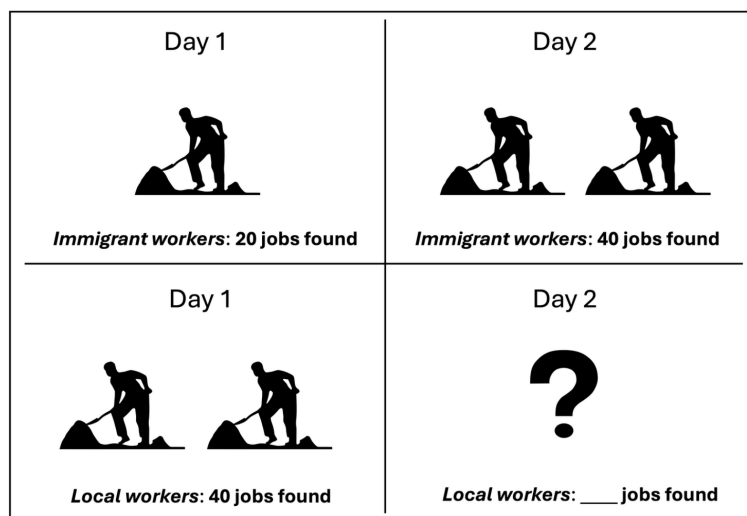
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<sup>23</sup>The analysis of utility is presented in [Section 5](#), and the results on the emotional components of utility are discussed in [Appendix C](#).

<sup>24</sup>The exact wording and layout of the question are provided in [Appendix D](#).

another IQ quiz in front of you right now, similar to the one you just completed but with a different set of fifty questions. You again have four minutes to answer as many as possible. If you get at least six questions<sup>25</sup> correct within the four minutes, you will receive a payment. For each amount (\$0.20, \$0.40, \$0.60, \$0.80, \$1.00, \$1.20, \$1.40, \$1.60, \$1.80, \$2.00, \$2.50, \$3.00, and \$5.00), please indicate ‘Yes’ if you would be willing to take the quiz for that payment, or ‘No’ otherwise.” As a complementary measure, respondents are also asked: “On a scale from 1 (not at all likely) to 7 (extremely likely), how likely would you be to voluntarily take another set of 10 IQ questions, similar to the ones you just completed, with no monetary rewards?”

Figure 3: Measure of Zero-Sum Thinking



☐ 20 jobs

☐ 80 jobs

*Notes:* The figure illustrates the vignette used to measure zero-sum thinking. Participants are presented with a simple scenario comparing the number of jobs found by immigrant and local workers over two consecutive days. On Day 1, immigrant workers find 20 jobs and local workers 40. On Day 2, immigrant workers find 40 jobs, and respondents are asked to determine how many jobs local workers find on Day 2, choosing between 20 and 80. Selecting 20 reflects a zero-sum view, in which the total number of available jobs is fixed and gains for one group imply losses for another. Selecting 80 reflects a non-zero-sum view, consistent with a positive-sum understanding that overall economic opportunities can expand.

<sup>25</sup>Six correct answers correspond to the median performance in the sample.

**Measuring Zero-Sum Thinking.** Following the approaches used in the World Values Survey (Inglehart et al., 2014) and Carvalho et al. (2023), I measure zero-sum thinking using two survey questions. The first asks respondents to position their views on a scale between two opposing statements: (1) “Wealth can grow so there’s enough for everyone” and (2) “People can only become wealthy at the expense of others.” Participants select a number from 1 to 10 that best represents their view, where 1 indicates complete agreement with the first statement (non-zero-sum thinking) and 10 indicates complete agreement with the second (zero-sum thinking), with intermediate values representing intermediate views. The second question adapts the measure from Carvalho et al. (2023). Respondents are presented with a scenario in which both immigrants and locals work as day laborers (see Figure 3). On Day 1, immigrant workers find 20 jobs while local workers find 40 jobs. On Day 2, immigrant workers find 40 jobs. Respondents are asked to estimate the number of jobs found by local workers on Day 2, choosing between 20 and 80. Choosing 20 reflects zero-sum thinking: if the total number of jobs is fixed, then when immigrants take more, fewer remain for locals. Choosing 80 reflects non-zero-sum thinking: if a common demand shock doubled the number of jobs for immigrants, then the number of jobs for locals would also double.

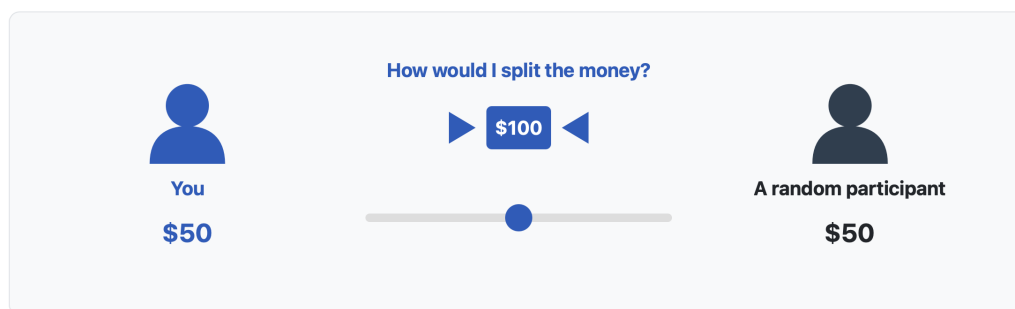


Figure 4: Dictator Game

*Notes:* This figure shows the experimental interface used for the dictator game. Participants are hypothetically given one hundred dollars and use a slider to decide how to divide it between themselves and a randomly selected participant. As they move the slider, the displayed amounts for each side adjust in real time to reflect the chosen allocation. The task measures altruistic behavior by capturing participants’ willingness to share resources in a non-strategic context.

**Measuring Altruistic Behavior.** I measure altruistic behavior using a dictator game and a public goods game. In the dictator game, I follow the approach used in Enke et al. (2023): participants are asked to split \$100 between themselves and a randomly selected

participant. Figure 4 shows the interface for this task, which provides a vivid and efficient measure of altruism. In the public goods game, participants are randomly paired. Each receives a \$0.50 endowment and must decide whether to keep it or contribute it to a common pool. Payoffs are determined as follows: if both keep, each earns \$0.50; if both contribute, each earns \$0.80; if one keeps while the other contributes, the keeper earns \$0.90 and the contributor \$0.40. The interface for this task is shown in Figure 5. All payments are real and paid as part of participants' final bonuses. After making their decision, participants are asked to estimate the share of others who chose to keep their endowment. This belief elicitation is incentivized: correct answers earn a monetary bonus.

\$ All payoffs are real and will be added as a bonus to your payment.

Please make your decision:



Drag your \$0.50 to one of the options below.

**Put into pool**

**Keep the money**

Figure 5: Public Goods Game

*Notes:* This figure shows the experimental interface used in the public goods game. Participants are given an endowment of fifty cents and decide whether to keep it or contribute it to a common pool. To make their decision, they drag the dollar representation into one of the two boxes labeled “Put into pool” or “Keep the money.” All payoffs are real and added to participants' final bonuses. This task measures cooperative behavior by observing contributions to a shared resource when personal and collective incentives conflict.

**Additional Measures.** As suggested by the theoretical framework in Section 2, beliefs about the probability of winning and perceived relative ability are important components of utility. I therefore elicit them both in the survey. To measure beliefs about winning, I

adapt the method used in [Niederle and Vesterlund \(2007\)](#). Participants in the competition treatments are asked whether they believe they were the top performer in the IQ Quiz Competition (yes/no). In the control treatments, they are asked whether they believe their score exceeded the threshold (yes/no). These questions are incentivized: participants receive a bonus for correct answers. Perceived relative ability is measured by asking participants to rate their performance relative to others on a 0–10 scale, where 0 indicates believing they performed worse than almost everyone and 10 indicates better than nearly everyone. For instance, a response of 7 corresponds to believing one performed better than about 70% of participants. As is standard practice, I also collect data on basic demographics including gender, race, income, and education. To ensure data quality, participants are also asked how well they understood the instructions, with response options “fully understand,” “almost,” “partly,” or “none.” The full survey is provided in [Section D.7](#).

### *C. Comprehension and Attention Checks*

Several quality control procedures are implemented to verify that participants clearly understand the experimental instructions and to maintain high data quality. These include comprehension questions, bot detection, and attention checks. Before the IQ quiz, participants are required to complete a practice question; only those who answer correctly can proceed. Comprehension questions are also administered before the public goods game to verify participants’ understanding of the payment rules. Participants must answer correctly to continue. I also record the number of attempts taken on each comprehension question, which serves as a screening criterion in the data analysis. To prevent automated participation, reCAPTCHA verification is included at the start of the experiment. Only participants who pass this verification can continue. For attention checks, I adopt the method used in [Bernheim et al. \(2024\)](#). Specifically, one survey item embedded among the main questions instructs participants not to select any option. Selecting an option indicates a failed attention check, and the participant’s data is excluded from the analysis.

## **3.2 Experimental Procedures**

***Participant Pool.*** Participants were recruited via Prolific, a widely used online platform for social science research ([Peer et al., 2022](#)). All participants resided in the United States and were at least 18 years old. To ensure high data quality, I applied several screening criteria. First, participants were required to have completed at least 100 prior submissions

on Prolific. Second, their prior approval rate had to be at least 95%. Third, the sample was balanced by gender within each experimental condition.<sup>26</sup> Fourth, each participant could access only one recruitment posting and participate in a single experimental condition.<sup>27</sup> Finally, to ensure smooth implementation, only desktop or laptop devices were permitted; responses from mobile phones and tablets were not allowed.

**Procedures.** The experiment was conducted on June 4, 2025, with the four experimental groups implemented over the course of the day in the following order: Gain Framing with Competition, Gain Framing without Competition, Loss Framing with Competition, and Loss Framing without Competition. In all groups, the study title read “A short quiz and a set of survey questions (15–25 minutes).” In the study description, participants in the Gain Framing condition were told the study paid \$2 with the possibility of earning a bonus, whereas participants in the Loss Framing condition were told the study paid \$6 with the possibility of a deduction.<sup>28</sup>

Upon entering the study, participants first viewed a welcome page outlining the study purpose and compensation. They then read the informed consent form, entered their Prolific ID, and completed a reCAPTCHA verification. Next, participants received the main experimental instructions, which varied slightly across treatment arms. The instructions described the IQ quiz, the follow-up survey, the expected completion time, and the payment rules. In the Gain Framing conditions, participants were informed that they would begin with \$2 and could earn up to a \$4 bonus based on their quiz performance. In the gain-framed competition treatment, the bonus required outperforming a randomly matched opponent. In the gain-framed control group, the bonus was tied to meeting a pre-specified performance threshold, which was the median quiz score from a prior session. In the Loss Framing conditions, participants were told they would begin with \$6 and could lose up to \$4. In the loss-framed competition treatment, avoiding the loss required outperforming a random opponent, while in the loss-framed control group, it required meeting a pre-specified threshold.<sup>29</sup>

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<sup>26</sup>This step was necessary because Prolific’s participant pool contains more females than males; simple random assignment would otherwise risk gender imbalance.

<sup>27</sup>Treatments were conducted sequentially, as the median IQ quiz score from competitive treatments served as the benchmark for non-competitive treatments. This sequential design necessitated strict prevention of multiple participation; accordingly, no participant was permitted to take part in more than one treatment.

<sup>28</sup>In the gain-framed treatments, the description stated: “The base payment for participating in the study is \$2.00. You can earn a bonus of up to \$4.00 in the first section, and additional bonuses in later sections.” In the loss-framed treatments, the description stated: “The base payment for participating in the study is \$6.00. You may lose up to \$4.00 of this amount in the first section. Bonuses are available in later sections.”

<sup>29</sup>The median quiz scores are 6.25 and 6.75 in the gain- and loss-framed competition treatments, respec-



All participants were required to answer comprehension questions after reading the instructions, and they had to correctly answer all questions before proceeding. Data indicate that 94.3% of participants passed within two attempts. Participants then completed a practice question to familiarize themselves with the quiz format, followed by a reminder page reiterating the payment rules. In the competition condition, participants were additionally reminded that they would be matched in real time with another participant and would observe their opponent's raw score bar updating live. On the matching page, a five-minute countdown timer was displayed. Participants who were not matched within this period received \$0.50 compensation for their time. In practice, 99.7% of participants were matched, with an average wait time of 19 seconds, and the minimum is 1.8 seconds.<sup>30</sup> Once matched, or proceeding directly in the non-competition condition, participants worked on the four-minute IQ quiz. Afterwards, they answered survey questions on utility, task preferences, zero-sum thinking, and made decisions in a dictator game and a public goods game. Questions on beliefs about winning, perceived relative ability, and demographic characteristics were asked at the end. On the final page, participants were shown their quiz performance and informed whether they had beaten their opponent or, in the non-competition condition, whether they had passed the threshold score.<sup>31</sup>

**Pre-registration.** The experiment was preregistered on the AEA RCT Registry<sup>32</sup>. The preregistration specified the experimental design, treatment interventions, planned sample sizes, and the primary hypotheses on competition and the main outcomes.

### 3.3 Data Description

A total of 798 participants enrolled in the study. Of these, 198 participated in the gain-framed competition treatments, and 200 participated in each of the other three treatments: gain-framed control, loss-framed competition, and loss-framed control. Eight participants

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tively, so the threshold is set at 6 for both non-competition groups.

<sup>30</sup>The experiment was programmed in oTree and hosted on Heroku. Matching was implemented in oTree. To maintain server stability during peak recruitment, participation slots were released in staged batches, for example, 50 positions at a time, until the target sample size was reached.

<sup>31</sup>In the Loss Framing conditions, participants were informed they might lose part of their payment. At the end of the experiment, a debrief clarified that, consistent with Prolific policy, advertised payments could not be reduced; all participants received the full \$6 regardless of performance. Participants in the competition groups also received an additional debrief because competition had not been disclosed in the consent form. While some participants may have known Prolific's policy, comments from study participants suggest this was rare. Random assignment ensures any such effect was balanced across conditions.

<sup>32</sup>ID: AEARCTR-0015723, <https://doi.org/10.1257/rct.15723-2.3>.

Table 1: Descriptive Statistics and Balance Tests

|                                                   | (1)                | (2)              | (3)              | (4)              | (5)              | (6)             | (7)                |
|---------------------------------------------------|--------------------|------------------|------------------|------------------|------------------|-----------------|--------------------|
|                                                   |                    | Gain Framing     |                  | Loss Framing     |                  | Difference      |                    |
|                                                   | <i>Full Sample</i> | Competition      | Control          | Competition      | Control          | (2)-(3)         | (4)-(5)            |
| <i>Gender: Female</i>                             | 0.50<br>(0.50)     | 0.51<br>(0.50)   | 0.52<br>(0.50)   | 0.50<br>(0.50)   | 0.50<br>(0.50)   | -0.01<br>[0.84] | 0.00<br>[0.96]     |
| <i>Age</i>                                        | 40.12<br>(13.41)   | 39.72<br>(13.64) | 40.21<br>(13.13) | 38.97<br>(13.75) | 41.55<br>(13.06) | -0.50<br>[0.71] | -2.58*<br>[0.06]   |
| <i>Race: White</i>                                | 0.60<br>(0.49)     | 0.62<br>(0.49)   | 0.59<br>(0.49)   | 0.56<br>(0.50)   | 0.61<br>(0.49)   | 0.03<br>[0.52]  | -0.05<br>[0.34]    |
| <i>Race: African American</i>                     | 0.28<br>(0.45)     | 0.30<br>(0.46)   | 0.27<br>(0.45)   | 0.30<br>(0.46)   | 0.25<br>(0.43)   | 0.03<br>[0.46]  | 0.05<br>[0.25]     |
| <i>Race: Hispanic</i>                             | 0.05<br>(0.22)     | 0.04<br>(0.20)   | 0.07<br>(0.25)   | 0.06<br>(0.23)   | 0.04<br>(0.18)   | -0.03<br>[0.27] | 0.02<br>[0.33]     |
| <i>Race: Asian</i>                                | 0.05<br>(0.21)     | 0.02<br>(0.14)   | 0.05<br>(0.22)   | 0.05<br>(0.21)   | 0.07<br>(0.26)   | -0.03<br>[0.11] | -0.02<br>[0.29]    |
| <i>Education: High school and less</i>            | 0.24<br>(0.43)     | 0.22<br>(0.41)   | 0.29<br>(0.45)   | 0.16<br>(0.37)   | 0.30<br>(0.46)   | -0.07<br>[0.12] | -0.14***<br>[0.00] |
| <i>Education: Bachelor's degree</i>               | 0.46<br>(0.50)     | 0.45<br>(0.50)   | 0.45<br>(0.50)   | 0.51<br>(0.50)   | 0.44<br>(0.50)   | 0.00<br>[0.99]  | 0.07<br>[0.15]     |
| <i>Education: Graduate or professional degree</i> | 0.29<br>(0.46)     | 0.33<br>(0.47)   | 0.26<br>(0.44)   | 0.33<br>(0.47)   | 0.26<br>(0.44)   | 0.07<br>[0.13]  | 0.07<br>[0.14]     |
| <i>Income: Less than \$10,000</i>                 | 0.03<br>(0.16)     | 0.03<br>(0.16)   | 0.03<br>(0.17)   | 0.03<br>(0.16)   | 0.03<br>(0.16)   | 0.00<br>[0.77]  | 0.00<br>[0.99]     |
| <i>Income: \$10,000 - \$34,999</i>                | 0.14<br>(0.35)     | 0.14<br>(0.35)   | 0.19<br>(0.39)   | 0.09<br>(0.29)   | 0.16<br>(0.36)   | -0.05<br>[0.19] | -0.06**<br>[0.05]  |
| <i>Income: \$35,000 - \$74,999</i>                | 0.33<br>(0.47)     | 0.29<br>(0.45)   | 0.31<br>(0.46)   | 0.35<br>(0.48)   | 0.36<br>(0.48)   | -0.02<br>[0.63] | 0.00<br>[0.95]     |
| <i>Income: \$75,000 - \$149,999</i>               | 0.37<br>(0.48)     | 0.40<br>(0.49)   | 0.33<br>(0.47)   | 0.38<br>(0.49)   | 0.35<br>(0.48)   | 0.07<br>[0.15]  | 0.03<br>[0.51]     |
| <i>Income: \$150,000 or more</i>                  | 0.13<br>(0.34)     | 0.14<br>(0.35)   | 0.14<br>(0.35)   | 0.15<br>(0.35)   | 0.11<br>(0.31)   | 0.01<br>[0.85]  | 0.04<br>[0.29]     |
| <i>Employment: Full time</i>                      | 0.65<br>(0.48)     | 0.64<br>(0.48)   | 0.57<br>(0.50)   | 0.69<br>(0.46)   | 0.69<br>(0.46)   | 0.08<br>[0.12]  | 0.00<br>[0.97]     |
| <i>Policy: Extreme or leaning left</i>            | 0.36<br>(0.48)     | 0.31<br>(0.47)   | 0.30<br>(0.46)   | 0.40<br>(0.49)   | 0.42<br>(0.50)   | 0.01<br>[0.77]  | -0.02<br>[0.64]    |
| <i>Policy: Extreme or leaning right</i>           | 0.42<br>(0.49)     | 0.46<br>(0.50)   | 0.48<br>(0.50)   | 0.36<br>(0.48)   | 0.37<br>(0.48)   | -0.02<br>[0.76] | 0.00<br>[0.95]     |
| <i>Policy: Center</i>                             | 0.22<br>(0.42)     | 0.22<br>(0.42)   | 0.22<br>(0.41)   | 0.24<br>(0.43)   | 0.21<br>(0.41)   | 0.00<br>[0.96]  | 0.03<br>[0.53]     |
| <i>Residence: U.S.</i>                            | 1.00<br>(0.06)     | 0.99<br>(0.10)   | 1.00<br>(0.00)   | 1.00<br>(0.00)   | 0.99<br>(0.07)   | -0.01<br>[0.15] | 0.01<br>[0.32]     |
| <i>Nationality: U.S.</i>                          | 0.99<br>(0.11)     | 0.99<br>(0.10)   | 0.98<br>(0.14)   | 0.99<br>(0.10)   | 0.99<br>(0.07)   | 0.01<br>[0.42]  | -0.01<br>[0.56]    |
| Observations                                      | 787                | 194              | 196              | 198              | 199              | 390             | 397                |

*Notes:* Means are reported with standard deviations in parentheses. P-val in brackets. All variables are coded as indicators and shown as proportions. Column (1) reports statistics for the full sample. Columns (2)–(3) present treatment and control groups under gain framing, and Columns (4)–(5) present treatment and control groups under loss framing. Column (6) shows mean differences between groups under gain framing, with p-values from two-sided t-tests in brackets, while Column (7) reports the same for loss framing. “Observations” indicates the number of responses in each condition.

failed the attention check and three submitted incomplete responses. Excluding these eleven participants leaves a final sample of 787.<sup>33</sup> Attrition was minimal. The median completion time was eighteen minutes.<sup>34</sup>

Demographic characteristics and balance tests are presented in [Table 1](#). The sample is well balanced across experimental conditions. Approximately half of the participants are female, with an average age of forty years. About sixty percent identify as White and twenty-eight percent as African American. Nearly one quarter of participants have a high school education or less, forty-six percent hold a bachelor’s degree, and twenty-nine percent have a graduate or professional degree. Approximately one third report annual household incomes between thirty-five thousand and seventy-five thousand dollars, and two thirds are employed full time. Demographic differences across the four experimental conditions are small. In the gain-framed treatments, mean values for age, gender, education, income, and political orientation are nearly identical between the competition and control groups. In the loss-framed treatments, two characteristics show modest differences: the proportion with a high school education or less and the proportion with incomes between ten thousand and thirty-five thousand dollars. However, these differences are not statistically concerning, as *F*-tests show no evidence of systematic imbalance across treatments (*F*-test,  $p = 0.61$  for gain-framing competition vs. control; *F*-test,  $p = 0.17$  for loss-framing competition vs. control).

## 4 Interpretation of the Competition Treatment

This section interprets what “competition” means in the context of the experiment. The competition treatment is characterized by three key features. First, it follows the standard definition of competition used in the literature: a situation where individuals’ payoffs depend not only on their own performance but also on the performance of others, and where a scarce resource must be competed for. This definition is widely used in the experimental studies for designing competitive environments ([Niederle and Vesterlund, 2007](#); [Bartling et al., 2012](#); [Gill and Prowse, 2012](#); [Delfgaauw et al., 2013](#)). Consistent with this definition, the payoff structure in my experiment is designed such that, in the competition

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<sup>33</sup>The excluded participants included three from gain-framed competition, four from gain-framed control, one from loss-framed Competition, and three from loss-framed Control.

<sup>34</sup>The target sample size was eight hundred, with two hundred participants per group. The shortfall of two resulted from minor technical issues on the Prolific platform. The median completion time was eighteen minutes in gain-framed Competition, seventeen minutes in gain-framed Control, nineteen minutes in loss-framed Competition, and seventeen minutes in loss-framed Control.

treatments, participants' monetary rewards depend on both their own quiz performance and their opponents' performance, and only one of the two could receive the reward. In the non-competition treatments, participants' monetary payoffs depend solely on their own performance. As long as their quiz score exceeds a pre-specified performance threshold, they receive the reward, regardless of others' outcomes; otherwise, they earn nothing.

The second feature is naturally embedded in the first. While individuals often perceive this as a default characteristic of competition without explicitly recognizing it, the comparison between competition and non-competition treatments makes this feature particularly salient. The feature is the awareness of other individuals also working on the task. In the competition treatments, participants were shown a matching screen while waiting to be paired, making it clear that another participant would be competing with them. In the non-competition treatments, this feature was absent. Participants completed the task without any explicit information indicating that others might also be working on it.

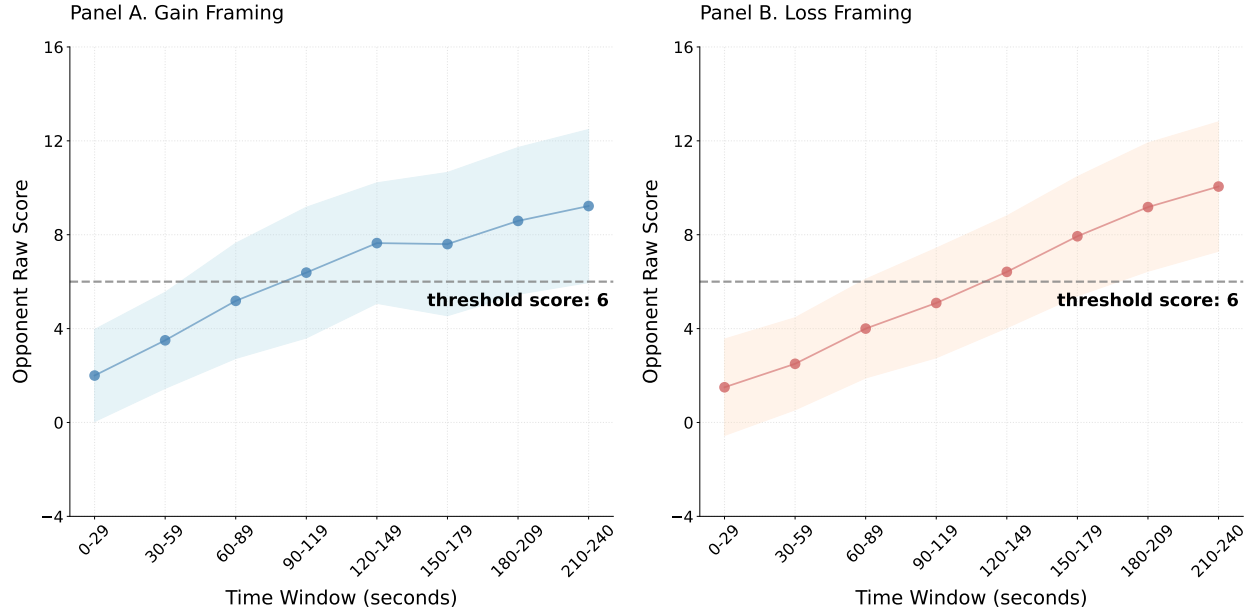
The third feature captures the information aspect of competitive contexts.<sup>35</sup> Although the information component is not a defining element of all competitive environments, it is commonly present in many forms of competition. Individuals often know something, but not everything, about other competitors, such as their ability, effort, or past performance. In the experiment, the opponent's raw score bar served this role, providing noisy information about the opponent's performance during the task. Clearly identifying this feature is important for interpreting the experimental findings, as it helps to uncover underlying mechanisms. It also suggests caution when generalizing the interpretation of the experimental results to other competitive contexts.

Figure 6 provides further detail regarding the information aspect. It shows the dynamics of the opponent raw score displayed to participants during the competition treatments. The four-minute quiz is divided into eight 30-second intervals, and for each interval I record the mean opponent score, effectively taking a "snapshot" of what participants observed on their opponent's score bar at each point in time. The figure reveals that participants in the gain-framed and loss-framed competition treatments observe similar patterns in the evolution of the opponent's score. On average, the opponent answers one question correctly every 30 seconds. Compared with the fixed threshold score of six in the non-competition treatments, where the target score remained constant throughout the entire four minutes, the competition treatments feature a continuously rising benchmark. The opponent's score in the competition treatments rose steadily over time. In the gain-framed

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<sup>35</sup>Here, "information" refers not to information about payoffs, but to any information related to the competitive setting, such as an opponent's performance.

Figure 6: Opponent Raw Scores across 30-Second Time Windows



*Notes:* This figure shows how the average opponent's raw score evolves over time in the competition treatments. The horizontal axis divides the four-minute quiz into 30-second intervals, and the vertical axis plots the mean opponent score within each window. The solid blue line in Panel A represents the gain-framed condition, and the solid red line in Panel B represents the loss-framed condition. The dashed horizontal line indicates the fixed threshold score of six used in the control treatments. In both framings, the opponent's score increases by approximately one point every 30 seconds, demonstrating that, unlike the fixed benchmark in the control condition, the competitive setting features a continuously rising reference point.

competition, participants see their opponent's raw score below six points during the first ninety seconds, then rise above six points thereafter. In the loss-framed competition, participants see their opponent's raw score below six points during the first two minutes, then rise above six points thereafter. If we view the threshold in the non-competition treatments as a fixed reference point for participants, then the opponent's raw score in the competition treatments functions as a moving reference point throughout the four-minute quiz. This figure provides detailed patterns of how exactly the "competition" treatment operates during the experiment.

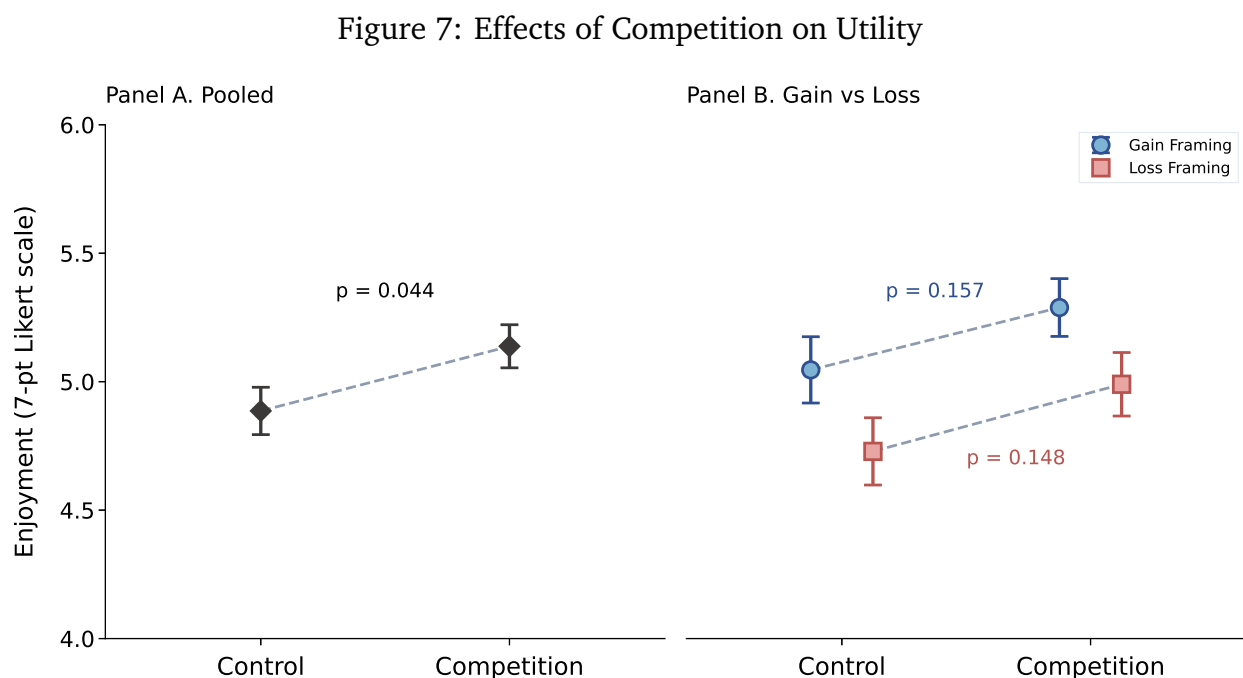
## 5 Results

This section provides experimental evidence that individuals derive utility directly from competition itself, independent of final outcomes. [Section 5.1](#) presents the average treatment effect of competition on utility. [Section 5.2](#) and [Section 5.3](#) investigate two un-

derlying mechanisms: a belief channel, in which competition reduces utility by lowering individuals' winning expectations, and a preference channel, through which individuals enjoy competition for its own sake. [Section 5.4](#) explores heterogeneity in these effects and discusses how they contribute to welfare evaluation.

## 5.1 Effects of Competition on Utility

[Figure 7](#) illustrates the average treatment effects of competition on utility. As shown in the figure, participants in the competition treatments had significantly higher levels of utility than those in the non-competition treatments.



*Notes:* This figure plots the treatment effects of competition on utility. Panel A presents pooled results, while Panel B shows results separately for the gain and loss framing conditions. In both panels, the horizontal axis indicates the treatment groups (Control vs. Competition), and the vertical axis presents the average response to the survey question: “On a scale from 1 (not at all enjoyable) to 7 (extremely enjoyable), how enjoyable was the IQ quiz (or IQ quiz competition)?” Each point in the figure represents the mean reported enjoyment within the corresponding group, with 95% confidence intervals indicated. Corresponding regression results are provided in [Table 2](#).

Panel B of [Figure 7](#) shows the average reported enjoyment of the experimental task in the gain- and loss-framing conditions, separately for the control and competition treatments. Under gain framing, average reported enjoyment rises from 5.05 to 5.29 ( $p = 0.157$ ); under loss framing, it increases from 4.73 to 4.99 ( $p = 0.148$ ). Panel A pools

observations across framing conditions, combining the control groups under gain and loss framing into a single control category, and the competition groups into a single competition category. Pooling is appropriate because the patterns under gain and loss framing are qualitatively similar, and the expected payoffs are identical by design across framing conditions. One advantage of pooling is that it increases statistical power. In the pooled sample, the mean reported enjoyment is 4.89 in the control group and 5.14 in the competition group, representing a statistically significant 5.1% increase ( $p = 0.044$ ). These findings suggest that competition increases participants' experienced utility of the task. Notably, as shown in panel B, average enjoyment under gain framing is consistently higher than that under loss framing, and this difference is statistically significant ( $p = 0.014$ ). Nevertheless, the difference in the treatment effects of competition on enjoyment under the two framing conditions is negligible ( $p = 0.941$ ).

**RESULT 1.** *Competition increases experienced utility, independently of final outcomes.*

## 5.2 Belief Channel

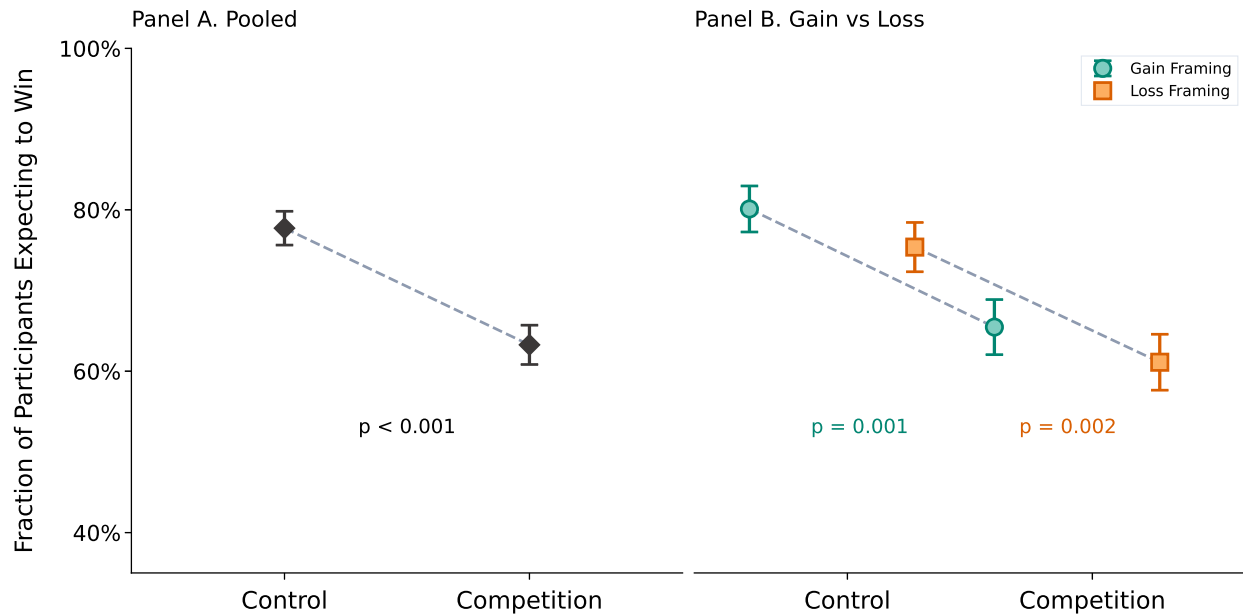
As outlined in [Section 2](#), competition affects utility through three channels: a belief channel that lowers utility, a preference channel that increases it, and an effort channel that is well-controlled in this experiment. This section focuses on the belief channel, while the preference and effort channels are discussed in [Section 5.3](#).

Through the belief channel, competition reduces utility by making participants less optimistic about their chances of winning. Lower expectations of success, in turn, lower their experienced utility. The negative effect of competition on beliefs about winning is illustrated in [Figure 8](#). The positive association between beliefs about winning and experienced utility is shown in [Figure 9](#).

Panel B of [Figure 8](#) shows the share of participants who believe they won the competition or met the pre-specified performance threshold, separately for the control and competition groups under gain and loss framing conditions. Under gain framing, 85% of participants in the control group believe they reached the median threshold score, whereas only 65% in the competition condition believe they outperformed their opponent. This represents a statistically significant decline of 20 percentage points ( $p = 0.001$ ). Under loss framing, 75% believe they reached the median threshold, while only 61% believe they won the competition, corresponding to a decline of 14 percentage points ( $p = 0.002$ ). Panel A aggregates observations across framing conditions, pooling the control groups into a sin-



Figure 8: Effects of Competition on Belief in Winning



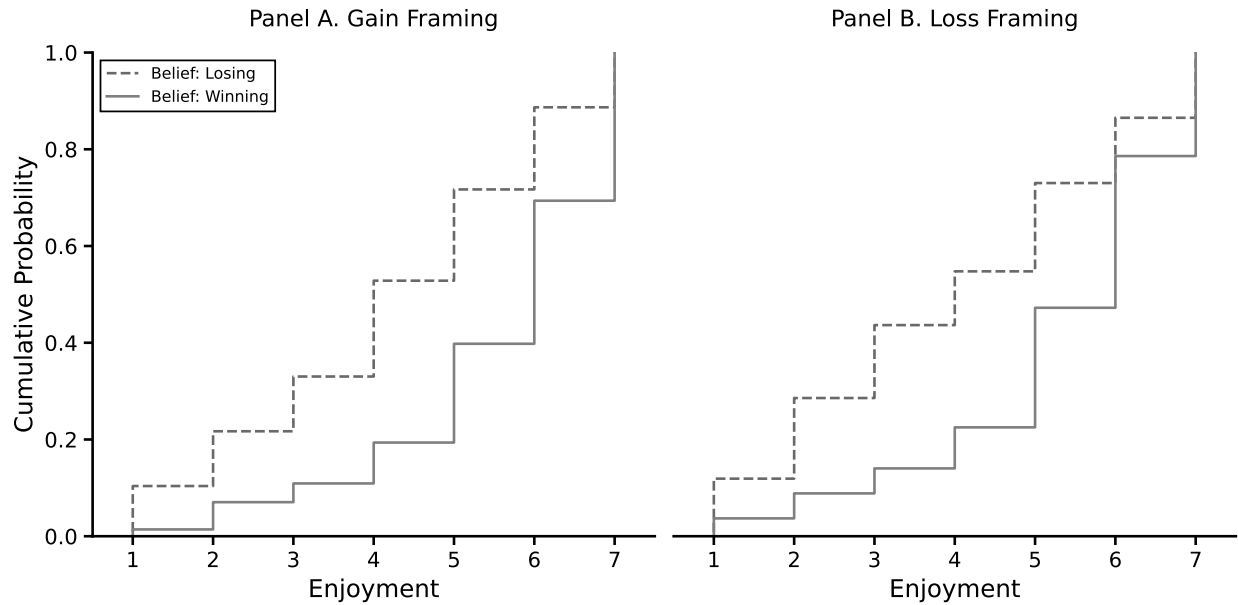
*Notes:* This figure illustrates how competition affects whether participants believe they can win. Panel A shows the pooled effect across all framing conditions, while Panel B separates results by gain versus loss framing. The y-axis represents the fraction of participants who expected to win. Across both panels, participants in the competition condition were significantly less likely to believe they would win compared to those in the control condition.

gle control sample and the competition groups into a single competition sample. In the pooled control group, 78% of participants believe they reached the threshold, while only 63% in the competition group believe they beat their opponent, representing a decline of 15 percentage points ( $p < 0.001$ ). These results consistently demonstrate that competition reduces participants' beliefs about winning.

As for the reasons behind this pattern, I propose two hypotheses. First, the opponent score bar, a key design feature of the experiment, may play an important role. This bar, shown only in the competition treatments, displayed the opponent's raw score, which rose monotonically over time. Because it counted only correct answers but not incorrect ones, it never declines. By contrast, participants in the control groups faced a fixed performance threshold that remained constant throughout the task. Given that participants received no feedback on their own performance, the ever-rising opponent score bar likely generated competitive pressure and biased participants toward pessimistic beliefs about their chances of winning. Second, competition inherently ties one's payoff to others' performance. In the competition treatments, outcomes depended on an unpredictable opponent's performance, which likely increased participants' perceived uncertainty and undermined their

confidence in winning. In contrast, the control group's fixed threshold provided a transparent and clearly defined target, enabling participants to calibrate effort effectively and therefore feel more confident about success. Together, the dynamic opponent score bar and the opponent-dependent payoff structure offer plausible explanations for why competition reduces participants' belief about winning.

Figure 9: Cumulative Distribution of Enjoyment by Winning Expectations



*Notes:* This figure plots the cumulative probability distributions of participants' self-reported enjoyment (1–7 scale) across two framing conditions: gain framing (Panel A) and loss framing (Panel B). Within each panel, the solid line represents participants who believed they were winning, and the dashed line represents those who believed they were losing. The figure shows that participants who believed they were winning reported higher enjoyment (distribution shifted to the right) compared to those who believed they were losing.

Figure 9 shows the cumulative distributions of reported enjoyment for participants who believed they won<sup>36</sup> and those who believed they lost.<sup>37</sup> Panel A presents results under gain framing and Panel B presents results under loss framing. In both framing conditions, the distribution for participants who believed they were winning lies to the right of those who believed they were losing (Kolmogorov–Smirnov test:  $p = 0.031$  under gain framing;  $p = 0.035$  under loss framing), indicating that participants who believed

<sup>36</sup>Winning refers to outperforming one's opponent in the competition treatments and to passing the performance threshold in the control groups.

<sup>37</sup>Losing refers to underperforming relative to one's opponent in the competition treatments and to failing to reach the performance threshold in the control groups.

they were winning reported greater enjoyment than those who believed they were losing throughout the distribution. For example, under gain framing, approximately 70% of those who believed they had lost reported enjoyment levels of six or below, while only about 40% among those who believed they had won did so. This implies that a larger share of winning believers experienced higher enjoyment levels. These patterns suggest a positive association between winning expectations and experienced utility.<sup>38</sup>

A plausible explanation for this positive relationship is belief-based utility, meaning that individuals derive utility from their beliefs about future outcomes (Caplin and Leahy, 2001). For instance, people may feel anxious before visiting a dentist; their utility changes as they anticipate the event, even before it actually occurs. In the experiment, participants may likewise derive utility from anticipating success. This utility may arise from the prospect of winning monetary rewards or from the psychological satisfaction of success itself, either the sense of superiority from outperforming others under competition or the feeling of achievement from meeting a performance threshold in the non-competitive setting.

Taken together, these findings indicate that participants who believed they were likely to win reported higher enjoyment than those who believed they were likely to lose. Combined with the earlier result that competition reduces participants' beliefs about their chances of winning, this suggests that competition lowers experienced utility through its negative effect on beliefs about winning, a mechanism that I refer to as the belief channel.

**RESULT 2.** *Competition affects utility through a belief channel: it reduces utility by lowering individuals' beliefs about winning.*

### 5.3 Preference Channel

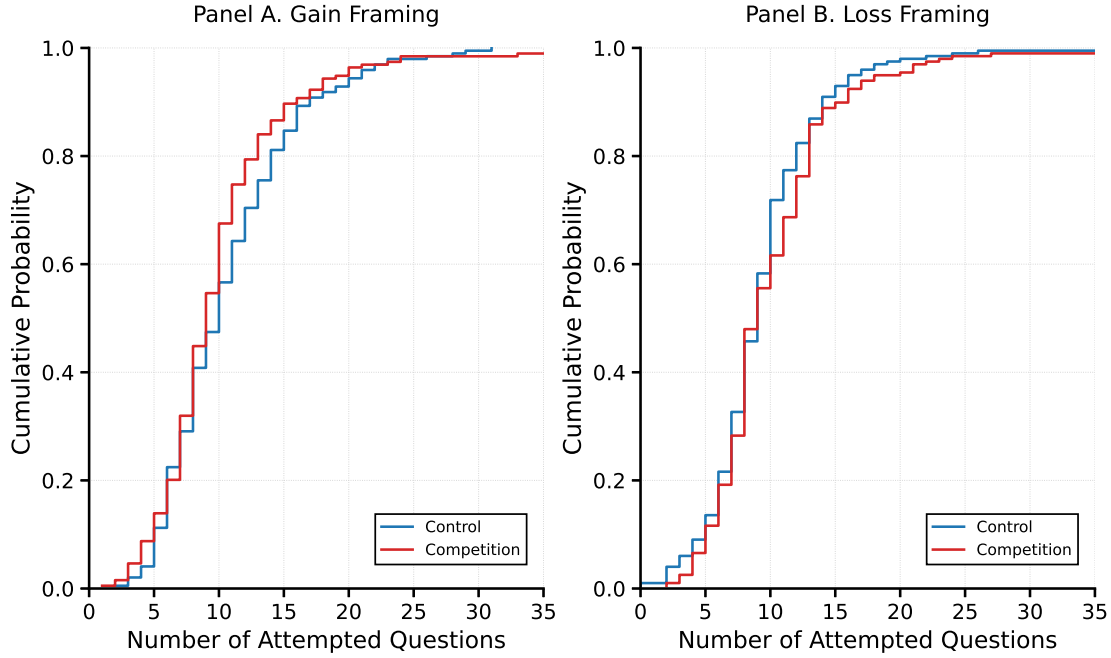
Competition affects utility through a preference channel: individuals derive enjoyment directly from the act of competing, independent of final outcomes. According to the theoretical framework in Section 2, competition impacts utility through three channels: belief, preference, and effort. To isolate the preference channel, I control for the beliefs and effort when estimating the overall impact of competition on utility. I proceed in two steps. First, I demonstrate that effort levels are nearly identical between competition and non-competition treatments (see Figure 10). Second, I run OLS regressions controlling for beliefs to obtain estimates for the preference channel (see Table 2).<sup>39</sup>

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<sup>38</sup>Further details and regression results are provided in Appendix Table B.3.

<sup>39</sup>Table 2 also includes specifications where effort is added as a control variable. I proceed with this two-step approach to avoid potential bad control problems. The findings are the same regardless of approach.

Figure 10: Cumulative Distribution of Effort Proxy: Attempted Questions



*Notes:* This figure shows the cumulative distribution of the effort proxy, measured by the total number of attempted questions in the IQ quiz, for the control and competition groups under gain and loss framings. Panel A corresponds to the gain-framed condition and Panel B to the loss-framed condition. In both framings, the distributions for competition and control groups almost perfectly overlap, indicating that participants exerted similar levels of effort regardless of whether they competed against another participant or faced a fixed threshold. Kolmogorov–Smirnov tests confirm no significant differences across treatments ( $p = 0.198$  under gain framing;  $p = 0.249$  under loss framing).

Figure 10 shows the cumulative distribution of the effort proxy, the number of attempted questions in the quiz, for the control and competition groups, separately under gain and loss framing conditions. Effort levels appear nearly identical across treatment groups under both framings (Kolmogorov–Smirnov test,  $p = 0.198$  under gain framing;  $p = 0.249$  under loss framing). In the gain frame, the competition group’s cumulative distribution lies slightly above that of the control group only for the number of attempted questions between ten and twenty, suggesting marginally lower effort in the competition group. Under the loss frame, the two distributions almost perfectly coincide, with the competition group’s curve slightly higher in the same range of ten to twenty attempts. However, none of these differences are statistically significant at the 5% significance level, which suggests that competition has little impact on effort regardless of the framing of reward. Consequently, the effort channel appears unlikely to be a major pathway through which competition affects utility in this experiment.

Table 2: Treatments Effects of Competition on Utility

|                   | <i>Dependent Variable: Enjoyment</i> |                   |                   |                |                   |                   |                |                   |                   |
|-------------------|--------------------------------------|-------------------|-------------------|----------------|-------------------|-------------------|----------------|-------------------|-------------------|
|                   | Pooled                               |                   |                   | Gain Framing   |                   |                   | Loss Framing   |                   |                   |
|                   | (1)                                  | (2)               | (3)               | (4)            | (5)               | (6)               | (7)            | (8)               | (9)               |
| Competition       | 0.25**<br>(0.12)                     | 0.45***<br>(0.12) | 0.45***<br>(0.12) | 0.24<br>(0.17) | 0.45***<br>(0.16) | 0.43***<br>(0.16) | 0.26<br>(0.18) | 0.45***<br>(0.17) | 0.48***<br>(0.17) |
| Belief in Winning |                                      | 1.36***<br>(0.14) | 1.37***<br>(0.14) |                | 1.39***<br>(0.20) | 1.39***<br>(0.20) |                | 1.31***<br>(0.20) | 1.33***<br>(0.20) |
| Effort proxy      |                                      |                   | ✓                 |                |                   | ✓                 |                |                   | ✓                 |
| Control mean      | 4.89                                 | 4.89              | 4.89              | 5.05           | 5.05              | 5.05              | 4.73           | 4.73              | 4.73              |
| Observations      | 787                                  | 787               | 787               | 390            | 390               | 390               | 397            | 397               | 397               |
| R <sup>2</sup>    | 0.01                                 | 0.13              | 0.13              | 0.01           | 0.13              | 0.14              | 0.01           | 0.12              | 0.13              |

*Notes:* This table summarizes OLS estimates of the effect of competition on utility, both overall and net of the belief (and effort) channels. The dependent variable is participants' responses to the question: "On a scale of 1 (not at all enjoyable) to 7 (extremely enjoyable), how enjoyable was the IQ Quiz Competition (or IQ Quiz)?" Belief in Winning is an indicator based on post-treatment responses to "Do you think you were the top performer in the IQ Quiz competition?" (or "Do you think you scored 6 or higher on the IQ Quiz?"). The cutoff of 6 corresponds to the median score in the competition group. Columns (1)–(3) use pooled data, while columns (4)–(6) and (7)–(9) report estimates separately for the gain and loss framing conditions. Control means refer to average enjoyment in the control group. Robust standard errors are in parentheses. The effort proxy in this table is the number of questions each participant attempted. Alternative proxies for effort, including quiz score, number of correctly answered questions, and average time spent per question, are reported in the Appendix Table B.1. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 2 presents the regression results for the impact of competition on utility for the pooled sample, gain-framed treatments, and loss-framed treatments. The baseline specifications (columns 1, 4, and 7) capture the overall effect of competition on utility, reflecting the combined influence through the belief, preference, and effort channels. In columns 2, 5, and 8, beliefs are included as control variables to shut down the belief channel. Given that the effort channel is negligible in this experiment (as shown in Figure 10), the coefficients of competition in these specifications therefore reflect the impact of competition on utility only through the preference channel in each corresponding sample. The estimated coefficients nearly double from column 1 to 2 (pooled sample), from column 4 to 5 (gain-framed), and from column 7 to 8 (loss-framed). This pattern suggests that, once differences in beliefs are accounted for, the remaining effect, which represents intrinsic utility from the act of competing, is both positive and substantial. As a further robustness check, columns 3, 6, and 9 include the effort proxy as an additional control variable, shutting down both the belief and effort channels simultaneously. The competition coefficients

in these specifications isolate the preference channel exclusively, capturing only the direct utility derived from competing.<sup>40</sup> The estimates from these specifications are nearly identical to those that control only for beliefs, confirming the robustness of the findings. The estimated preference-channel effects are stable across both gain and loss framings, further confirming the robustness of this finding: individuals derive positive intrinsic utility solely from the act of competing.

**RESULT 3.** *Competition affects utility through a preference channel: individuals derive enjoyment from the act of competing itself.*

## 5.4 Discussion

### A. Heterogeneity

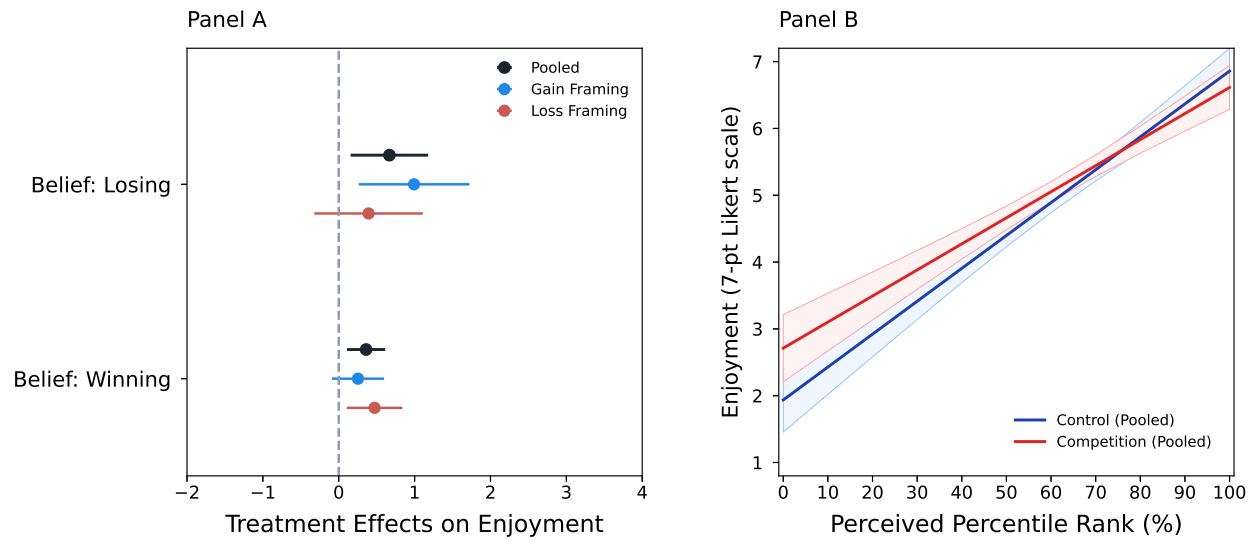
Figure 11 shows heterogeneous treatment effects of competition on utility across participants with different beliefs about winning and different perceived relative ability. Panel A illustrates that competition increases enjoyment for both participants who believe they won and those who believe they lost. In the pooled sample, the effect is slightly larger among those who believe they lost, although the difference between the two groups is not statistically significant ( $p = 0.242$ ). The pattern differs across framing conditions. Under gain framing, competition increases reported enjoyment for participants who believe they lost but not for those who believe they won. Under loss framing, the pattern is the opposite: competition raises reported enjoyment for participants who believe they won but not for those who believe they lost.

Panel B shows that enjoyment increases monotonically with perceived performance rank, indicating that individuals who perceive themselves as more capable derive greater task utility overall. Comparing the control and competition groups, the increase in enjoyment under competition is mainly driven by participants below the 70th percentile of perceived rank, suggesting that competition enhances enjoyment particularly among those who view themselves as relatively less capable. Notably, the competition line (red) starts at a higher intercept than the control line (blue), demonstrating that low-ability individuals experience greater utility gains from competitive environments. Above the 70th percentile, enjoyment levels are nearly identical between the two groups, suggesting that competition generates little utility gains for participants who perceive themselves as highly capable.

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<sup>40</sup>Alternative effort proxies, such as quiz scores, the number of correctly answered questions, and the average time spent per question, are also included in the robustness analysis reported in Appendix Table B.1. The results remain highly consistent.

Figure 11: Heterogeneity in Utility by Beliefs in Winning and by Perceived Ability

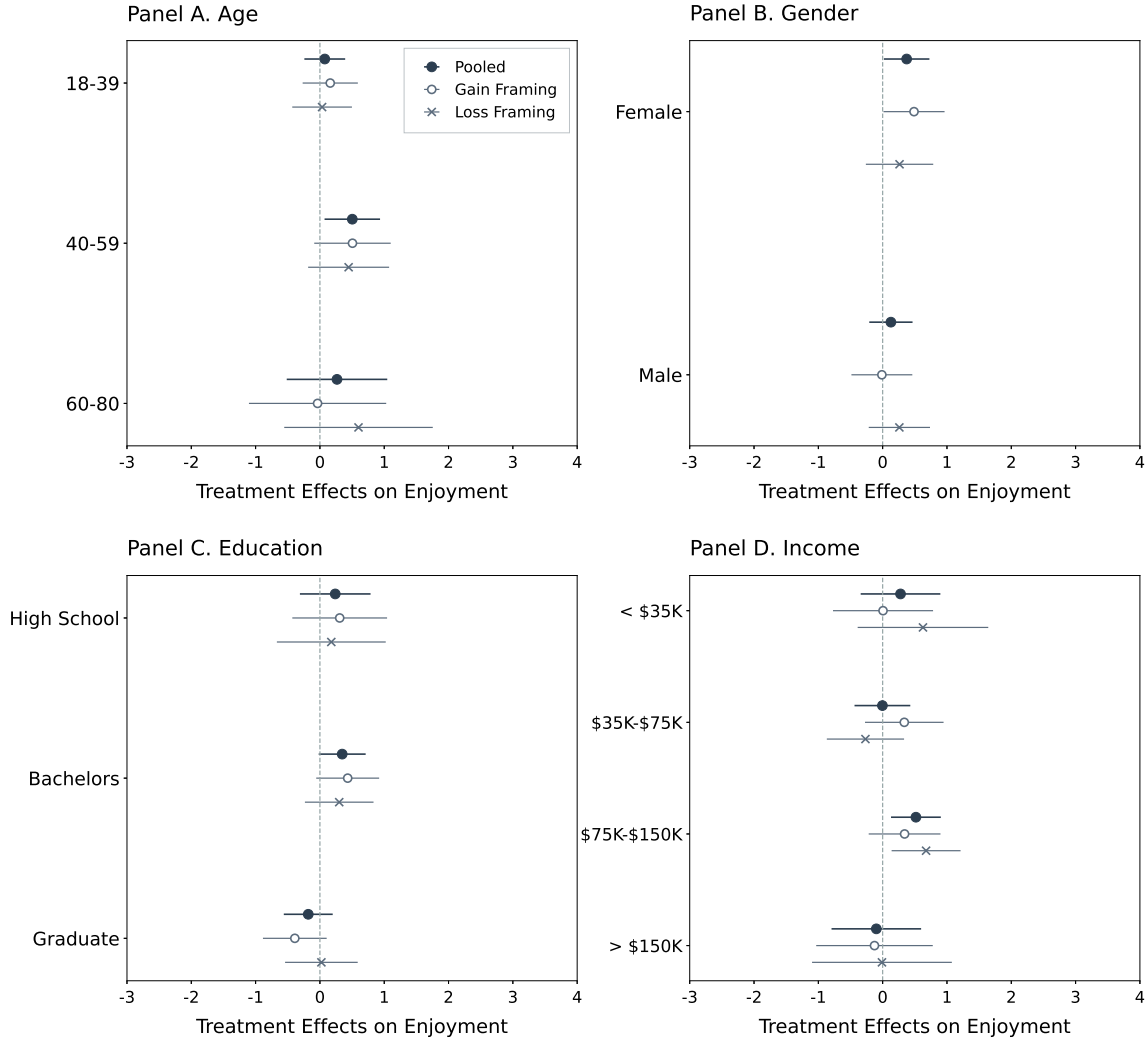


*Notes:* This figure shows the heterogeneous effects of competition on enjoyment by participants' beliefs about winning and by their perceived performance rank. Panel A presents the estimated treatment effects of competition on reported enjoyment, conditional on participants' beliefs about whether they believe they won or lost. The results are shown separately for the pooled sample (black), the gain framing condition (blue), and the loss framing condition (red). Each point corresponds to the estimated mean, with horizontal bars denoting 95% confidence intervals. Panel B illustrates the relationship between perceived percentile rank (as a proxy for perceived ability) and self-reported enjoyment on a 7-point Likert scale. The blue and red lines represent the pooled control and competition conditions, respectively, with shaded regions indicating 95% confidence intervals.

Figure 12 presents heterogeneity in the effect of competition on utility across demographic groups. In terms of age, the statistically significant impact of competition on utility is concentrated among individuals aged 40–59 ( $p = 0.053$ , pooled sample), whereas the effects for younger participants (18–39) and older participants (60–80) are small and statistically insignificant. By gender, competition significantly increases enjoyment among women ( $p = 0.040$ , pooled sample), but not among men; however, the gender difference in treatment effects is not statistically distinguishable ( $p = 0.327$ , pooled sample). Education patterns show that the treatment effect is strongest among respondents with a bachelor's degree ( $p = 0.063$ , pooled sample), while those with only a high school degree or with graduate training exhibit smaller and insignificant effects. Finally, heterogeneity by income indicates that competition significantly increases enjoyment for individuals with household incomes between \$75,000 and \$150,000 ( $p = 0.009$ , pooled sample), but not for those with lower or higher income levels.



Figure 12: Heterogeneity in Utility Across Demographic Groups



*Notes:* This figure shows the heterogeneous effects of competition on enjoyment across demographic groups. Enjoyment is measured by asking participants to rate, on a scale from 1 (not at all enjoyable) to 7 (extremely enjoyable), how enjoyable they found the IQ quiz in the control group or the IQ quiz competition in the treatment group. The filled circle denotes the treatment effect for the pooled sample, the hollow circle the gain framing condition, and the star the loss framing condition. All estimates are reported with 95% confidence intervals. Panel A presents results by age groups (18–39, 40–59, and 60–80), Panel B by gender, Panel C by education level (high school, bachelor's, and graduate), and Panel D by income categories (<\$35K, \$35K–\$75K, \$75K–\$150K, and >\$150K).

### B. Welfare Evaluation with Utility from Competition

A complete welfare assessment of competition should account for not only the utility derived from final outcomes but also for the direct utility costs and benefits generated by the act of competing. Economists often view competition as an effective means to incen-

tivize effort, improve efficiency, and promote welfare, but this instrumental perspective overlooks the fact that competition itself can directly affect individuals' welfare. Experimental evidence has shown that individuals derive utility directly from the experience of competition, independent of final outcomes.

While the experimental results suggest that competition can generate utility gains, it is equally possible that competition induces utility losses. The experimental findings reveal three channels through which competition affects utility: belief, preference, and effort. Holding the effort channel constant, competition becomes a source of enjoyment when the positive preference channel dominates the negative belief channel. This pattern may correspond to recreational forms of competition in everyday life, where competitive elements make activities more engaging. Conversely, when both the belief and preference channels are negative,<sup>41</sup> competition can feel oppressive and unpleasant. This scenario may correspond to highly competitive labor markets or academic environments, where competition itself generates enormous pressure and emotional strain. At the same time, the perceived probability of success declines sharply in such environments, further diminishing utility.

These direct utility benefits and costs play a non-negligible role in overall welfare. Even if competition leads to an efficient allocation of resources in outcome terms, ignoring the welfare effects from competition itself may bias the overall welfare evaluation downward when competition generates enjoyment, and upward when competition brings displeasure.

## 6 Implications

### 6.1 Does competition affect future choices?

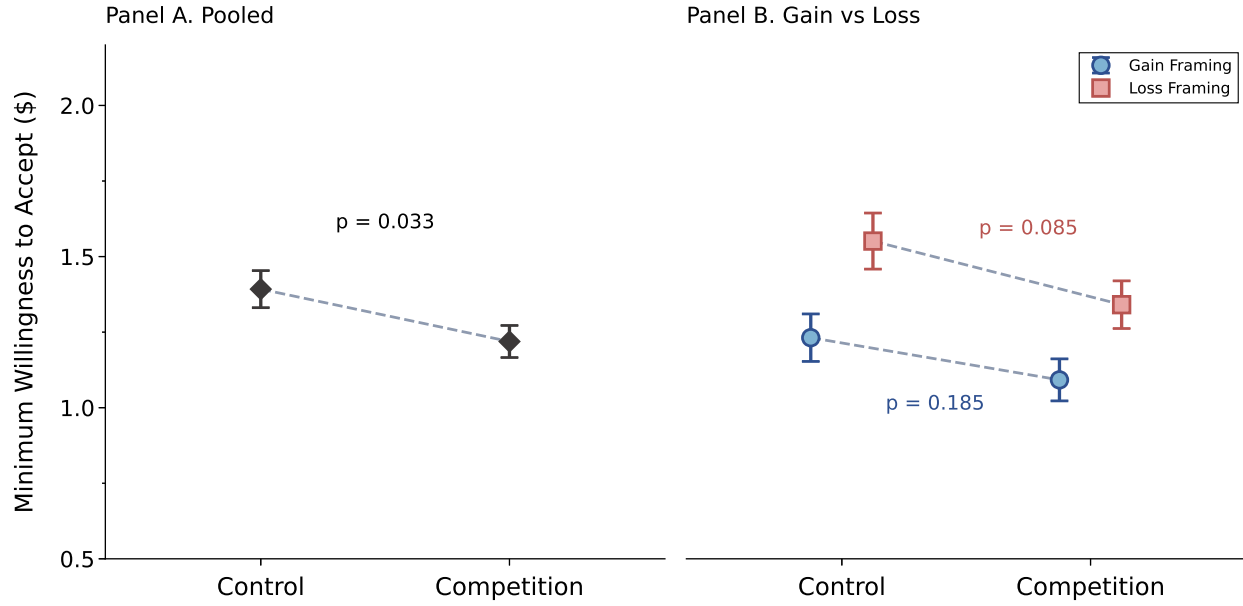
The experiment demonstrates that competition enhances individuals' enjoyment at work. I now investigate whether this increase in utility shapes their subsequent decisions, specifically whether it affects their willingness to repeat the task in the future. According to the concept of attribution bias,<sup>42</sup> people may develop a more favorable attitude toward an activity when they experience it in a particularly satisfying environment. In the experiment, participants may misattribute the enjoyment generated by competition to the inherent characteristics of the task itself.

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<sup>41</sup>In this experiment, the intrinsic utility from competing is positive. In theory, however, intrinsic utility could be negative, driven primarily by stress or competitive pressure.

<sup>42</sup>Attribution bias refers to the tendency to misattribute the influence of a temporary state to a stable property of an object or activity. For example, individuals who visit a new restaurant while very hungry may later judge the restaurant as high quality, even if the food itself is mediocre (Haggag et al., 2019).

Figure 13: Effects of Competition on the Task Choice



*Notes:* This figure shows how competition affects participants' willingness to undertake the same task again in the future. The vertical axis plots the average minimum willingness to accept (WTA) payment required to perform the IQ quiz again, where a higher value indicates a lower preference for the task. Panel A pools data across gain- and loss-framed conditions, while Panel B presents the results separately by framing. In both framings, participants in the competition condition require a lower WTA than those in the control condition, implying a greater willingness to repeat the task after experiencing competition. The difference is statistically significant when data are pooled across framings ( $p = 0.033$ ) and marginally significant under the loss framing ( $p = 0.085$ ).

Consistent with this hypothesis, I find that participants exposed to enjoyable competition are more likely to perceive the underlying task as pleasant and express a stronger willingness to engage in it again, even in the absence of competition. Figure 13 provides supporting evidence. The figure displays the average minimum amount of money participants would require to undertake the task again. A higher required amount indicates a lower preference for the task and a lower willingness to repeat it in the future. Panel B of Figure 13 presents results by framing condition. Under the gain framing, participants in the control group require at least \$1.23 on average to undertake the task again, whereas those in the competition group require \$1.09. This difference corresponds to a decline of \$0.14 ( $p = 0.185$ ). Under the loss framing, the control group requires at least \$1.55 to repeat the task, while the competition group requires \$1.34, representing a decline of \$0.21 ( $p = 0.085$ ). Panel A pools data across framing conditions. On average, control group participants require at least \$1.39 to undertake the task again, whereas those in the competition group require \$1.22 ( $p = 0.033$ ). These findings indicate that participants in the

competition group find the task itself more appealing and are more willing to re-engage, even after the competitive element has been removed.

**RESULT 4.** *Enjoyable competitive experiences increase participants' preference for the underlying task and their willingness to undertake it again, even in the absence of competition. This pattern indicates the presence of attribution bias in a strategic context of competition.*

## 6.2 Does competition affect subsequent social interactions?

The utility gain from competition not only affects individuals' preferences for specific tasks, but also impacts their social behavior. This subsection examines a second implication of competition-induced enjoyment: its impact on people's attitudes toward social interactions. In particular, I consider the impact on zero-sum thinking and altruism.

### A. Zero-sum Thinking

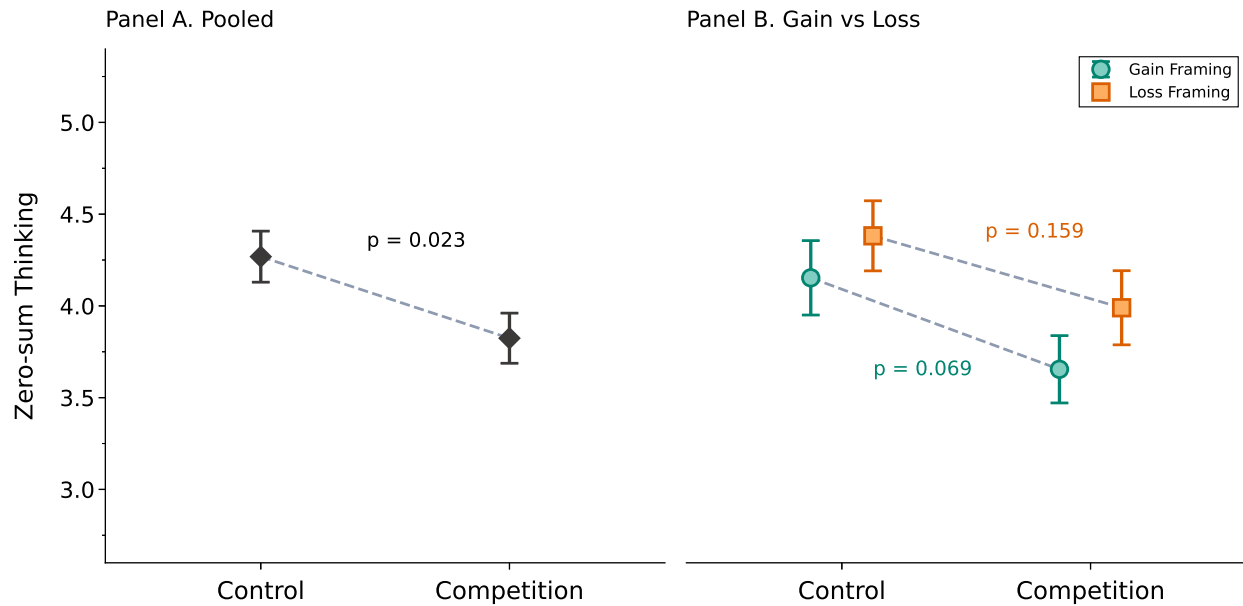
Figure 14 shows that competition reduces individuals' zero-sum mindsets. The measure of zero-sum thinking, based on the World Values Survey question described in Section 3.1, is normalized on a 1-10 scale, where 10 indicates a highly zero-sum view and 1 represents a non-zero-sum view.<sup>43</sup> Panel B of Figure 14 reports the mean value of zero-sum thinking for the control and competition groups, separately under both gain and loss framing conditions. Under the gain framing, the mean declines from 4.12 in the control group to 3.69 in the competition group ( $p = 0.069$ ). Under the loss framing, the decline is from 4.36 in the control group to 3.99 in the competition group ( $p = 0.159$ ). Panel A presents the pooled results, combining both gain and loss framing conditions. The mean zero-sum score in the pooled control group is 4.24, while in the pooled competition group it is 3.85 ( $p = 0.023$ ). These results indicate that participants exposed to competition exhibit lower levels of zero-sum thinking.

This finding runs counter to the pre-registered hypothesis that exposure to competition would increase zero-sum thinking, given that the competition's payoff structure in the experiment is zero-sum by design. This counterintuitive experimental result likely stems

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<sup>43</sup>A caveat is that although the World Values Survey item suggests competition reduces zero-sum thinking, this conclusion may not generalize. An alternative measure based on the immigrant-local job tradeoff, as described in Section 3.1, shows no significant treatment effects. The corresponding results are presented in Appendix Figure B.8, with regression estimates in Appendix Table B.9.

Figure 14: Effects of Competition on Zero-sum Thinking



*Notes:* This figure shows the effect of competition on zero-sum thinking. Zero-sum thinking is measured using a World Values Survey question (Inglehart et al., 2014) in which respondents rate their agreement, on a 1–10 scale, between two statements: “Wealth can grow so there’s enough for everyone” (1, indicating minimal zero-sum thinking) and “People can only become wealthy at the expense of others” (10, indicating strong zero-sum thinking). Higher values on the y-axis therefore represent stronger zero-sum thinking. The figure displays the average response values for the four treatment groups (gain framing vs. loss framing; control vs. competition), with 95% confidence intervals. Panel A presents results for the pooled sample, while Panel B shows results by gain and loss framing separately. Regression estimates are reported in Appendix Table B.8.

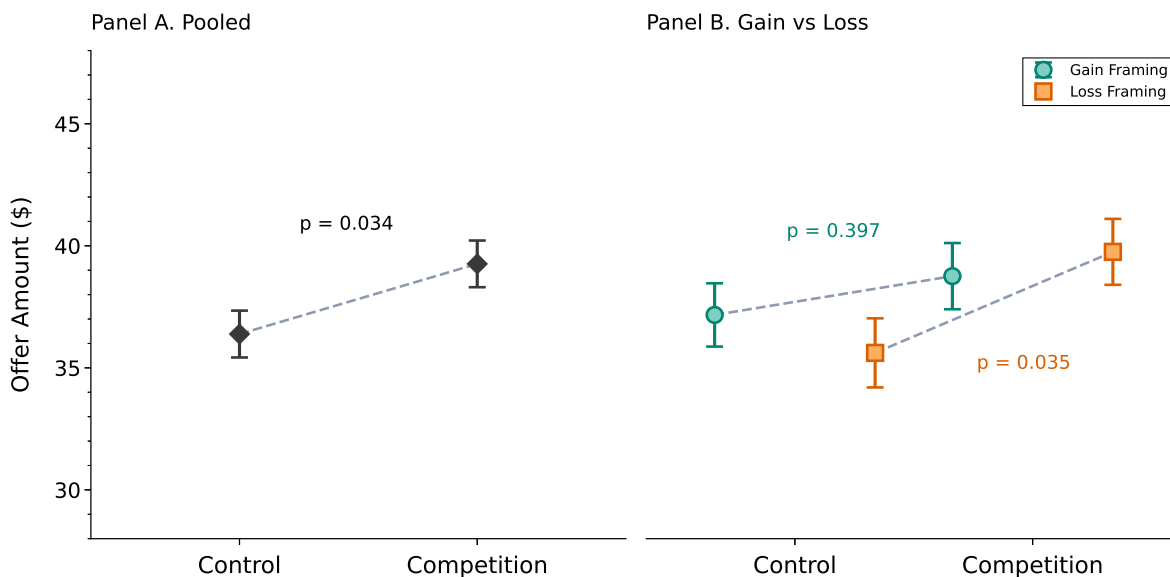
from a key feature of the competitive experience: participants found the competition enjoyable. This positive experience may reduce zero-sum thinking through two distinct but related pathways. First, the affective mechanism: enjoyable competition may induce positive emotions and elevated mood, which in turn affect how individuals perceive social and strategic interactions more broadly. Studies in psychology show that positive affective states tend to foster more optimistic and cooperative worldviews (Rand et al., 2015; Oriol et al., 2020; Mesurado et al., 2021). When participants feel good after competing, they may be more inclined to see the world as offering opportunities for mutual gain rather than fixed-pie tradeoffs. Second, the social connection mechanism: even though competition in the experiment has a zero-sum structure, the engaging nature of the interactive competing experience may strengthen participants’ sense of social connection. Humans are inherently social beings, and positive social interactions can generate feelings of connection and interpersonal bonds (Brown and Fredrickson, 2021; Delgado et al., 2023). These positive social emotions may counteract the competitive tension arising from the

task's zero-sum nature, leading participants to adopt less zero-sum views about society at large. Importantly, both explanations point to a common underlying factor: how the competitive experience is perceived. This suggests that the effect could reverse if competition were experienced as hostile rather than enjoyable. In such cases, competition might indeed increase zero-sum thinking.

## B. Altruism

Competition also influences individuals' altruistic behavior. Altruism is measured by the amount of money participants choose to offer to their randomly matched partner in a dictator game.

Figure 15: Effects of Competition on Altruism



*Notes:* This figure shows the effect of competition on altruism, measured using a dictator game in which participants allocated \$100 between themselves and another participant. The outcome is the amount offered to the partner. The figure illustrates mean offers across the four treatment groups (gain framing vs. loss framing; control vs. competition) with 95% confidence intervals. Panel A shows results for the pooled sample, and Panel B shows results separately by gain and loss framing. Corresponding regression estimates are provided in Appendix Table B.8.

Figure 15 shows the average offer amount from a total endowment of \$100, where larger offers indicate greater altruism. Panel A presents results for the pooled sample. Participants in the control group offered an average of \$36.38 to their partner, while those in the competition group offered \$39.26. Competition increased the average offer by 7.92 percent ( $p = 0.034$ ). Panel B shows the results by framing condition. Under gain

framing, average offers increased from \$37.16 to \$38.75 ( $p = 0.397$ ), while under loss framing, offers rose from \$35.61 to \$39.75 ( $p = 0.035$ ). Although the treatment effects of competition on altruism differ in magnitude across the two framing conditions, the difference is not statistically significant ( $p = 0.347$ ).

This positive effect of competition on altruism may arise from the same underlying mechanisms discussed in the zero-sum thinking section. An enjoyable competitive experience may induce a more positive affective state, thereby fostering other-regarding behavior. Alternatively, the interactive and socially engaging nature of competition may enhance feelings of connectedness, leading participants to treat others more generously and with greater kindness. To examine whether this pattern extends to other forms of prosocial behavior, I also analyze decisions in a public goods game, where participants decided whether to contribute their endowment to a common pool or keep it for themselves. Participants also reported their beliefs about the percentage of others who chose not to contribute, as described in [Section 3.1](#). The results show no statistically significant effects of competition on either contribution decisions or beliefs about the share of non-contributors.

**RESULT 6.** *Enjoyable competitive experiences reduce zero-sum thinking and foster altruism.*

## 7 Conclusion

Economists often treat competition as an instrument for incentivizing effort, improving efficiency, and ultimately enhancing welfare ([Smith, 1937](#); [Stigler, 1957](#); [Dhingra and Morrow, 2019](#); [Englmaier et al., 2024](#)). However, this perspective overlooks an important dimension: competition itself can be a direct source of welfare. This paper develops a simple conceptual framework and conducts a series of carefully designed experiments to explore this idea. By evaluating the utility costs and benefits of competition per se, it demonstrates that the intrinsic welfare value of competition warrants greater attention.

The experimental evidence indicates that competition generates positive utility. This finding, however, comes with a caveat: the laboratory setting uses relatively low-stakes rewards, leaving open the question of whether competition might instead generate disutility in high-stakes environments. Prior research suggests that individuals' performance can differ substantially between low- and high-stakes contexts ([Ariely et al., 2009](#); [Apesteguia and Palacios-Huerta, 2010](#)). Competition may be enjoyable and engaging in low-stakes settings, yet it can become stressful and welfare-reducing in high-pressure situations such

as elite university admissions or intensely competitive labor markets. While the experiments in this paper cannot fully resolve this question, they offer valuable insights into the mechanisms through which competition affects utility.

First, there is a belief channel: competition can lower individuals' perceived probability of success, which in turn reduces their utility. Second, there is a preference channel, which captures the intrinsic utility derived from act of competing. In the experiment, positive emotions such as excitement and satisfaction dominated negative feelings such as pressure and anxiety, resulting in net positive intrinsic utility. Third, there is an effort channel, which is well-controlled in the experiment. An interesting observation is that when rewards are gain-framed, performance in the non-competitive setting, where individuals work independently to surpass a threshold, is better than in the competitive setting. This runs counter to the common expectation that competition motivates greater effort, assuming performance serves as a reasonable proxy for effort.

These findings about how competition affects utility also have important implications for individuals' future choices and social attitudes. The enjoyable nature of the competitive experience, shapes subsequent behavior through multiple pathways. A positive competitive experience can increase individuals' intrinsic interest in the underlying task and their willingness to engage in it again. In practical terms, someone who finds competition enjoyable may be more inclined to participate in similar activities in the future, even in the absence of competition. A potential hypothesis generated from this finding is that if students experience distress under intense academic competition, they may develop an aversion to studying itself, potentially hindering their long-term human capital accumulation. The character of competition also influences social interactions. A pleasant competitive experience reduces zero-sum thinking and fosters greater altruism. By extension, it is plausible that an unpleasant competitive experience might have the opposite effect. These hypotheses require further empirical investigation and merit future research.

In today's economic and social systems, competition is pervasive. A complete assessment of its welfare effects should consider not only the indirect benefits of competition as an incentive mechanism influencing outcomes, but also the direct utility costs and benefits of the competitive experience itself. Competition should not be viewed only as a tool for achieving other objectives; the welfare it generates directly is also important. This distinction is crucial for both welfare analysis and policy design. Moreover, understanding how these direct utility effects shape individuals' preferences, beliefs, and social behavior is also crucial and constitutes a promising direction for future research.



## References

- Abbink, K., J. Brandts, B. Herrmann, and H. Orzen (2010). Intergroup conflict and intra-group punishment in an experimental contest game. *American Economic Review* 100(1), 420–447.
- Alan, S., T. Boneva, and S. Ertac (2019). Ever failed, try again, succeed better: Results from a randomized educational intervention on grit. *The Quarterly Journal of Economics* 134(3), 1121–1162.
- Apesteguia, J. and I. Palacios-Huerta (2010). Psychological pressure in competitive environments: Evidence from a randomized natural experiment. *American Economic Review* 100(5), 2548–2564.
- Ariely, D., U. Gneezy, G. Loewenstein, and N. Mazar (2009). Large stakes and big mistakes. *The review of economic studies* 76(2), 451–469.
- Balafoutas, L. and M. Sutter (2019). How uncertainty and ambiguity in tournaments affect gender differences in competitive behavior. *European Economic Review* 118, 1–13.
- Bartling, B., E. Fehr, and H. Herz (2014). The intrinsic value of decision rights. *Econometrica* 82(6), 2005–2039.
- Bartling, B., E. Fehr, and K. M. Schmidt (2012). Screening, competition, and job design: Economic origins of good jobs. *American Economic Review* 102(2), 834–864.
- Benjamin, D. J., O. Heffetz, M. S. Kimball, and N. Szembrot (2014). Beyond happiness and satisfaction: Toward well-being indices based on stated preference. *American economic review* 104(9), 2698–2735.
- Benz, M. and B. S. Frey (2008). The value of doing what you like: Evidence from the self-employed in 23 countries. *Journal of Economic Behavior & Organization* 68(3-4), 445–455.
- Bernheim, B. D., K. Kim, and D. Taubinsky (2024). Welfare and the act of choosing. Technical report, National Bureau of Economic Research.
- Bhatt, V. and A. M. Smith (2025). Overconfidence and performance: Evidence from a simple real-effort task. *Journal of Behavioral and Experimental Economics* 114, 102330.
- Brown, A. L., T. Imai, F. M. Vieider, and C. F. Camerer (2024). Meta-analysis of empirical estimates of loss aversion. *Journal of Economic Literature* 62(2), 485–516.
- Brown, C. L. and B. L. Fredrickson (2021). Characteristics and consequences of co-experienced positive affect: Understanding the origins of social skills, social bonds, and caring, healthy communities. *Current Opinion in Behavioral Sciences* 39, 58–63.
- Brunnermeier, M. K. and J. A. Parker (2005). Optimal expectations. *American Economic Review* 95(4), 1092–1118.
- Bruttel, L., M. Bulutay, C. Cornand, F. Heinemann, and A. Zylbersztejn (2023). Measuring strategic-uncertainty attitudes. *Experimental Economics* 26(3), 522–549.
- Buser, T., M. Niederle, and H. Oosterbeek (2014). Gender, competitiveness, and career choices. *The quarterly journal of economics* 129(3), 1409–1447.

- Bushong, B. and T. Gagnon-Bartsch (2023). Reference dependence and attribution bias: evidence from real-effort experiments. *American Economic Journal: Microeconomics* 15(2), 271–308.
- Caplin, A. and J. Leahy (2001). Psychological expected utility theory and anticipatory feelings. *The Quarterly Journal of Economics* 116(1), 55–79.
- Carpenter, J., P. H. Matthews, and J. Schirm (2010). Tournaments and office politics: Evidence from a real effort experiment. *American Economic Review* 100(1), 504–517.
- Carvalho, J.-P., A. Bergeron, J. Henrich, N. Nunn, and J. Weigel (2023). *Zero-sum thinking, the evolution of effort-suppressing beliefs, and economic development*. Number w31663. National Bureau of Economic Research.
- Cason, T. N., W. A. Masters, and R. M. Sheremeta (2010). Entry into winner-take-all and proportional-prize contests: An experimental study. *Journal of Public Economics* 94(9–10), 604–611.
- Cason, T. N., W. A. Masters, and R. M. Sheremeta (2020). Winner-take-all and proportional-prize contests: theory and experimental results. *Journal of Economic Behavior & Organization* 175, 314–327.
- Chen, D. L., M. Schonger, and C. Wickens (2016). otree—an open-source platform for laboratory, online, and field experiments. *Journal of Behavioral and Experimental Finance* 9, 88–97.
- Civelli, A. and C. Deck (2017). A flexible and customizable method for assessing cognitive abilities.
- Croson, R. and U. Gneezy (2009). Gender differences in preferences. *Journal of Economic literature* 47(2), 448–474.
- Dato, S., A. Grunewald, and D. Müller (2018). Expectation-based loss aversion and rank-order tournaments. *Economic Theory* 66(4), 901–928.
- Deaves, R., E. Lüders, and M. Schröder (2010). The dynamics of overconfidence: Evidence from stock market forecasters. *Journal of Economic Behavior & Organization* 75(3), 402–412.
- Delfgaauw, J., R. Dur, J. Sol, and W. Verbeke (2013). Tournament incentives in the field: Gender differences in the workplace. *Journal of Labor Economics* 31(2), 305–326.
- Delgado, M. R., D. S. Fareri, and L. J. Chang (2023). Characterizing the mechanisms of social connection. *Neuron* 111(24), 3911–3925.
- Dhingra, S. and J. Morrow (2019). Monopolistic competition and optimum product diversity under firm heterogeneity. *Journal of Political Economy* 127(1), 196–232.
- Dohmen, T. and A. Falk (2011). Performance pay and multidimensional sorting: Productivity, preferences, and gender. *American economic review* 101(2), 556–590.
- Drobner, C. and S. J. Goerg (2024). Motivated belief updating and rationalization of information. *Management Science* 70(7), 4583–4592.
- Englmaier, F., S. Grimm, D. Grothe, D. Schindler, and S. Schudy (2024). The efficacy of tournaments for nonroutine team tasks. *Journal of Labor Economics* 42(4), 921–948.

- Enke, B., R. Rodríguez-Padilla, and F. Zimmermann (2023). Moral universalism and the structure of ideology. *The Review of economic studies* 90(4), 1934–1962.
- Fallucchi, F., D. Nosenzo, and E. Reuben (2020). Measuring preferences for competition with experimentally-validated survey questions. *Journal of Economic Behavior & Organization* 178, 402–423.
- Frey, B. S., M. Benz, and A. Stutzer (2004). Introducing procedural utility: Not only what, but also how matters. *Journal of Institutional and Theoretical Economics (JITE)/Zeitschrift für die gesamte Staatswissenschaft*, 377–401.
- Frey, B. S. and A. Stutzer (2005). Beyond outcomes: measuring procedural utility. *Oxford Economic Papers* 57(1), 90–111.
- Gill, D. and V. Prowse (2012). A structural analysis of disappointment aversion in a real effort competition. *American Economic Review* 102(1), 469–503.
- Gill, D. and V. Prowse (2016). Cognitive ability, character skills, and learning to play equilibrium: A level-k analysis. *Journal of Political Economy* 124(6), 1619–1676.
- Gneezy, U., M. Niederle, and A. Rustichini (2003). Performance in competitive environments: Gender differences. *The quarterly journal of economics* 118(3), 1049–1074.
- Haggag, K., R. W. Patterson, N. G. Pope, and A. Feudo (2021). Attribution bias in major decisions: Evidence from the united states military academy. *Journal of Public Economics* 200, 104445.
- Haggag, K., D. G. Pope, K. B. Bryant-Lees, and M. W. Bos (2019). Attribution bias in consumer choice. *The Review of Economic Studies* 86(5), 2136–2183.
- Harbring, C. and B. Irlenbusch (2003). An experimental study on tournament design. *Labour Economics* 10(4), 443–464.
- Harbring, C. and B. Irlenbusch (2011). Sabotage in tournaments: Evidence from a laboratory experiment. *Management Science* 57(4), 611–627.
- Hauge, K. E., A. Kotsadam, and A. Riege (2023). Culture and gender differences in willingness to compete. *The Economic Journal* 133(654), 2403–2426.
- Hong, F., T. Hossain, and J. A. List (2015). Framing manipulations in contests: a natural field experiment. *Journal of Economic Behavior & Organization* 118, 372–382.
- Inglehart, R., C. Haerpfer, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, and B. Puranen (2014). World Values Survey: Round Six - Country-Pooled Datafile Version. Available at <http://www.worldvaluessurvey.org/WVSDocumentationWV6.jsp>.
- Kahneman, D. and A. Deaton (2010). High income improves evaluation of life but not emotional well-being. *Proceedings of the national academy of sciences* 107(38), 16489–16493.
- Kahneman, D., E. Diener, and N. Schwarz (1999). *Well-being: Foundations of hedonic psychology*. Russell Sage Foundation.
- Kahneman, D., J. L. Knetsch, and R. H. Thaler (1991). The endowment effect, loss aversion, and status quo bias.

- Kahneman, D. and A. B. Krueger (2006). Developments in the measurement of subjective well-being. *Journal of Economic perspectives* 20(1), 3–24.
- Kahneman, D., P. P. Wakker, and R. Sarin (1997). Back to bentham? explorations of experienced utility. *The quarterly journal of economics* 112(2), 375–406.
- Kőszegi, B. and M. Rabin (2006). A model of reference-dependent preferences. *The Quarterly Journal of Economics* 121(4), 1133–1165.
- Krishna, K., S. Lychagin, W. Olszewski, R. Siegel, and C. Tergiman (2025). Pareto improvements in the contest for college admissions. *Review of Economic Studies*, rdaf033.
- Lazear, E. P. and S. Rosen (1981). Rank-order tournaments as optimum labor contracts. *Journal of political Economy* 89(5), 841–864.
- Malamud, O. and C. Pop-Eleches (2011). Home computer use and the development of human capital. *The Quarterly journal of economics* 126(2), 987–1027.
- Mankiw, N. G. (2018). Principles of microeconomics, (2007).
- Marzilli Ericson, K. M. and A. Fuster (2011). Expectations as endowments: Evidence on reference-dependent preferences from exchange and valuation experiments. *The Quarterly Journal of Economics* 126(4), 1879–1907.
- Mesurado, B., S. Resett, M. Tezón, and C. E. Vanney (2021). Do positive emotions make you more prosocial? direct and indirect effects of an intervention program on prosociality in colombian adolescents during social isolation due to covid-19. *Frontiers in Psychology* 12, 710037.
- Möbius, M. M., M. Niederle, P. Niehaus, and T. S. Rosenblat (2022). Managing self-confidence: Theory and experimental evidence. *Management Science* 68(11), 7793–7817.
- Niederle, M. and L. Vesterlund (2007). Do women shy away from competition? do men compete too much? *The quarterly journal of economics* 122(3), 1067–1101.
- Niederle, M. and L. Vesterlund (2011). Gender and competition. *Annu. Rev. Econ.* 3(1), 601–630.
- Oriol, X., R. Miranda, C. Bazán, and E. Benavente (2020). Distinct routes to understand the relationship between dispositional optimism and life satisfaction: Self-control and grit, positive affect, gratitude, and meaning in life. *Frontiers in Psychology* 11, 907.
- Peer, E., D. Rothschild, A. Gordon, Z. Evernden, and E. Damer (2022). Data quality of platforms and panels for online behavioral research. *Behavior research methods* 54(4), 1643–1662.
- Rand, D. G., G. Kraft-Todd, and J. Gruber (2015). The collective benefits of feeling good and letting go: Positive emotion and (dis) inhibition interact to predict cooperative behavior. *PloS one* 10(1), e0117426.
- Reuben, E., P. Sapienza, and L. Zingales (2024). Overconfidence and preferences for competition. *The Journal of Finance* 79(2), 1087–1121.
- Smith, A. (1937). *The wealth of nations [1776]*, Volume 11937. na.

- Stigler, G. J. (1957). Perfect competition, historically contemplated. *Journal of political economy* 65(1), 1–17.
- Tversky, A. and D. Kahneman (1979). Prospect theory: An analysis of decision under risk. Technical report, David K. Levine.
- Willadsen, H., S. Zaccagni, M. Piovesan, and E. Wengström (2024). Measures of cognitive ability and choice inconsistency. *Journal of Economic Behavior & Organization* 220, 495–506.

# APPENDIX

## A Theoretical Appendix

### A.1 Optimal Effort

This subsection derives the optimal effort  $e_{i,s}^{*,f}$  for any incentive structure  $s \in \{C, NC\}$  and frame  $f \in \{G, L\}$ . For brevity, suppress the subscript  $i$ . Under  $(s, f)$ , the utility  $U^{s,f}(e)$ <sup>44</sup> is,

$$P^s(e) \cdot v(w^{win,f}) + (1 - P^s(e)) \cdot v(w^{lose,f}) - c(e) + u(P^s(e)) + \psi^f \mathbf{1}_{\{s=T\}} \quad (\text{A.1})$$

Collect the two terms that are linear in  $P^s(e)$  and rearrange the equation,

$$P^s(e) \cdot (v(w^{win,f}) - v(w^{lose,f})) - c(e) + u(P^s(e)) + \psi^f \mathbf{1}_{\{s=T\}} + v(w^{lose,f}) \quad (\text{A.2})$$

The last two terms do not depend on  $e$ . Assume  $P^s$  is continuously differentiable and strictly increasing (i.e.,  $P^{s'}(e) > 0$ ),  $u$  is continuously differentiable, and  $c$  is continuously differentiable, strictly increasing, and convex, with  $c(0) = 0$ ,  $c'(\cdot) > 0$  and  $c''(\cdot) \geq 0$ . The first order condition for an optimal effort level  $e^*$  is

$$c'(e^*) = P^{s'}(e^*) \left( v(w^{win,f}) - v(w^{lose,f}) + u'(P^s(e^*)) \right) \quad (\text{A.3})$$

Optimal effort equalizes marginal cost with marginal improvement in success probability multiplied by the sum of the monetary payoff spread and the marginal belief utility. Consider the reference-dependent value function

$$v(x) = \begin{cases} x, & x \geq 0, \\ \lambda x, & x < 0, \end{cases} \quad (\lambda > 1)$$

and the payoff structure  $(w^{win,G}, w^{lose,G}) = (w^{win}, 0)$ ,  $(w^{win,L}, w^{lose,L}) = (0, -w^{win})$ , the optimal effort under a gain frame satisfies

$$c'(e^*) = P^{s'}(e^*) \left( w^{win} + u'(P^s(e^*)) \right) \quad (\text{A.4})$$

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<sup>44</sup>The notation used in the appendix differs slightly from that in the main text and will be updated in the next version of the paper.

and the optimal effort under a loss frame satisfies

$$c'(e^*) = P^{s'}(e^*) \left( \lambda w^{\text{win}} + u'(P^s(e^*)) \right) \quad (\text{A.5})$$

## A.2 Belief Channel

This subsection proves Prediction 1. Fix  $f \in \{G, L\}$ . Under Assumption 1, I show that, when evaluated at the respective optima, the competitive setting yields (weakly) lower expected utility from monetary payoffs and belief utility, as stated in Section 2.

**Proposition 1.** *Fix a frame  $f \in \{G, L\}$ . Under Assumptions 1, the expected utility from monetary payoffs is (weakly) lower in the competitive setting than in the non-competitive setting when evaluated at the respective optima:*

$$U_{i,\text{pec}}^{C,f}(e_C^{*,f}) \leq U_{i,\text{pec}}^{NC,f}(e_{NC}^{*,f}).$$

**Proof.** Fix  $f \in \{G, L\}$  and define, for  $s \in \{C, NC\}$ ,

$$U_{\text{pec}}^{s,f}(e) = \Delta v^f P^s(e) - c(e) + v(w^{\text{lose},f}),$$

$$\Delta v^f := v(w^{\text{win},f}) - v(w^{\text{lose},f}) > 0.$$

By Assumption 1, for every effort level  $e \geq 0$ ,  $P^C(e) \leq P^{NC}(e)$ . Hence, for the same  $e$ ,

$$U_{\text{pec}}^{C,f}(e) - U_{\text{pec}}^{NC,f}(e) = \Delta v^f (P^C(e) - P^{NC}(e)) \leq 0,$$

because the cost term  $c(e)$  and the constant  $v(w^{\text{lose},f})$  cancel. Therefore  $U_{\text{pec}}^{C,f}(e) \leq U_{\text{pec}}^{NC,f}(e)$  for all  $e \geq 0$ . Moreover, evaluating at  $e = e_C^{*,f}$ , and using the optimality of  $e_{NC}^{*,f}$  for the non-competitive setting gives

$$U_{\text{pec}}^{NC,f}(e_{NC}^{*,f}) \geq U_{\text{pec}}^{NC,f}(e_C^{*,f}) \geq U_{\text{pec}}^{C,f}(e_C^{*,f}),$$

where the first inequality is by optimality of  $e_{NC}^{*,f}$  and the second by the pointwise dominance  $U_{\text{pec}}^{C,f}(e) \leq U_{\text{pec}}^{NC,f}(e)$  at  $e = e_C^{*,f}$ . Hence,

$$U_{\text{pec}}^{C,f}(e_C^{*,f}) \leq U_{\text{pec}}^{NC,f}(e_{NC}^{*,f}).$$

### A.3 preference channel

This subsection proves Prediction 2. The non-pecuniary component differs across settings only through  $\psi_i^f$ : it enters additively in the competitive setting and is zero otherwise. Holding the belief channel fixed, I show that under Assumption 2, this yields a positive intrinsic premium in the gain frame and a negative one in the loss frame, as stated in Section 2.

**Proposition 3.** *Fix  $f \in \{G, L\}$  and hold the belief channel fixed. Under Assumption 2, the competitive setting, relative to the non-competitive benchmark, yields higher intrinsic utility in the gain frame,*

$$U_{i,\text{nonpec}}^{C,G} \geq U_{i,\text{nonpec}}^{NC,G},$$

*and lower nonpec utility in the loss frame,*

$$U_{i,\text{nonpec}}^{C,L} \leq U_{i,\text{nonpec}}^{NC,L}.$$

**Proof.** Fix a frame  $f \in \{G, L\}$  and hold the belief channel fixed. By construction, the non-pecuniary term enters additively in both settings, with potentially different levels:

$$U_{i,\text{nonpec}}^{C,f} = \psi_i^{C,f} \quad \text{and} \quad U_{i,\text{nonpec}}^{NC,f} = \psi_i^{NC,f}.$$

The comparison reduces to a setting-specific constant. Under Assumption 2, we have

$$\psi_i^{C,G} \geq \psi_i^{NC,G} \quad \text{and} \quad \psi_i^{C,L} \leq \psi_i^{NC,L}.$$

These inequalities imply

$$U_{i,\text{nonpec}}^{C,G} \geq U_{i,\text{nonpec}}^{NC,G} \quad \text{and} \quad U_{i,\text{nonpec}}^{C,L} \leq U_{i,\text{nonpec}}^{NC,L}.$$



## A.4 Supplemental Proofs

This subsection analyzes optimal effort under the competitive and non-competitive settings. A further assumption is introduced to deliver a directional prediction.

**Assumption 3** *Fix a frame  $f \in \{G, L\}$ . At the optimal effort in each setting, perceived success is (weakly) less sensitive to effort in the competitive setting than in the non-competitive setting:*

$$P^T'(e_C^{*,G}) \leq P^{C'}(e_{NC}^{*,G})$$

Complementing Assumption 1's level effect, the competitive setting makes the success benchmark endogenous and moving. This strategic uncertainty reduces how effectively a marginal increase in own effort translates into perceived success, making the belief–effort map locally flatter at the competitive optimum than at the non-competitive optimum. Intuitively, when the target can shift with opponents' actions, marginal effort "buys" less perceived progress.

**Proposition 4.** *Fix a frame  $f \in \{G, L\}$ . The optimal effort is (weakly) lower in the competitive setting than in the non-competitive setting:*

$$e_C^{f,*} \leq e_{NC}^{f,*}.$$

**Proof.** Consider first the gain frame ( $f = G$ ). The optimal effort in setting  $s \in \{C, NC\}$  solves

$$c'(e_s^{*,G}) = P^{s'}(e_s^{*,G}) \left( w^{win} + u'(P^s(e_s^{*,G})) \right). \quad (\text{A.6})$$

Define the marginal–benefit for any effort level  $e$  by

$$MB_s(e) := P^{s'}(e) \left( w^{win} + u'(P^s(e)) \right). \quad (\text{A.7})$$

By Assumption 1,  $P^C(e) \leq P^{NC}(e)$  at all effort  $e$ . Since  $u''(\cdot) \geq 0$ , we have

$$u'(P^C(e)) \leq u'(P^{NC}(e)) \quad (\text{A.8})$$

Moreover, Assumption 3 yields

$$P^{C'}(e_C^{*,G}) \leq P^{NC'}(e_{NC}^{*,G})$$

Combining these inequalities with (A.8), we obtain

$$\begin{aligned}
MB_C(e_C^{*,G}) &= P^C'(e_C^{*,G}) \left( w^{win} + u'(P^C(e_C^{*,G})) \right) \\
&\leq P^C'(e_C^{*,G}) \left( w^{win} + u'(P^{NC}(e_{NC}^{*,G})) \right) \\
&\leq P^{NC}'(e_C^{*,G}) \left( w^{win} + u'(P^{NC}(e_{NC}^{*,G})) \right) = MB_{NC}(e_{NC}^{*,G}).
\end{aligned} \tag{A.11}$$

From (A.6), optimal effort satisfies  $c'(e_s^{*,G}) = MB_s(e_s^{*,G})$ . Since  $c(\cdot)$  is strictly increasing and convex,  $c'(\cdot)$  is strictly increasing; and with  $P$  and  $u$  differentiable,  $MB_{NC}(\cdot)$  is continuous. Hence,

$$c'(e_C^{*,G}) = MB_C(e_C^{*,G}) \leq MB_{NC}(e_{NC}^{*,G}) = c'(e_{NC}^{*,G}). \tag{A.12}$$

This implies

$$c'(e_C^{*,G}) \leq c'(e_{NC}^{*,G}). \tag{A.13}$$

Therefore,

$$e_C^{*,G} \leq e_{NC}^{*,G}.$$

The loss frame ( $f = L$ ) is analogous: replace  $w^{win}$  by  $\lambda w^{win}$  in (A.6)–(A.7); the remainder of the argument is unchanged. Hence, for either frame  $f \in \{G, L\}$ ,

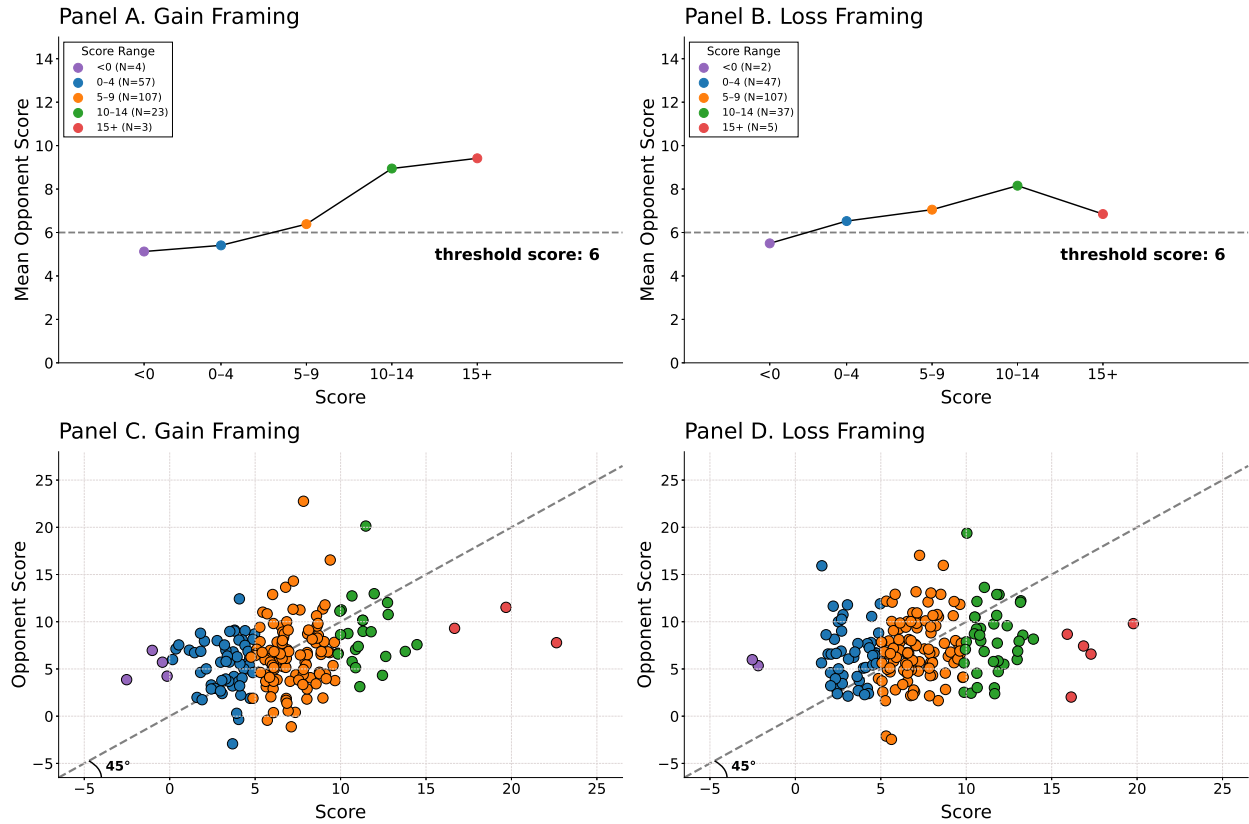
$$e_C^{*,f} \leq e_{NC}^{*,f}$$

as claimed.

## B Additional Figures and Tables

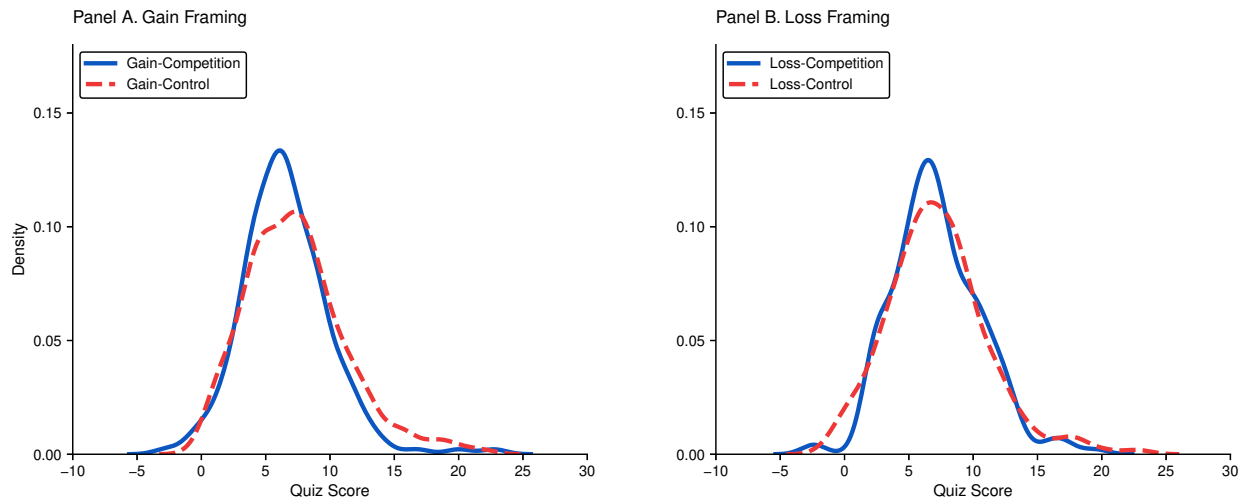
### B.1 Additional Figures

Appendix Figure B.1: Matching Patterns



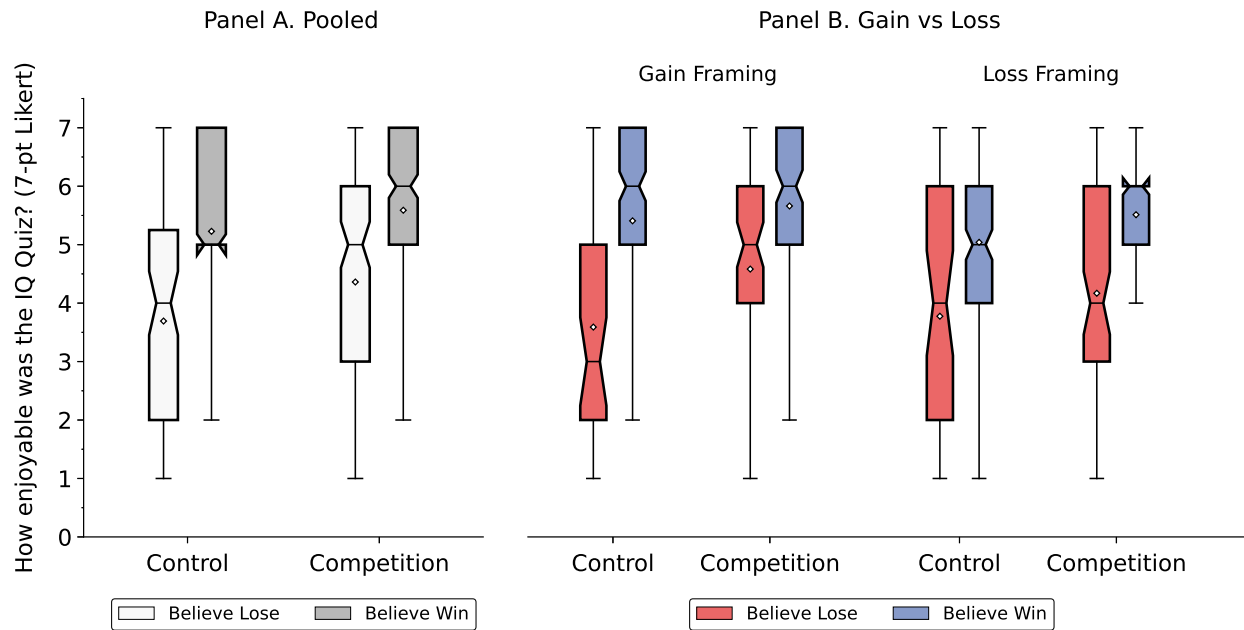
*Notes:* This figure shows the realized matching patterns among participants in the competition treatments. Panels A and B present binned scatter plots of participants' quiz scores (horizontal axis) and the mean scores of their matched opponents (vertical axis) under gain and loss framings. The dashed gray line marks the threshold score of six used in the control treatments. In both framings, the mean opponent score stays close to six, indicating similar average difficulty across treatments. Panels C and D display individual-level scatter plots, with each point representing a participant–opponent pair. The dotted 45-degree line denotes equal scores. Participants with lower scores tend to face stronger opponents, while higher-scoring participants tend to face weaker ones. Matching patterns are broadly similar across framings.

Appendix Figure B.2: Distribution of Quiz Scores



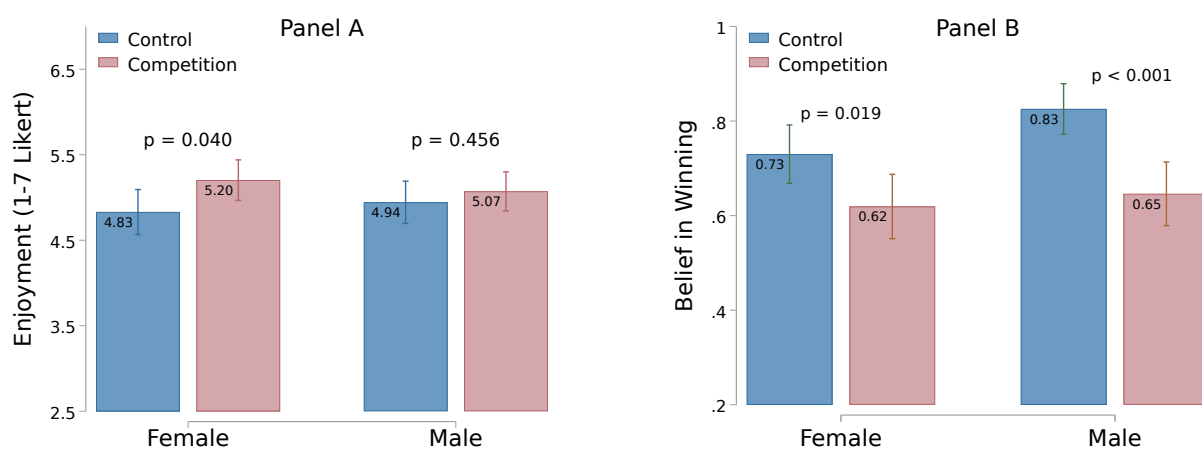
*Notes:* This figure displays the distribution of participants' quiz scores across framing and competition conditions. Panel A compares Gain–Competition (solid blue) with Gain–Control (dashed red); Panel B compares Loss–Competition (solid blue) with Loss–Control (dashed red). Scores are measured on the –10 to 30 scale, and densities are normalized to integrate to one within each panel. The distributions are closely aligned across treatments, indicating comparable task difficulty and performance dispersion between competition and control conditions.

Appendix Figure B.3: Belief in Winning and Utility



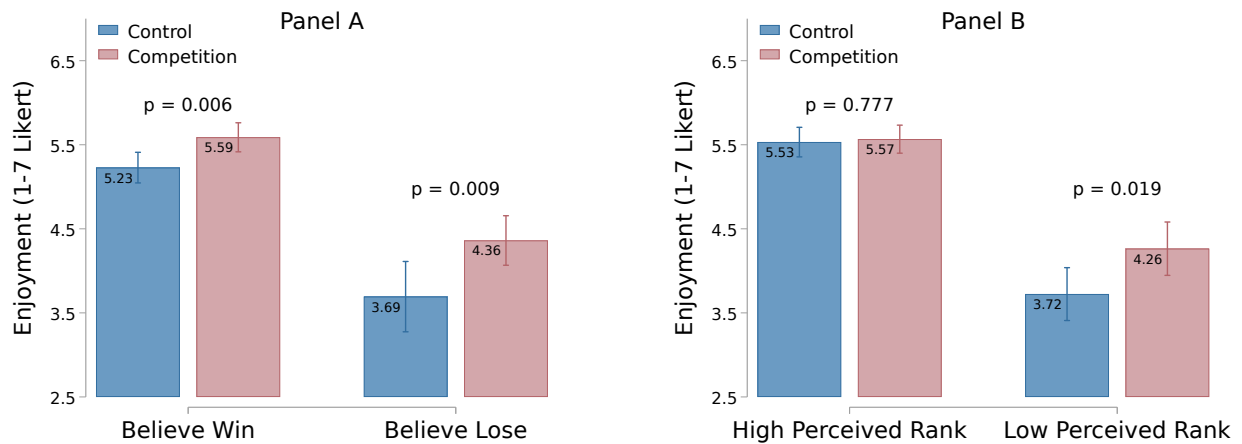
*Notes:* This figure plots self-reported enjoyment by winning beliefs (“believe lose” vs. “believe win”) across experimental conditions (competition vs. control). Enjoyment was measured on a 7-point Likert scale in response to the question: “On a scale of 1 (not at all enjoyable) to 7 (extremely enjoyable), how enjoyable was the IQ Quiz Competition (or IQ Quiz)?” Winning belief is defined as an indicator equal to 1 if the participant expects to win and 0 otherwise. Panel A pools observations across gain and loss frames, while Panel B reports them separately. Diamonds indicate mean values.

Appendix Figure B.4: Effects of Competition by Gender: Utility and Confidence



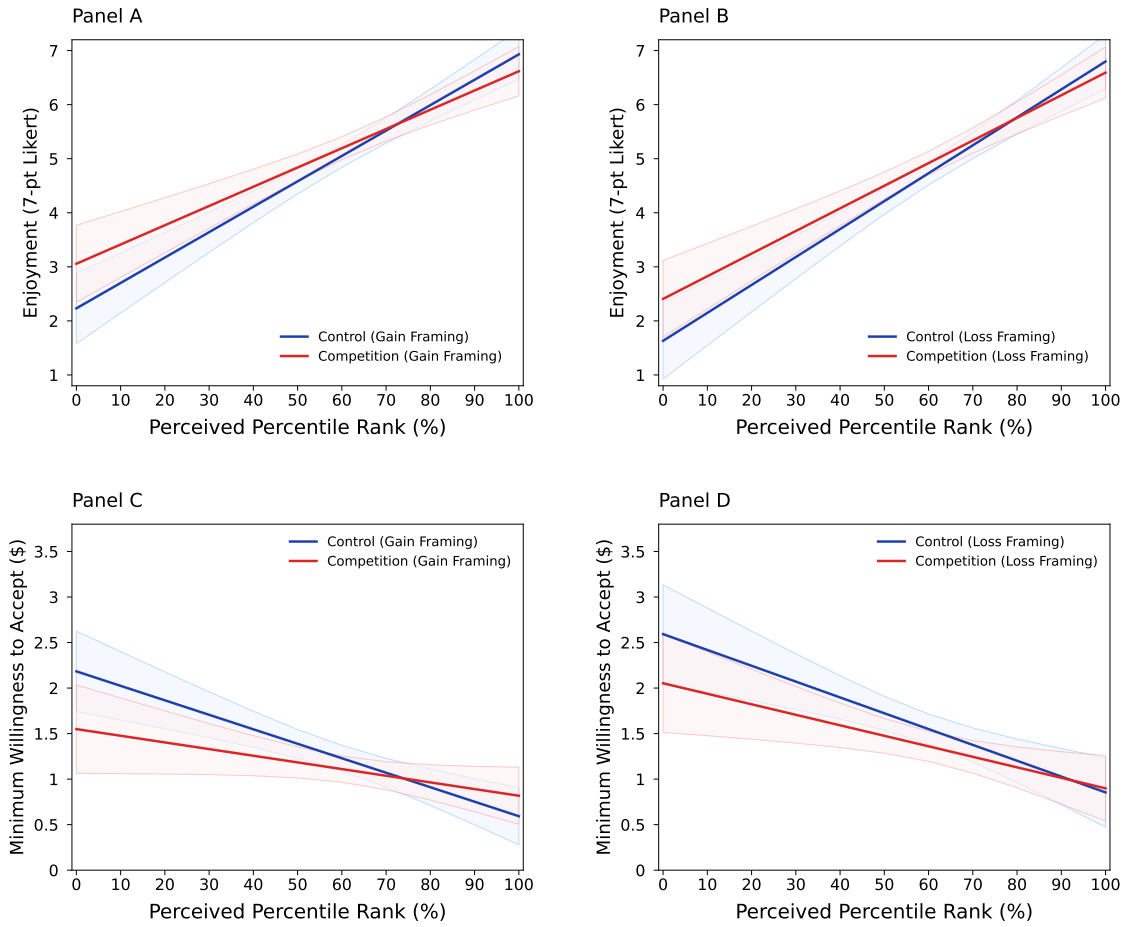
*Notes:* This figure illustrates the effects of competition by gender on enjoyment and belief in winning. Enjoyment is measured using a 7-point Likert scale question, where a larger number indicates greater enjoyment. Belief in winning is a binary indicator; the bar height represents the percentage of participants who believe they will win. Error bars denote 95% confidence intervals. p-values correspond to two-sample t-tests comparing the Competition and Control treatments within each gender.

Appendix Figure B.5: Effects on Utility by Winning Beliefs and Perceived Rank



*Notes:* This figure illustrates the effects of competition on enjoyment by participants' winning beliefs and perceived rank. Enjoyment is measured on a 7-point Likert scale, with higher values indicating greater enjoyment. Bars represent mean enjoyment within each subgroup. "Believe Win" denotes participants who expected to win, while "Believe Lose" refers to those who expected to lose. "High Perceived Rank" refers to participants who believed their performance was above the median performance of all participants in the study, while "Low Perceived Rank" corresponds to those who believed their performance was at or below that median. Error bars indicate 95 percent confidence intervals. p-values are based on two-sample t-tests comparing the Competition and Control treatments within each subgroup.

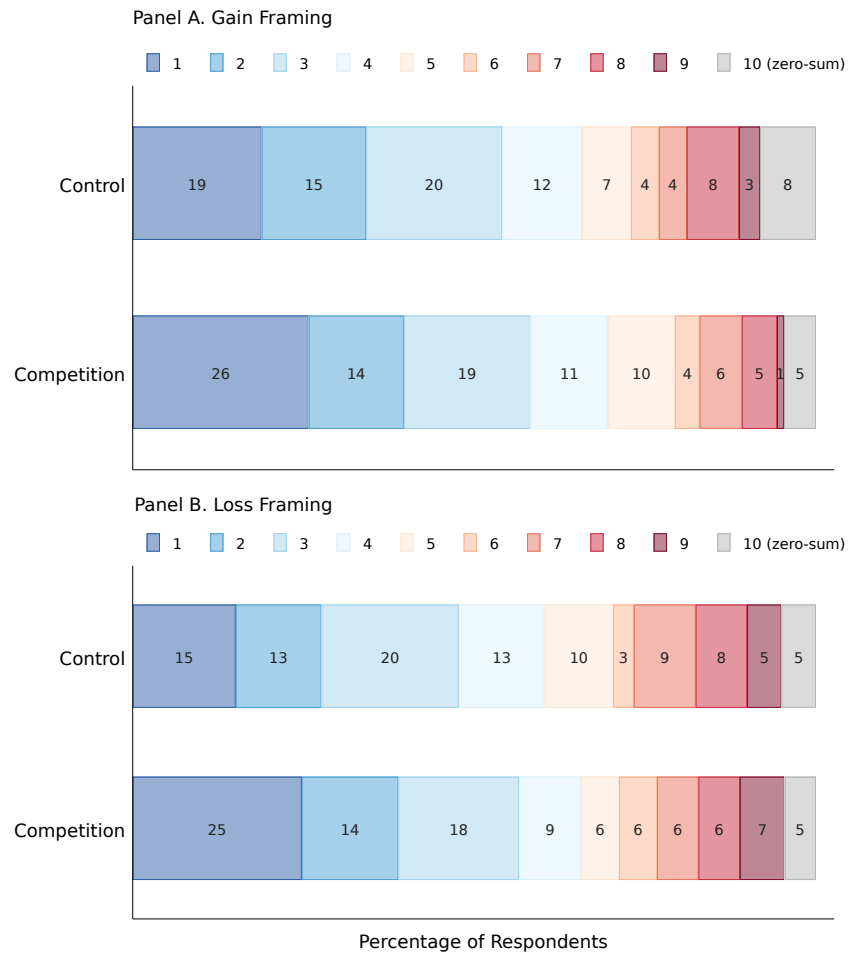
Appendix Figure B.6: Heterogeneity by Perceived Ability



*Notes:* This figure shows the heterogeneous effects of competition by framing condition and perceived performance rank. Panels A and B plot predicted enjoyment (measured on a 7-point Likert scale) against perceived percentile rank for the gain- and loss-framing conditions, respectively. Panels C and D report predicted minimum willingness to accept (WTA) to perform the task again, also by perceived percentile rank, for the gain- and loss-framing conditions. Blue lines represent the control group and red lines represent the competition group, with shaded areas denoting 95% confidence intervals.

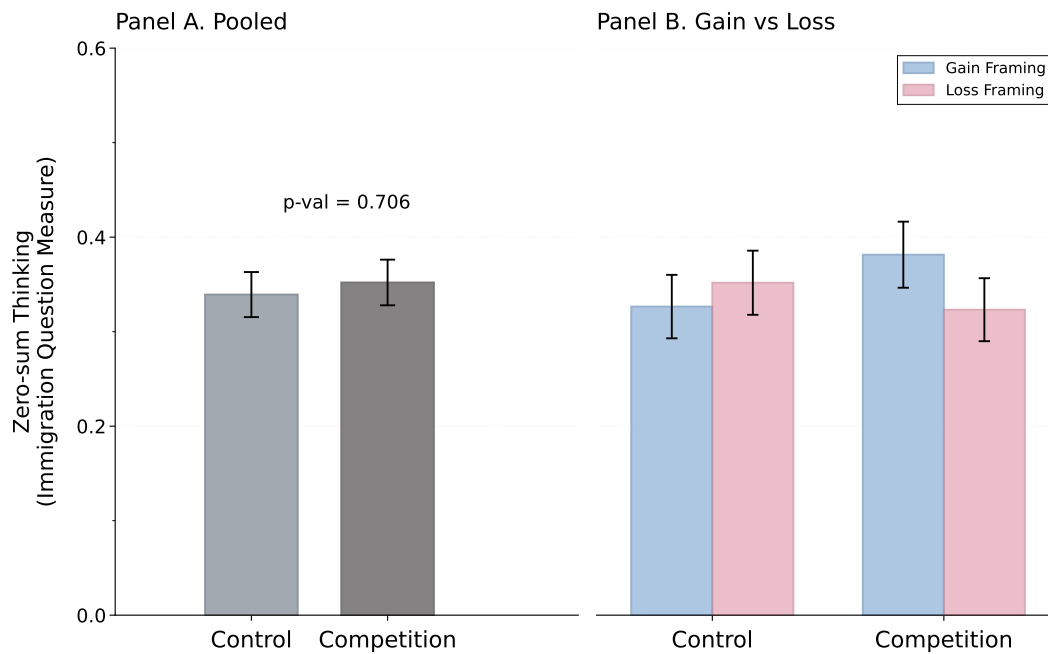


Appendix Figure B.7: Zero-Sum Thinking Distribution by Treatment



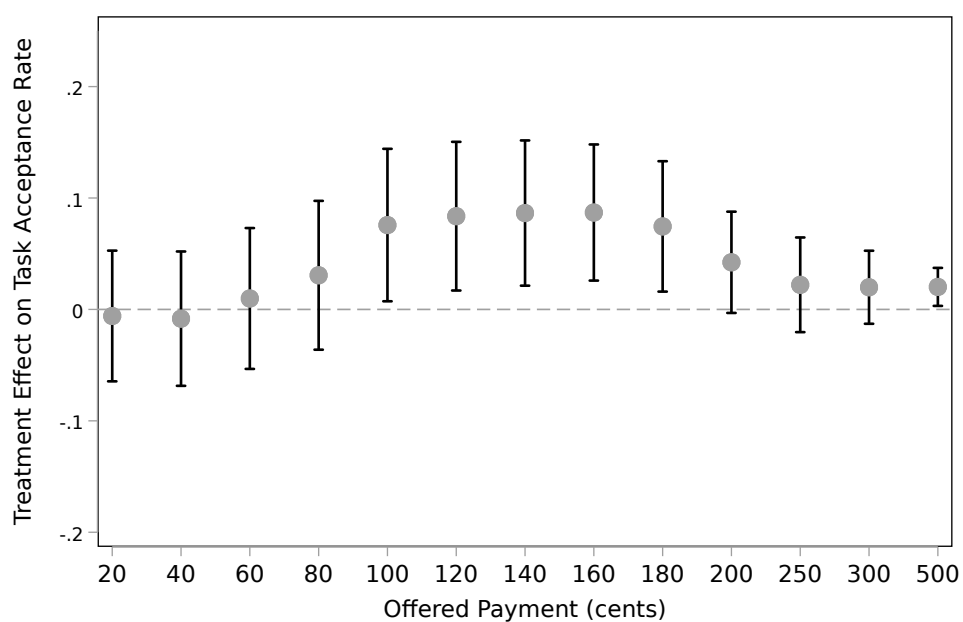
*Notes:* The figure displays the distribution of responses on the zero-sum scale under gain framing (Panel A) and loss framing (Panel B). Zero-sum beliefs are measured using a World Values Survey item (Inglehart et al., 2014), where respondents rate their agreement on a 1–10 scale between the statements: “Wealth can grow so there’s enough for everyone” (1, indicating minimal zero-sum thinking) and “People can only become wealthy at the expense of others” (10, indicating strong zero-sum thinking). Higher values correspond to stronger endorsement of zero-sum beliefs. Percentages indicate the share of respondents selecting each scale point within the control and competition groups.

Appendix Figure B.8: Treatment Effects on Zero-sum Thinking



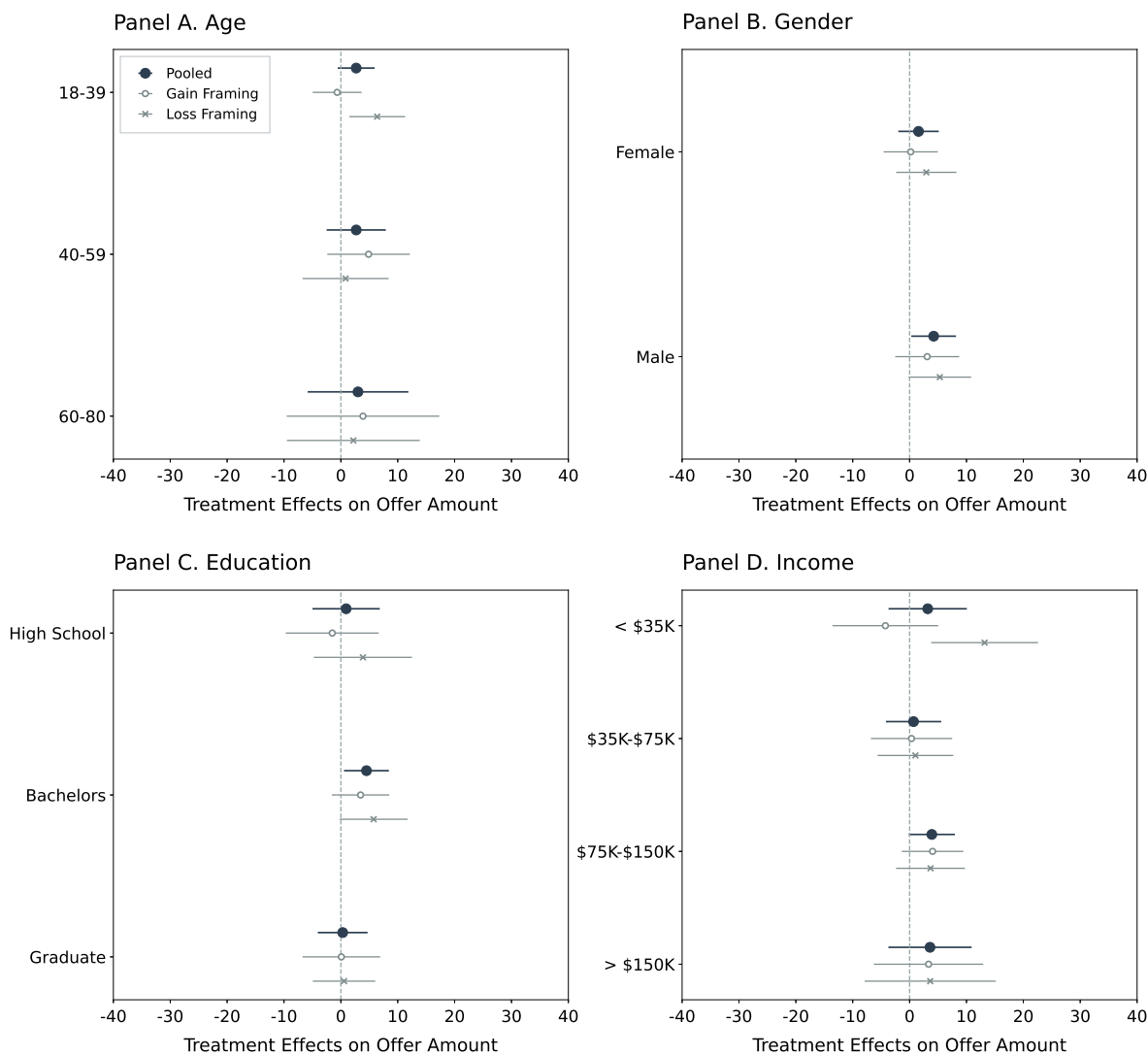
*Notes:* This figure illustrates the estimated treatment effects of competition on zero-sum thinking using the immigration measure. Zero-sum thinking is measured with a scenario-choice question in which both immigrants and locals work as day laborers. On Day 1, immigrant workers find 20 jobs and local workers 40 jobs. On Day 2, immigrant workers again find 40 jobs, and respondents then choose whether local workers find 20 or 80 jobs. Choosing 20 reflects zero-sum thinking (coded as 1), while choosing 80 reflects non-zero-sum thinking (coded as 0). Panel A presents results for the pooled sample, and Panel B shows results separately for gain- and loss-framing conditions. Bars indicate mean values with error bars representing robust standard errors.

Appendix Figure B.9: Effects on Task Preference through Attribution Bias



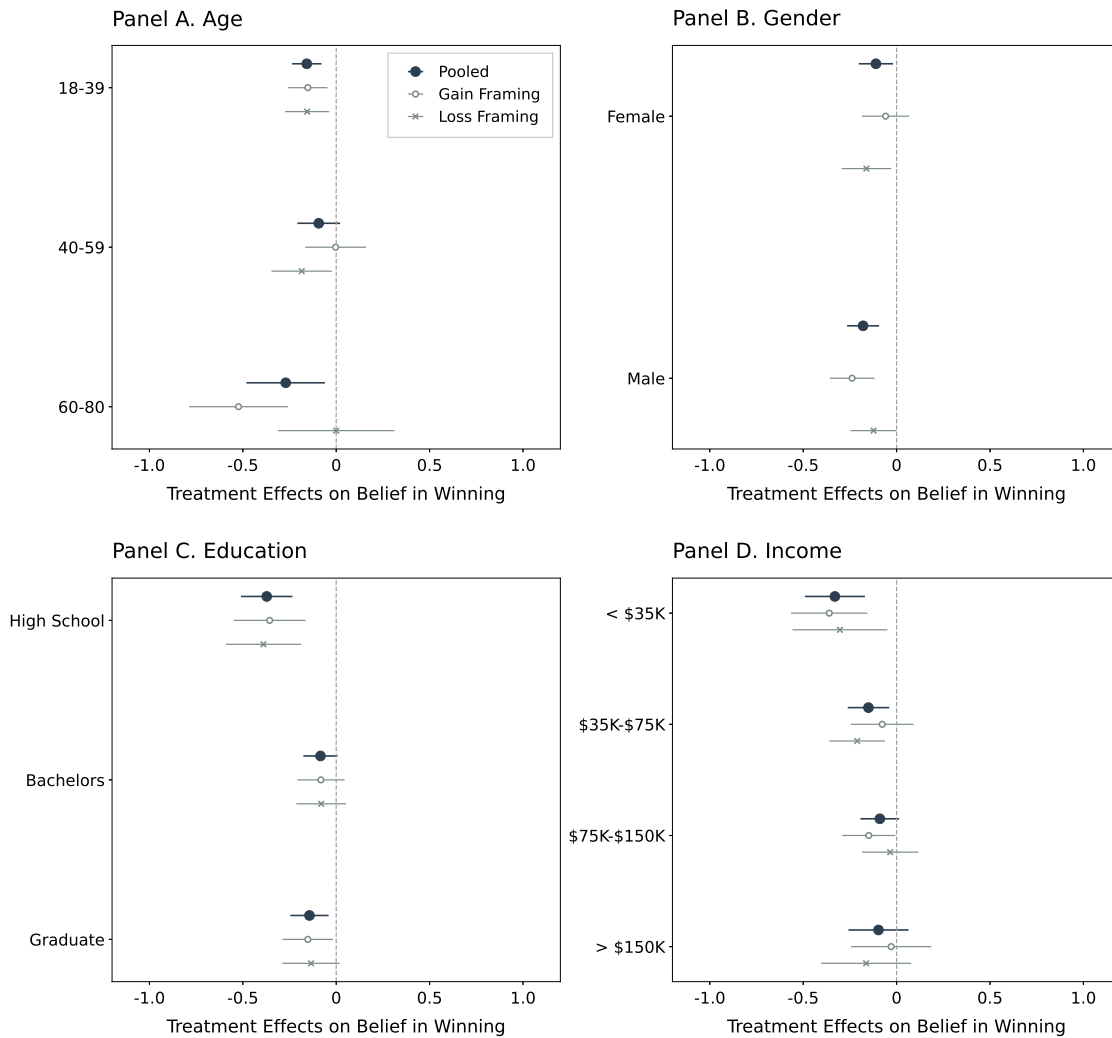
*Notes:* This figure shows the effect of competition on willingness to perform the experimental task again at different monetary amounts. For each listed amount, the outcome is a binary indicator equal to one if participants stated that they would accept performing the task again for that payment and zero otherwise. The offered amounts are \$0.20, \$0.40, \$0.60, \$0.80, \$1.00, \$1.20, \$1.40, \$1.60, \$1.80, \$2.00, \$2.50, \$3.00, and \$5.00. The figure reports results for the pooled sample.

Appendix Figure B.10: Effects of Competition on Altruism across Demographic Groups



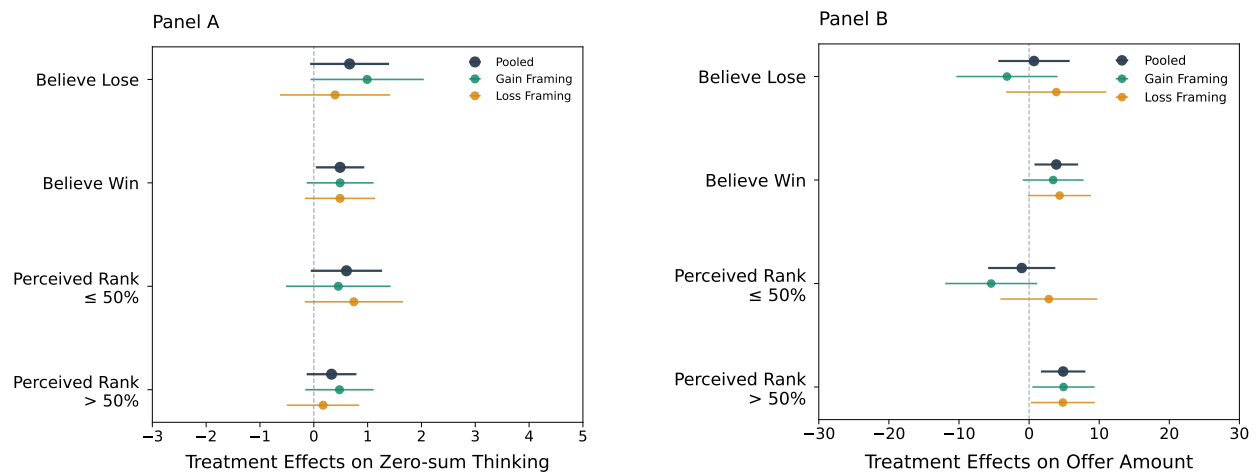
*Notes:* This figure shows the heterogeneous effects of competition on altruism by participants' beliefs about winning and their perceived performance rank under gain and loss framing. Panel A displays results for the control and competition groups under gain framing, and Panel B displays the corresponding results under loss framing. The figure plots predicted altruism, measured by the offer amount in the dictator game, as a function of perceived percentile rank (0–100, in increments of 10). Shaded areas indicate 95% confidence intervals.

Appendix Figure B.11: Effect of Competition on Belief in Winning by Demographic Group



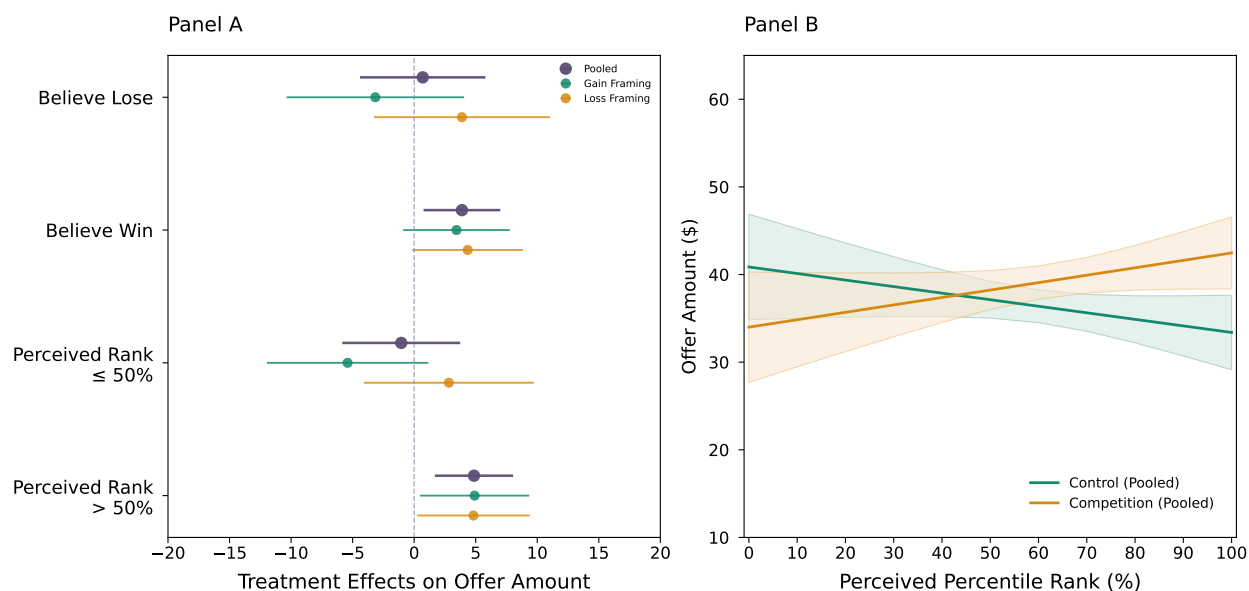
*Notes:* This figure presents the heterogeneous effects of competition on the belief in winning across demographic groups. Belief in winning is a binary variable equal to 1 if participants report expecting to beat their paired opponent in the competition treatment, or to surpass the threshold score in the control group, and 0 otherwise. The filled circle represents the treatment effect for the pooled sample, the hollow circle for the gain-framing condition, and the star for the loss-framing condition. All estimates are shown with 95% confidence intervals. Panel A reports results by age group (18–39, 40–59, and 60–80), Panel B by gender, Panel C by educational attainment (high school, bachelor’s, and graduate), and Panel D by income categories (<\$35K, \$35K–\$75K, \$75K–\$150K, and >\$150K).

Appendix Figure B.12: Heterogeneous Effects on Zero-sum Thinking and Altruism



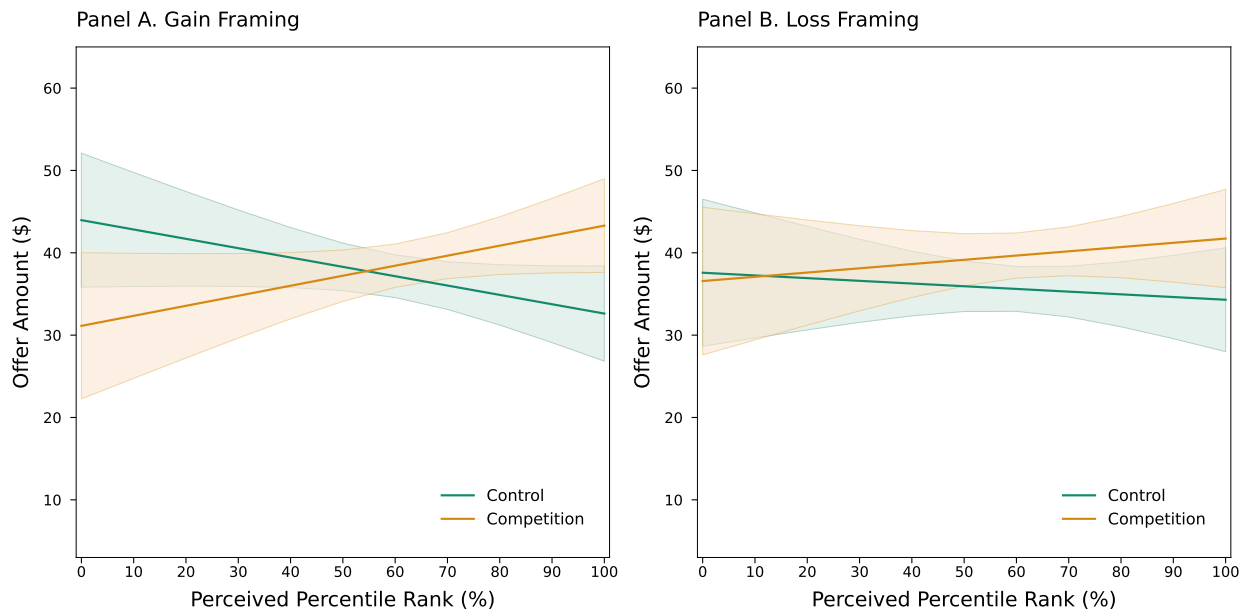
*Notes:* This figure shows heterogeneous treatment effects of competition on zero-sum thinking and altruism. Panel A shows effects on zero-sum thinking, and Panel B shows effects on offer amounts. Each point represents the estimated treatment effect, and the horizontal lines indicate 95% confidence intervals. Results are presented separately for the pooled sample, the gain framing condition, and the loss framing condition, as well as across subgroups based on participants' perceived rank (above or below the median) and beliefs about winning or losing.

Appendix Figure B.13: Heterogeneous Effects on Altruism



*Notes:* This figure shows the heterogeneous effects of competition on altruism by participants' beliefs about winning and their perceived performance rank. Panel A presents treatment effects for the pooled sample, the gain-framing condition, and the loss-framing condition. Black markers indicate pooled estimates, green markers indicate gain framing, and orange markers indicate loss framing. Each marker shows the estimated mean, with horizontal bars denoting 95 percent confidence intervals. Panel B plots predicted altruism, measured by the offer amount in the dictator game, as a function of perceived percentile rank (0–100, in increments of 10). Shaded areas indicate 95% confidence intervals. Results for the gain- and loss-framing conditions are reported in Appendix Figure B.14.

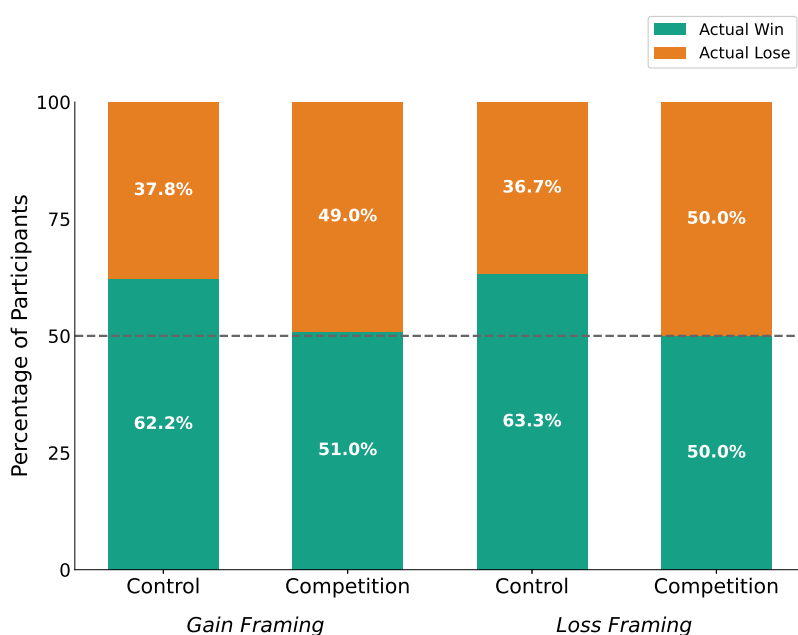
Appendix Figure B.14: Effects of Competition on Altruism by Perceived Ability



*Notes:* This figure shows the heterogeneous effects of competition on altruism by participants' beliefs about winning and their perceived performance rank under gain and loss framing. Panel A displays results for the control and competition groups under gain framing, and Panel B displays the corresponding results under loss framing. The figure plots predicted altruism, measured by the offer amount in the dictator game, as a function of perceived percentile rank (0–100, in increments of 10). Shaded areas indicate 95% confidence intervals.

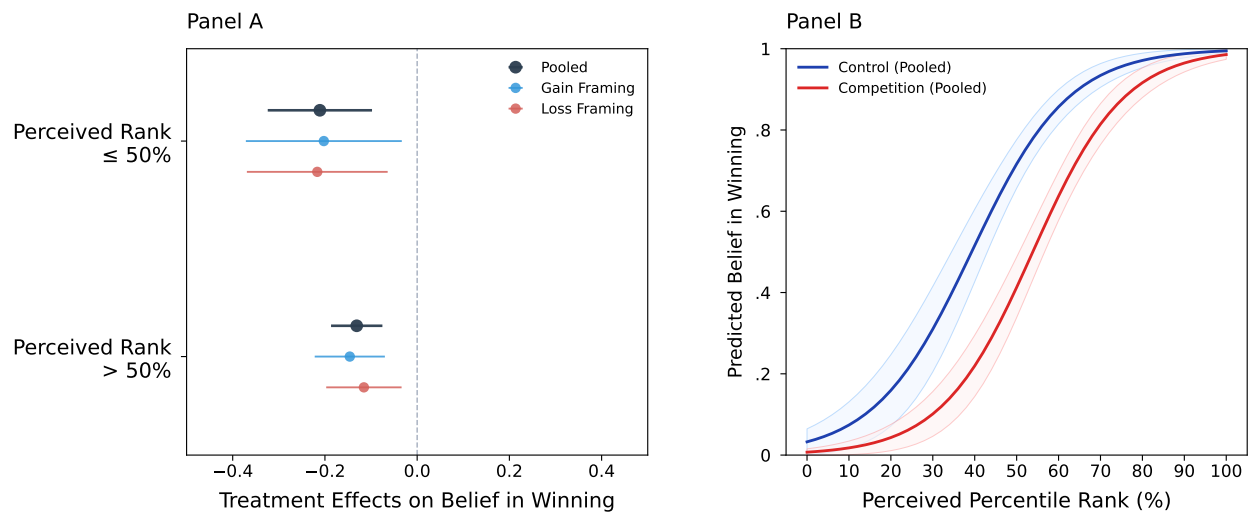


Appendix Figure B.15: Actual Win–Loss Outcomes by Framing and Treatment Condition



*Notes:* This figure shows the distribution of actual win–loss outcomes across control and competition conditions under gain and loss framing. In the control condition, a “win” is defined as achieving a score on the IQ quiz above the threshold score, while a “loss” indicates performance below the threshold. In the competition condition, a “win” denotes that a participant outperformed their paired opponent on the IQ quiz, while a “loss” indicates the opposite. Under gain framing, participants were endowed with \$2; winners earned an additional \$2 (totaling \$4), while losers received no additional reward. Under loss framing, participants were endowed with \$6; winners retained the full endowment, while losers forfeited \$4 from the initial amount.

Appendix Figure B.16: Heterogeneity in Belief in Winning by Perceived Ability



*Notes:* This figure presents the effect of competition on belief in winning by perceived ability. Panel A shows estimated mean treatment effects with 95% confidence intervals for the pooled sample, the gain-framing condition, and the loss-framing condition. Results are reported separately for individuals who perceived their performance rank to be below or above the median. Belief in winning is a binary variable equal to one if participants believed they would win the IQ quiz competition in the treatment group or pass the threshold score in the control group. Perceived rank is elicited by asking participants to estimate the rank of their performance relative to others. Panel B plots predicted probabilities of belief in winning against perceived percentile rank for the pooled sample, based on estimates from a logit model.

## B.2 Additional Tables

Appendix Table B.1: Robustness Using Alternative Effort Proxies

|                            | <i>Dependent Variable: Enjoyment</i> |                   |                   |                   |                   |                   |                   |                   |                   |
|----------------------------|--------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                            | Pooled                               |                   |                   | Gain Framing      |                   |                   | Loss Framing      |                   |                   |
|                            | (1)                                  | (2)               | (3)               | (4)               | (5)               | (6)               | (7)               | (8)               | (9)               |
| Competition                | 0.43***<br>(0.12)                    | 0.44***<br>(0.12) | 0.46***<br>(0.12) | 0.42**<br>(0.16)  | 0.42**<br>(0.17)  | 0.43***<br>(0.16) | 0.45***<br>(0.17) | 0.46***<br>(0.17) | 0.50***<br>(0.17) |
| Belief in Winning          | 1.40***<br>(0.14)                    | 1.39***<br>(0.14) | 1.37***<br>(0.14) | 1.40***<br>(0.20) | 1.40***<br>(0.20) | 1.37***<br>(0.20) | 1.37***<br>(0.20) | 1.36***<br>(0.20) | 1.33***<br>(0.20) |
| Effort (score)             | ✓                                    |                   |                   | ✓                 |                   |                   | ✓                 |                   |                   |
| Effort (correct answers)   |                                      | ✓                 |                   |                   | ✓                 |                   |                   | ✓                 |                   |
| Effort (time per question) |                                      |                   | ✓                 |                   |                   | ✓                 |                   |                   | ✓                 |
| Control mean               | 4.89                                 | 4.89              | 4.89              | 5.05              | 5.05              | 5.05              | 4.73              | 4.73              | 4.73              |
| Observations               | 787                                  | 787               | 787               | 390               | 390               | 390               | 397               | 397               | 397               |
| R <sup>2</sup>             | 0.14                                 | 0.14              | 0.13              | 0.14              | 0.14              | 0.14              | 0.13              | 0.13              | 0.13              |

*Notes:* This table reports OLS estimates of the effect of competition on utility, both overall and net of the belief-updating channel, with alternative effort proxies as controls. The check mark (✓) indicates which effort proxy is included in each specification. “Score” denotes the total number of points achieved on the IQ quiz; “Correct answers” is the number of questions answered correctly; and “Time per question” measures the average time spent on each question. Control means refer to average enjoyment in the control group. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.2: Heterogeneity by Beliefs and Perceived Ability

|                                          | <i>Dependent Variable: Enjoyment</i> |                     |                     |
|------------------------------------------|--------------------------------------|---------------------|---------------------|
|                                          | Pooled<br>(1)                        | Gain Framing<br>(2) | Loss Framing<br>(3) |
| <b>Panel A: Belief in Winning</b>        |                                      |                     |                     |
| Competition                              | 0.67**<br>(0.26)                     | 0.99***<br>(0.37)   | 0.39<br>(0.36)      |
| Believe Win                              | 1.53***<br>(0.23)                    | 1.82***<br>(0.33)   | 1.26***<br>(0.32)   |
| Competition $\times$ Believe Win         | -0.31<br>(0.29)                      | -0.74*<br>(0.41)    | 0.08<br>(0.41)      |
| Control mean                             | 3.69                                 | 3.59                | 3.78                |
| Observations                             | 787                                  | 390                 | 397                 |
| R <sup>2</sup>                           | 0.13                                 | 0.14                | 0.12                |
| <b>Panel B: Perceived Rank</b>           |                                      |                     |                     |
| Competition                              | 0.54**<br>(0.23)                     | 0.70**<br>(0.32)    | 0.40<br>(0.33)      |
| Perceived High Rank                      | 1.81***<br>(0.18)                    | 2.07***<br>(0.25)   | 1.55***<br>(0.27)   |
| Competition $\times$ Perceived High Rank | -0.51*<br>(0.26)                     | -0.77**<br>(0.36)   | -0.26<br>(0.37)     |
| Control mean                             | 3.72                                 | 3.69                | 3.76                |
| Observations                             | 787                                  | 390                 | 397                 |
| R <sup>2</sup>                           | 0.19                                 | 0.24                | 0.15                |

*Notes:* This table shows heterogeneous treatment effects of competition on utility. The dependent variable is enjoyment, measured on a 7-point Likert scale, with higher values indicating greater enjoyment. Competition is an indicator equal to one if the participant was assigned to the competition condition. In Panel A, heterogeneity is examined by whether participants believed they would win (Believe Win = 1 if the participant expected to win, 0 otherwise). In Panel B, heterogeneity is based on perceived relative rank (Perceived High Rank = 1 if the participant perceived their performance to be above the median, 0 otherwise). Interaction terms capture differential treatment effects by beliefs and perceived ability. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.3: Correlation between Belief in Winning and Utility

| <i>Dependent Variable: Enjoyment</i> |                   |                    |                   |                    |                   |                    |
|--------------------------------------|-------------------|--------------------|-------------------|--------------------|-------------------|--------------------|
|                                      | Pooled            |                    | Gain Framing      |                    | Loss Framing      |                    |
|                                      | Control<br>(1)    | Competition<br>(2) | Control<br>(3)    | Competition<br>(4) | Control<br>(5)    | Competition<br>(6) |
| Belief in Winning                    | 1.53***<br>(0.23) | 1.23***<br>(0.17)  | 1.82***<br>(0.33) | 1.08***<br>(0.24)  | 1.26***<br>(0.32) | 1.34***<br>(0.25)  |
| Control mean                         | 4.89              | 5.14               | 5.05              | 5.29               | 4.73              | 4.99               |
| Observations                         | 395               | 392                | 196               | 194                | 199               | 198                |
| R <sup>2</sup>                       | 0.12              | 0.13               | 0.16              | 0.11               | 0.09              | 0.14               |

*Notes:* This table summarizes OLS estimates of the relationship between belief in winning and enjoyment. The dependent variable is participants' responses to the question: "On a scale of 1 (not at all enjoyable) to 7 (extremely enjoyable), how enjoyable was the IQ Quiz Competition (or IQ Quiz)?" Belief in Winning is an indicator based on post-treatment responses to "Do you think you were the top performer in the IQ Quiz competition?" (or "Do you think you scored 6 or higher on the IQ Quiz?"). Columns (1)–(2) pool all framing conditions, while columns (3)–(4) and (5)–(6) report results separately for the gain and loss framing treatments. Within each framing condition, the first column corresponds to the Control group and the second to the Treatment (Competition) group. Reported coefficients are based on robust standard errors shown in parentheses. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.4: Effects on Task Preference through Attribution Bias

|                                    | Competition<br>(1) | Control<br>(2) | Difference<br>[(1)-(2)] | p-value<br>(4) |
|------------------------------------|--------------------|----------------|-------------------------|----------------|
| Minimum Willingness to Accept (\$) | 1.22<br>(0.05)     | 1.39<br>(0.06) | -0.17**<br>(0.08)       | 0.03           |
| Willing to Accept for 20 cents     | 0.22<br>(0.02)     | 0.23<br>(0.02) | -0.01<br>(0.03)         | 0.84           |
| Willing to Accept for 40 cents     | 0.24<br>(0.02)     | 0.25<br>(0.02) | -0.01<br>(0.03)         | 0.79           |
| Willing to Accept for 60 cents     | 0.29<br>(0.02)     | 0.28<br>(0.02) | 0.01<br>(0.03)          | 0.76           |
| Willing to Accept for 80 cents     | 0.37<br>(0.02)     | 0.34<br>(0.02) | 0.03<br>(0.03)          | 0.37           |
| Willing to Accept for 100 cents    | 0.64<br>(0.02)     | 0.56<br>(0.03) | 0.08**<br>(0.03)        | 0.03           |
| Willing to Accept for 120 cents    | 0.69<br>(0.02)     | 0.60<br>(0.02) | 0.09**<br>(0.03)        | 0.01           |
| Willing to Accept for 140 cents    | 0.72<br>(0.02)     | 0.63<br>(0.02) | 0.09**<br>(0.03)        | 0.01           |
| Willing to Accept for 160 cents    | 0.78<br>(0.02)     | 0.70<br>(0.02) | 0.08**<br>(0.03)        | 0.01           |
| Willing to Accept for 180 cents    | 0.81<br>(0.02)     | 0.73<br>(0.02) | 0.08**<br>(0.03)        | 0.01           |
| Willing to Accept for 200 cents    | 0.90<br>(0.02)     | 0.86<br>(0.02) | 0.04*<br>(0.02)         | 0.07           |
| Willing to Accept for 250 cents    | 0.91<br>(0.01)     | 0.89<br>(0.02) | 0.02<br>(0.02)          | 0.31           |
| Willing to Accept for 300 cents    | 0.95<br>(0.01)     | 0.93<br>(0.01) | 0.02<br>(0.02)          | 0.23           |
| Willing to Accept for 500 cents    | 0.99<br>(0.00)     | 0.97<br>(0.01) | 0.02**<br>(0.01)        | 0.02           |

*Notes:* This table reports the effects of competition on willingness to perform the experimental task again at different monetary amounts for the pooled sample. Columns (1) and (2) present mean acceptance rates in the competition and control groups, respectively. Column (3) reports the difference between the two groups, and column (4) provides the associated p-values. Standard errors are shown in parentheses. The outcome variable is a binary indicator equal to one if participants accepted performing the task again at the specified payment amount and zero otherwise. The first row reports the estimated minimum willingness to accept (WTA) in dollars. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.5: Treatment Effects on Task Preference (Gain Framing)

| <i>Sample: Gain Framing</i>        | Competition    | Control        | Difference      | p-value |
|------------------------------------|----------------|----------------|-----------------|---------|
|                                    | (1)            | (2)            | [(1)-(2)]       | (4)     |
| Minimum Willingness to Accept (\$) | 1.09<br>(0.07) | 1.23<br>(0.08) | -0.14<br>(0.11) | 0.18    |
| Willing to Accept for 20 cents     | 0.26<br>(0.03) | 0.27<br>(0.03) | -0.01<br>(0.04) | 0.78    |
| Willing to Accept for 40 cents     | 0.28<br>(0.03) | 0.29<br>(0.03) | -0.01<br>(0.05) | 0.87    |
| Willing to Accept for 60 cents     | 0.34<br>(0.03) | 0.32<br>(0.03) | 0.02<br>(0.05)  | 0.78    |
| Willing to Accept for 80 cents     | 0.41<br>(0.04) | 0.38<br>(0.03) | 0.03<br>(0.05)  | 0.62    |
| Willing to Accept for 100 cents    | 0.69<br>(0.03) | 0.60<br>(0.04) | 0.09*<br>(0.05) | 0.09    |
| Willing to Accept for 120 cents    | 0.73<br>(0.03) | 0.66<br>(0.03) | 0.07<br>(0.05)  | 0.11    |
| Willing to Accept for 140 cents    | 0.76<br>(0.03) | 0.68<br>(0.03) | 0.08*<br>(0.05) | 0.06    |
| Willing to Accept for 160 cents    | 0.82<br>(0.03) | 0.74<br>(0.03) | 0.08*<br>(0.04) | 0.06    |
| Willing to Accept for 180 cents    | 0.85<br>(0.03) | 0.78<br>(0.03) | 0.07*<br>(0.04) | 0.08    |
| Willing to Accept for 200 cents    | 0.94<br>(0.02) | 0.90<br>(0.02) | 0.04<br>(0.03)  | 0.20    |
| Willing to Accept for 250 cents    | 0.93<br>(0.02) | 0.93<br>(0.02) | 0.00<br>(0.03)  | 0.86    |
| Willing to Accept for 300 cents    | 0.96<br>(0.01) | 0.95<br>(0.02) | 0.01<br>(0.02)  | 0.65    |
| Willing to Accept for 500 cents    | 0.99<br>(0.01) | 0.98<br>(0.01) | 0.01<br>(0.01)  | 0.42    |

*Notes:* This table reports the effects of competition on willingness to accept (WTA) performing the experimental task again, restricted to the gain-framing condition. Column (1) shows mean acceptance rates in the competition group, and column (2) shows mean acceptance rates in the control group. Column (3) reports the difference between the two groups, and column (4) provides the associated p-values. Standard errors are reported in parentheses. The outcome variable is a binary indicator equal to one if participants accepted performing the task again at the specified payment amount and zero otherwise. The first row reports the estimated minimum WTA in dollars. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.6: Treatment Effects on Task Preference (Loss Framing)

| <i>Sample: Loss Framing</i>        | Competition    | Control        | Difference       | p-value |
|------------------------------------|----------------|----------------|------------------|---------|
|                                    | (1)            | (2)            | [(1)-(2)]        | (4)     |
| Minimum Willingness to Accept (\$) | 1.34<br>(0.08) | 1.55<br>(0.09) | -0.21*<br>(0.12) | 0.08    |
| Willing to Accept for 20 cents     | 0.19<br>(0.03) | 0.19<br>(0.03) | 0.00<br>(0.04)   | 0.98    |
| Willing to Accept for 40 cents     | 0.21<br>(0.03) | 0.22<br>(0.03) | -0.01<br>(0.04)  | 0.83    |
| Willing to Accept for 60 cents     | 0.25<br>(0.03) | 0.24<br>(0.03) | 0.01<br>(0.04)   | 0.88    |
| Willing to Accept for 80 cents     | 0.33<br>(0.03) | 0.29<br>(0.03) | 0.04<br>(0.05)   | 0.43    |
| Willing to Accept for 100 cents    | 0.59<br>(0.04) | 0.52<br>(0.04) | 0.07<br>(0.05)   | 0.17    |
| Willing to Accept for 120 cents    | 0.64<br>(0.03) | 0.55<br>(0.04) | 0.09*<br>(0.05)  | 0.06    |
| Willing to Accept for 140 cents    | 0.68<br>(0.03) | 0.59<br>(0.03) | 0.09*<br>(0.05)  | 0.07    |
| Willing to Accept for 160 cents    | 0.74<br>(0.03) | 0.65<br>(0.03) | 0.09**<br>(0.05) | 0.04    |
| Willing to Accept for 180 cents    | 0.77<br>(0.03) | 0.69<br>(0.03) | 0.08*<br>(0.04)  | 0.08    |
| Willing to Accept for 200 cents    | 0.86<br>(0.02) | 0.81<br>(0.03) | 0.05<br>(0.04)   | 0.18    |
| Willing to Accept for 250 cents    | 0.88<br>(0.02) | 0.84<br>(0.03) | 0.04<br>(0.03)   | 0.25    |
| Willing to Accept for 300 cents    | 0.94<br>(0.02) | 0.91<br>(0.02) | 0.03<br>(0.03)   | 0.25    |
| Willing to Accept for 500 cents    | 1.00<br>(0.00) | 0.97<br>(0.01) | 0.03**<br>(0.01) | 0.01    |

*Notes:* This table reports the effects of competition on willingness to accept (WTA) performing the experimental task again, restricted to the loss-framing condition. Column (1) shows mean acceptance rates in the competition group, and column (2) shows mean acceptance rates in the control group. Column (3) reports the difference between the two groups, and column (4) provides the associated p-values. Standard errors are reported in parentheses. The outcome variable is a binary indicator equal to one if participants accepted performing the task again at the specified payment amount and zero otherwise. The first row reports the estimated minimum WTA in dollars. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .



Appendix Table B.7: Treatment Effects on Task Preference (Self-Report Measure)

|                | Dependent Variable:<br>Another 10 questions without reward |                     |                     |
|----------------|------------------------------------------------------------|---------------------|---------------------|
|                | Pooled<br>(1)                                              | Gain Framing<br>(2) | Loss Framing<br>(3) |
| Competition    | 0.26<br>(0.16)                                             | 0.12<br>(0.22)      | 0.39*<br>(0.23)     |
| Control mean   | 4.36                                                       | 4.74                | 3.98                |
| Observations   | 787                                                        | 390                 | 397                 |
| R <sup>2</sup> | 0.00                                                       | 0.00                | 0.01                |

*Notes:* This table shows the effect of competition on task preference, measured by an unincentivized question asking participants, on a scale from 1 to 7, how likely they were to take another 10 IQ questions, similar to the ones in the IQ quiz, but without monetary rewards. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.8: Treatment Effects on Zero-sum Thinking

| <i>Dependent Variable: Zero-sum Thinking<br/>(World Values Survey Measure)</i> |                   |                  |                 |
|--------------------------------------------------------------------------------|-------------------|------------------|-----------------|
|                                                                                | Pooled            | Gain Framing     | Loss Framing    |
|                                                                                | (1)               | (2)              | (3)             |
| Competition                                                                    | -0.44**<br>(0.20) | -0.50*<br>(0.27) | -0.39<br>(0.28) |
| Control mean                                                                   | 4.27              | 4.15             | 4.38            |
| Observations                                                                   | 787               | 390              | 397             |
| R <sup>2</sup>                                                                 | 0.01              | 0.01             | 0.01            |

*Notes:* This table reports the estimated treatment effects of competition on zero-sum thinking. Zero-sum thinking is measured using a World Values Survey question (Inglehart et al., 2014) in which respondents rate their agreement, on a 1–10 scale, between two statements: “Wealth can grow so there’s enough for everyone” (1, indicating minimal zero-sum thinking) and “People can only become wealthy at the expense of others” (10, indicating strong zero-sum thinking). Higher values on the y-axis therefore represent stronger zero-sum thinking. Each column presents results from an OLS regression of zero-sum thinking on a treatment indicator for competition. Column (1) pools all observations, while Columns (2) and (3) report results separately for gain framing and loss framing. Robust standard errors in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.9: Treatment Effects on Zero-sum Thinking (Immigration Measure)

|                | <i>Dependent Variable: Zero-sum Thinking<br/>(Immigration Measure)</i> |                |                 |
|----------------|------------------------------------------------------------------------|----------------|-----------------|
|                | Pooled                                                                 | Gain Framing   | Loss Framing    |
|                | (1)                                                                    | (2)            | (3)             |
| Competition    | 0.01<br>(0.03)                                                         | 0.05<br>(0.05) | -0.03<br>(0.05) |
| Control mean   | 0.34                                                                   | 0.33           | 0.35            |
| Observations   | 787                                                                    | 390            | 397             |
| R <sup>2</sup> | 0.00                                                                   | 0.00           | 0.00            |

*Notes:* This table shows the estimated treatment effects of competition on zero-sum thinking. Zero-sum thinking is measured with a scenario-choice question in which both immigrants and locals work as day laborers. On Day 1, immigrant workers find 20 jobs and local workers 40 jobs. On Day 2, immigrant workers find 40 jobs. Respondents then choose whether local workers find 20 or 80 jobs on Day 2. Choosing 20 reflects zero-sum thinking (coded as 1), while choosing 80 reflects non-zero-sum thinking (coded as 0). Column (1) presents results for the pooled sample, while Columns (2) and (3) present results for gain and loss framing, respectively. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.10: Effects on Altruism by Belief in Winning

|                                        | <i>Dependent Variable: Altruism</i> |        |              |        |              |        |
|----------------------------------------|-------------------------------------|--------|--------------|--------|--------------|--------|
|                                        | Pooled                              |        | Gain Framing |        | Loss Framing |        |
|                                        | (1)                                 | (2)    | (3)          | (4)    | (5)          | (6)    |
| Competition                            | 2.88**                              | 0.70   | 1.59         | -3.15  | 4.14**       | 3.88   |
|                                        | (1.36)                              | (2.59) | (1.88)       | (3.66) | (1.96)       | (3.64) |
| Belief in Winning                      |                                     | -1.13  |              | -2.93  |              | 0.19   |
|                                        |                                     | (2.25) |              | (3.20) |              | (3.17) |
| Competition $\times$ Belief in Winning |                                     | 3.19   |              | 6.59   |              | 0.47   |
|                                        |                                     | (3.04) |              | (4.28) |              | (4.30) |
| Control mean                           | 36.38                               | 37.26  | 37.17        | 39.51  | 35.61        | 35.47  |
| Observations                           | 787                                 | 787    | 390          | 390    | 397          | 397    |
| R <sup>2</sup>                         | 0.01                                | 0.01   | 0.00         | 0.01   | 0.01         | 0.01   |

*Notes:* The table shows the effect of competition on altruism, measured by the amount offered to a randomly matched partner in the dictator game. It also presents heterogeneous effects by Belief in Winning, a binary variable equal to 1 if the participant expected to win and 0 otherwise. Columns (1)–(2) presents results for the pooled sample, while Columns (3)–(4) for gain framing and Columns (5)–(6) for loss framing. Perceived Rank is the self-assessed percentile rank in the IQ quiz. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.11: Treatment Effects on Altruism by Perceived Percentile Rank

|                                     | <i>Dependent Variable: Altruism</i> |        |              |         |              |        |
|-------------------------------------|-------------------------------------|--------|--------------|---------|--------------|--------|
|                                     | Pooled                              |        | Gain Framing |         | Loss Framing |        |
|                                     | (1)                                 | (2)    | (3)          | (4)     | (5)          | (6)    |
| Competition                         | 2.88**                              | -1.05  | 1.59         | -5.41   | 4.14**       | 2.82   |
|                                     | (1.36)                              | (2.44) | (1.88)       | (3.34)  | (1.96)       | (3.52) |
| Perceived Rank                      |                                     | -1.45  |              | -4.82*  |              | 1.64   |
|                                     |                                     | (2.02) |              | (2.61)  |              | (3.03) |
| Competition $\times$ Perceived Rank |                                     | 5.91** |              | 10.32** |              | 2.00   |
|                                     |                                     | (2.93) |              | (4.03)  |              | (4.22) |
| Control mean                        | 36.38                               | 37.32  | 37.17        | 40.34   | 35.61        | 34.58  |
| Observations                        | 787                                 | 787    | 390          | 390     | 397          | 397    |
| R <sup>2</sup>                      | 0.01                                | 0.01   | 0.00         | 0.02    | 0.01         | 0.02   |

*Notes:* The table shows the effect of competition on altruism, measured by the amount offered to a randomly matched partner in the dictator game. It also shows heterogeneous effects by perceived percentile rank. Columns (1)–(2) presents results for the pooled sample, while Columns (3)–(4) for gain framing and Columns (5)–(6) for loss framing. Perceived Rank is the self-assessed percentile rank in the IQ quiz. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.12: Effects on Contribution Decisions and Beliefs in a Public Goods Game

|                | <i>Contribution to Public Pool</i> |                        |                        | <i>Belief: Others Not Contributing (%)</i> |                        |                        |
|----------------|------------------------------------|------------------------|------------------------|--------------------------------------------|------------------------|------------------------|
|                | Pooled<br>(1)                      | Gain<br>Framing<br>(2) | Loss<br>Framing<br>(3) | Pooled<br>(4)                              | Gain<br>Framing<br>(5) | Loss<br>Framing<br>(6) |
| Competition    | -0.05<br>(0.04)                    | -0.05<br>(0.05)        | -0.05<br>(0.05)        | 0.23<br>(0.15)                             | 0.37*<br>(0.21)        | 0.09<br>(0.22)         |
| Control mean   | 0.60                               | 0.57                   | 0.63                   | 6.17                                       | 6.25                   | 6.09                   |
| Observations   | 787                                | 390                    | 397                    | 787                                        | 390                    | 397                    |
| R <sup>2</sup> | 0.00                               | 0.00                   | 0.00                   | 0.00                                       | 0.01                   | 0.00                   |

*Notes:* This table reports the effect of competition on contribution decisions and beliefs in a public goods game. Competition is a binary indicator equal to 1 for participants in the competition group and 0 for those in the control group. The dependent variable in Columns (1)–(3) is a binary indicator for contributing to the public pool, while the dependent variable in Columns (4)–(6) is the participant’s belief (in percent) about the share of others not contributing (i.e., keeping the money for themselves). Columns (1) and (4) pool all treatments, whereas Columns (2)–(3) and (5)–(6) report results separately for the gain- and loss-framing conditions. Robust standard errors are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.13: Treatment Effects on Positive Emotions

| <i>Panel A. Dependent Variable: Excitement</i> |        |         |              |         |              |         |
|------------------------------------------------|--------|---------|--------------|---------|--------------|---------|
|                                                | Pooled |         | Gain Framing |         | Loss Framing |         |
|                                                | (1)    | (2)     | (3)          | (4)     | (5)          | (6)     |
| Competition                                    | 0.24*  | 0.42*** | 0.23         | 0.41**  | 0.26         | 0.42**  |
|                                                | (0.13) | (0.12)  | (0.18)       | (0.17)  | (0.19)       | (0.18)  |
| Believe Win                                    |        | 1.21*** |              | 1.24*** |              | 1.14*** |
|                                                |        | (0.15)  |              | (0.22)  |              | (0.20)  |
| Control mean                                   | 4.65   | 4.65    | 4.83         | 4.83    | 4.47         | 4.47    |
| Observations                                   | 787    | 787     | 390          | 390     | 397          | 397     |
| R <sup>2</sup>                                 | 0.00   | 0.09    | 0.00         | 0.10    | 0.00         | 0.08    |

| <i>Panel B. Dependent Variable: Satisfaction</i> |        |         |              |         |              |         |
|--------------------------------------------------|--------|---------|--------------|---------|--------------|---------|
|                                                  | Pooled |         | Gain Framing |         | Loss Framing |         |
|                                                  | (1)    | (2)     | (3)          | (4)     | (5)          | (6)     |
| Competition                                      | 0.21   | 0.41*** | 0.24         | 0.45*** | 0.18         | 0.37**  |
|                                                  | (0.13) | (0.12)  | (0.18)       | (0.17)  | (0.19)       | (0.18)  |
| Believe Win                                      |        | 1.37*** |              | 1.41*** |              | 1.30*** |
|                                                  |        | (0.14)  |              | (0.21)  |              | (0.20)  |
| Control mean                                     | 4.33   | 4.33    | 4.48         | 4.48    | 4.19         | 4.19    |
| Observations                                     | 787    | 787     | 390          | 390     | 397          | 397     |
| R <sup>2</sup>                                   | 0.00   | 0.12    | 0.00         | 0.13    | 0.00         | 0.11    |

Notes: This table reports the treatment effects of competition on positive emotions. Panel A presents results for excitement, and Panel B presents results for satisfaction. Each panel shows estimates for the pooled sample (columns 1–2), the gain-framing condition (columns 3–4), and the loss-framing condition (columns 5–6). Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.14: Treatment Effects of Competition on Negative Emotions

| <i>Panel A. Dependent Variable: Stress</i> |        |          |              |          |              |          |
|--------------------------------------------|--------|----------|--------------|----------|--------------|----------|
|                                            | Pooled |          | Gain Framing |          | Loss Framing |          |
|                                            | (1)    | (2)      | (3)          | (4)      | (5)          | (6)      |
| Competition                                | -0.23  | -0.32**  | -0.43**      | -0.53*** | -0.02        | -0.11    |
|                                            | (0.14) | (0.14)   | (0.19)       | (0.20)   | (0.20)       | (0.20)   |
| Believe Win                                |        | -0.65*** |              | -0.67*** |              | -0.60*** |
|                                            |        | (0.16)   |              | (0.22)   |              | (0.22)   |
| Control mean                               | 3.90   | 3.90     | 3.84         | 3.84     | 3.96         | 3.96     |
| Observations                               | 787    | 787      | 390          | 390      | 397          | 397      |
| R <sup>2</sup>                             | 0.00   | 0.03     | 0.01         | 0.04     | 0.00         | 0.02     |

| <i>Panel B. Dependent Variable: Anxiety</i> |        |          |              |        |              |          |
|---------------------------------------------|--------|----------|--------------|--------|--------------|----------|
|                                             | Pooled |          | Gain Framing |        | Loss Framing |          |
|                                             | (1)    | (2)      | (3)          | (4)    | (5)          | (6)      |
| Competition                                 | -0.02  | -0.09    | -0.26        | -0.32  | 0.22         | 0.13     |
|                                             | (0.14) | (0.14)   | (0.20)       | (0.20) | (0.20)       | (0.20)   |
| Believe Win                                 |        | -0.53*** |              | -0.40* |              | -0.62*** |
|                                             |        | (0.16)   |              | (0.23) |              | (0.21)   |
| Control mean                                | 3.95   | 3.95     | 3.94         | 3.94   | 3.96         | 3.96     |
| Observations                                | 787    | 787      | 390          | 390    | 397          | 397      |
| R <sup>2</sup>                              | 0.00   | 0.01     | 0.00         | 0.01   | 0.00         | 0.02     |

| <i>Panel C. Dependent Variable: Embarrassment</i> |        |          |              |          |              |          |
|---------------------------------------------------|--------|----------|--------------|----------|--------------|----------|
|                                                   | Pooled |          | Gain Framing |          | Loss Framing |          |
|                                                   | (1)    | (2)      | (3)          | (4)      | (5)          | (6)      |
| Competition                                       | -0.04  | -0.15    | -0.24        | -0.35**  | 0.15         | 0.05     |
|                                                   | (0.12) | (0.12)   | (0.16)       | (0.16)   | (0.18)       | (0.18)   |
| Believe Win                                       |        | -0.75*** |              | -0.74*** |              | -0.74*** |
|                                                   |        | (0.14)   |              | (0.19)   |              | (0.21)   |
| Control mean                                      | 2.50   | 2.50     | 2.47         | 2.47     | 2.53         | 2.53     |
| Observations                                      | 787    | 787      | 390          | 390      | 397          | 397      |
| R <sup>2</sup>                                    | 0.00   | 0.04     | 0.01         | 0.05     | 0.00         | 0.04     |

Notes: This table reports the treatment effects of competition on negative emotions. Panel A presents results for stress, Panel B presents results for anxiety, and Panel C for embarrassment. Each panel shows estimates for the pooled sample (columns 1–2), the gain-framing condition (columns 3–4), and the loss-framing condition (columns 5–6). Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .



Appendix Table B.15: Treatment Effects of Competition on Positive and Negative Emotions

|                   | <i>Positive Emotion Index</i> |                   |                |                   | <i>Negative Emotion Index</i> |                    |                |                    |
|-------------------|-------------------------------|-------------------|----------------|-------------------|-------------------------------|--------------------|----------------|--------------------|
|                   | Gain Framing                  |                   | Loss Framing   |                   | Gain Framing                  |                    | Loss Framing   |                    |
|                   | (1)                           | (2)               | (3)            | (4)               | (5)                           | (6)                | (7)            | (8)                |
| Competition       | 0.18<br>(0.13)                | 0.33***<br>(0.12) | 0.17<br>(0.14) | 0.31**<br>(0.13)  | -0.29**<br>(0.14)             | -0.37***<br>(0.14) | 0.11<br>(0.15) | 0.02<br>(0.15)     |
| Belief in Winning |                               | 1.03***<br>(0.15) |                | 0.95***<br>(0.14) |                               | -0.56***<br>(0.17) |                | -0.60***<br>(0.17) |
| Control mean      | 0.04                          | 0.04              | -0.21          | -0.21             | 0.04                          | 0.04               | -0.21          | -0.21              |
| Observations      | 390                           | 390               | 397            | 397               | 390                           | 390                | 397            | 397                |
| R <sup>2</sup>    | 0.01                          | 0.13              | 0.00           | 0.11              | 0.01                          | 0.04               | 0.00           | 0.04               |

*Notes:* This table reports the treatment effects of competition on the emotion index. The positive emotion index (columns 1–6) is constructed from measures of excitement and satisfaction, while the negative emotion index (columns 7–10) is constructed from measures of stress, anxiety, and embarrassment, using principal component analysis (PCA). Estimates are reported separately for the gain-framing and loss-framing conditions. “Competition” is an indicator for assignment to the competition treatment. “Believe Win” captures participants’ belief that they would win the competition. Robust standard errors are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.16: Heterogeneity in Intrinsic Enjoyment by Perceived Ability

|                                     | <i>Dependent Variable: Enjoyment</i> |                   |                   |
|-------------------------------------|--------------------------------------|-------------------|-------------------|
|                                     | Pooled                               | Gain Framing      | Loss Framing      |
|                                     | (1)                                  | (2)               | (3)               |
| Competition                         | 0.69***<br>(0.23)                    | 0.84***<br>(0.31) | 0.56*<br>(0.33)   |
| Perceived Rank                      | 1.49***<br>(0.19)                    | 1.77***<br>(0.26) | 1.21***<br>(0.28) |
| Competition $\times$ Perceived Rank | -0.56**<br>(0.26)                    | -0.81**<br>(0.35) | -0.33<br>(0.37)   |
| Belief in Winning                   | 0.70***<br>(0.15)                    | 0.66***<br>(0.20) | 0.73***<br>(0.23) |
| Observations                        | 787                                  | 390               | 397               |
| R <sup>2</sup>                      | 0.21                                 | 0.26              | 0.18              |

*Notes:* This table reports OLS regressions of enjoyment on competition, perceived performance rank, their interaction, and belief in winning. The dependent variable is enjoyment, measured on a 7-point Likert scale, with larger values indicating greater utility. Competition is a treatment indicator equal to one for participants assigned to the competition group and zero for those in the control group. Perceived Rank denotes participants' self-assessed percentile rank of performance. Columns (1), (2), and (3) present results for the pooled sample, the gain-framing condition, and the loss-framing condition, respectively. Heterogeneity is captured by the interaction between competition and perceived rank. Robust standard errors are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.17: Heterogeneity in Intrinsic Enjoyment Across Demographic Groups

|                                     | <i>Dependent Variable: Enjoyment</i> |                  |                   |                  |
|-------------------------------------|--------------------------------------|------------------|-------------------|------------------|
|                                     | (1)                                  | (2)              | (3)               | (4)              |
| Competition                         | 0.28*<br>(0.15)                      | 0.37**<br>(0.17) | 0.70***<br>(0.26) | 0.72**<br>(0.29) |
| Competition × Age (40-59)           | 0.34<br>(0.26)                       |                  |                   |                  |
| Competition × Age ( $\geq 60$ )     | 0.34<br>(0.41)                       |                  |                   |                  |
| Competition × Female                |                                      | 0.15<br>(0.23)   |                   |                  |
| Competition × Bachelor Degree       |                                      |                  | -0.24<br>(0.31)   |                  |
| Competition × Graduate Degree       |                                      |                  | -0.69**<br>(0.32) |                  |
| Competition × Income (\$35K-\$75K)  |                                      |                  |                   | -0.51<br>(0.36)  |
| Competition × Income (\$75K-\$150K) |                                      |                  |                   | -0.10<br>(0.34)  |
| Competition × Income ( $> \$150K$ ) |                                      |                  |                   | -0.69<br>(0.44)  |
| Observations                        | 787                                  | 787              | 787               | 787              |
| R <sup>2</sup>                      | 0.13                                 | 0.13             | 0.15              | 0.14             |

*Notes:* This table reports heterogeneous treatment effects of competition on utility using the pooled sample. Results by gain and loss framing are presented in the Appendix Tables B.18 and B.19. The dependent variable is enjoyment, measured on a 7-point scale where higher values indicate greater enjoyment. Competition is a treatment indicator equal to one for participants assigned to the competition group and zero otherwise. Each column corresponds to a separate regression where competition is interacted with the indicated subgroup characteristic. All specifications include belief in winning as a control. The omitted categories are age 18–39, male, high school or less, and income below \$35K, which are coded as zero within their respective groups. For example, age is coded as 0 for ages 18–39, 1 for ages 40–59, and 2 for ages 60 and above; other subgroup variables follow a similar coding scheme. Robust standard errors are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.18: Heterogeneity in Intrinsic Enjoyment Across Demographic Groups

| <i>Sample: Gain Framing</i>         | <i>Dependent Variable: Enjoyment</i> |                |                   |                 |
|-------------------------------------|--------------------------------------|----------------|-------------------|-----------------|
|                                     | (1)                                  | (2)            | (3)               | (4)             |
| Competition                         | 0.36*<br>(0.22)                      | 0.32<br>(0.24) | 0.71**<br>(0.34)  | 0.50<br>(0.38)  |
| Competition × Age (40-59)           | 0.15<br>(0.34)                       |                |                   |                 |
| Competition × Age ( $\geq 60$ )     | 0.29<br>(0.57)                       |                |                   |                 |
| Competition × Female                |                                      | 0.25<br>(0.33) |                   |                 |
| Competition × Bachelor Degree       |                                      |                | -0.17<br>(0.41)   |                 |
| Competition × Graduate Degree       |                                      |                | -0.91**<br>(0.42) |                 |
| Competition × Income (\$35K-\$75K)  |                                      |                |                   | -0.06<br>(0.47) |
| Competition × Income (\$75K-\$150K) |                                      |                |                   | 0.04<br>(0.46)  |
| Competition × Income ( $> \$150K$ ) |                                      |                |                   | -0.59<br>(0.56) |
| Observations                        | 390                                  | 390            | 390               | 390             |
| R <sup>2</sup>                      | 0.15                                 | 0.14           | 0.18              | 0.15            |

*Notes:* This table reports heterogeneous treatment effects of competition on utility using the gain framing sample. The dependent variable is enjoyment, measured on a 7-point scale where higher values indicate greater enjoyment. Competition is a treatment indicator equal to one for participants assigned to the competition group and zero otherwise. Each column corresponds to a separate regression where competition is interacted with the indicated subgroup characteristic. The omitted categories are age 18–39, male, high school or less, and income below \$35K, which are coded as zero within their respective groups. For example, age is coded as 0 for ages 18–39, 1 for ages 40–59, and 2 for ages 60 and above; other subgroup variables follow a similar coding scheme. Robust standard errors are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table B.19: Heterogeneity in Intrinsic Enjoyment Across Demographic Groups

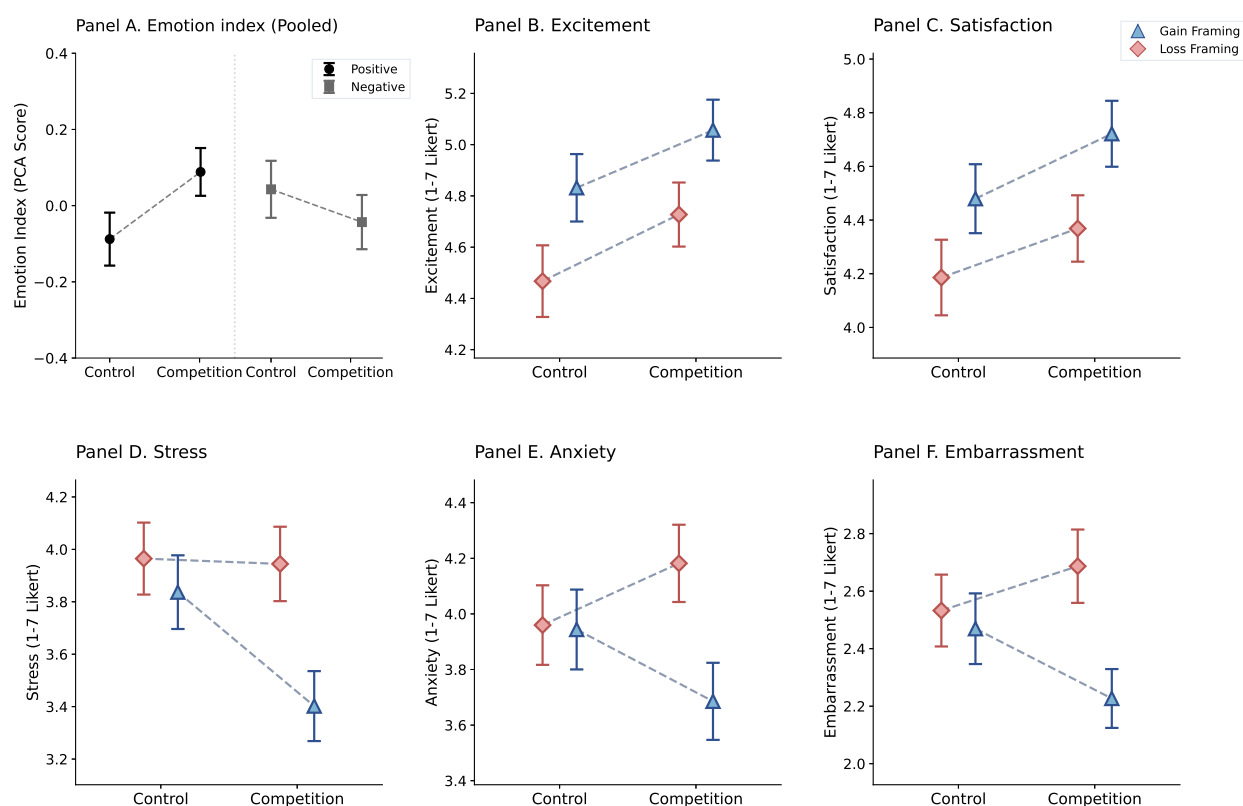
| <i>Sample: Loss Framing</i>                | <i>Dependent Variable: Enjoyment</i> |                 |                 |                  |
|--------------------------------------------|--------------------------------------|-----------------|-----------------|------------------|
|                                            | (1)                                  | (2)             | (3)             | (4)              |
| Competition                                | 0.24<br>(0.22)                       | 0.42*<br>(0.23) | 0.67*<br>(0.41) | 1.02**<br>(0.46) |
| Competition $\times$ Age (40-59)           | 0.45<br>(0.38)                       |                 |                 |                  |
| Competition $\times$ Age ( $\geq 60$ )     | 0.35<br>(0.59)                       |                 |                 |                  |
| Competition $\times$ Female                |                                      | 0.05<br>(0.34)  |                 |                  |
| Competition $\times$ Bachelor Degree       |                                      |                 | -0.27<br>(0.48) |                  |
| Competition $\times$ Graduate Degree       |                                      |                 | -0.48<br>(0.49) |                  |
| Competition $\times$ Income (\$35K-\$75K)  |                                      |                 |                 | -1.00*<br>(0.55) |
| Competition $\times$ Income (\$75K-\$150K) |                                      |                 |                 | -0.32<br>(0.53)  |
| Competition $\times$ Income ( $>$ \$150K)  |                                      |                 |                 | -0.82<br>(0.71)  |
| Observations                               | 397                                  | 397             | 397             | 397              |
| R <sup>2</sup>                             | 0.12                                 | 0.12            | 0.13            | 0.13             |

*Notes:* This table reports heterogeneous treatment effects of competition on utility using the loss framing sample. The dependent variable is enjoyment, measured on a 7-point scale where higher values indicate greater enjoyment. Competition is a treatment indicator equal to one for participants assigned to the competition group and zero otherwise. Each column corresponds to a separate regression where competition is interacted with the indicated subgroup characteristic. The omitted categories are age 18–39, male, high school or less, and income below \$35K, which are coded as zero within their respective groups. For example, age is coded as 0 for ages 18–39, 1 for ages 40–59, and 2 for ages 60 and above; other subgroup variables follow a similar coding scheme. Robust standard errors are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## C Supplemental Analyses

**Decomposing Utility into Emotional Components.** To better understand the mechanisms underlying utility changes, I decompose subjective well-being into specific emotional components. Appendix Figure C.17 shows how competition affects excitement, satisfaction, stress, anxiety, and embarrassment. I classify the first two as positive emotions and the latter three as negative. All emotions were measured on a 7-point Likert scale, with higher values indicating stronger intensities.

Appendix Figure C.17: Effects of Competition on Emotional Components of Utility



*Notes:* This figure shows the treatment effects of competition on the emotional components of utility, including positive emotions (excitement, satisfaction) and negative emotions (stress, anxiety, embarrassment). These emotions are measured using 7-point Likert scale questions administered after the experimental task. Higher values indicate stronger experiences of the respective emotion. Panels B–F display mean values of participants' responses to each emotion question across the four experimental treatments (control vs. competition, under gain vs. loss framing), with 95% confidence intervals. Panel A summarizes these measures into an overall emotion index constructed using principal component analysis (PCA). Detailed regression estimates of the effects of competition on each emotional component of utility are provided in [Appendix Table B.15](#), and [Appendix Table B.13](#) and [B.14](#).

Panel A of Appendix Figure C.17 documents the general pattern that competition increases positive emotions while reducing negative ones. Panels B and C show that compe-

tition raises excitement and satisfaction in the IQ quiz, regardless of whether the payoff is framed as a gain or a loss. Panels D, E, and F indicate that competition reduces stress, anxiety, and embarrassment under gain framing, while potentially increasing them under loss framing. Individuals are generally more excited and satisfied, and less stressed, anxious, and embarrassed when payoffs are framed as gains rather than losses. While these findings provide valuable insights into the emotional consequences of competition, it should be noted that some effects are estimated with limited statistical power.

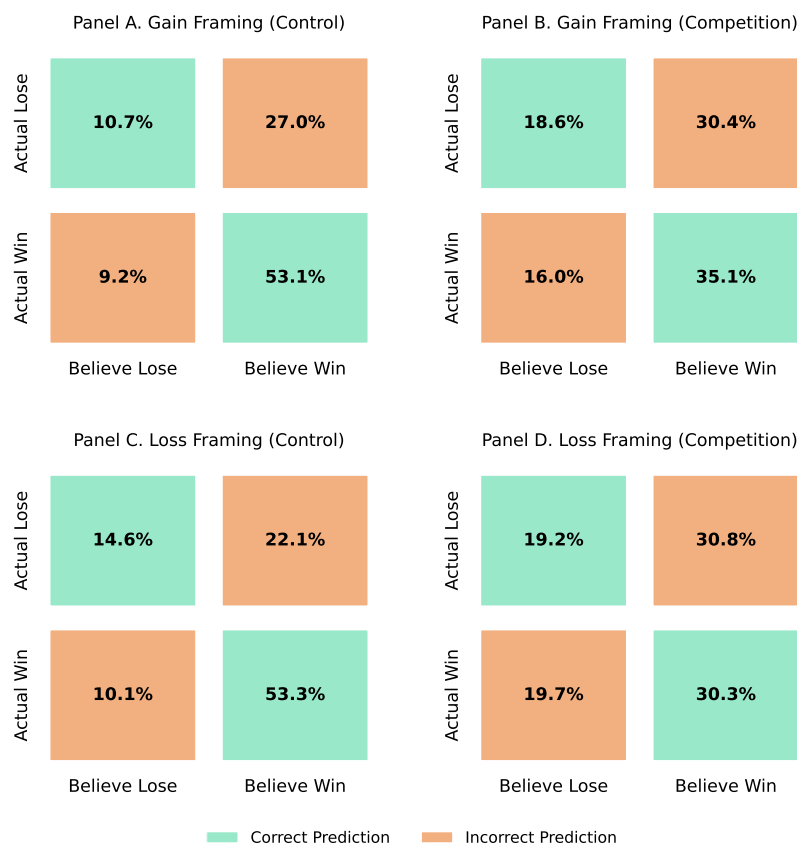
Appendix Table B.15 present the treatment effects of competition on positive and negative emotions under gain and loss framing, respectively. Introducing the "belief in winning" variable shuts down the belief channel and isolates the intrinsic emotional effect of competition on specific emotional components. The change in the competition coefficients after controlling for belief in winning suggests that competition, through the belief channel, lowers expectations of winning, which reduces positive emotions and increases negative emotions. The intrinsic utility channel, however, indicates that competition itself generates positive emotions under both gain and loss framing and mitigates negative emotions under gain framing, but has a weaker mitigating effect under loss framing.

These results reveal a dual mechanism through which competition affects emotional components of utility. Through the belief channel, competition reduces positive emotions and increases negative ones by lowering individuals' expectations of winning. Conversely, through the preference channel, competition does the opposite: it generates positive emotions and reduces negative ones, particularly under gain framing. This distinction highlights the importance of separating the psychological impact of competitive environments from the effects of winning expectations when evaluating the overall emotional consequences of competition.

**Expected versus Actual Outcomes.** Welfare assessments should account not only for the utility derived from actual outcomes, but also for the utility generated by the act of competing itself. The utility participants derive from the competitive process may differ substantially from the utility they obtain from actual outcomes. Understanding this distinction is crucial for evaluating the overall welfare implications of competition. If people hold biased expectations about competitive outcomes or their emotional responses to these outcomes, their ex-ante willingness to compete may diverge from their ex-post experienced utility. Appendix Figure C.18 illustrates this distinction by comparing anticipated versus actual outcomes: believing one would win vs. actually winning, and believing one would lose vs. actually losing. In the competition group, a correct prediction means participants believed they would beat their opponent and indeed did so; in the control group, it means

they expected their score to exceed the threshold and it did. Incorrect predictions capture the opposite cases. Green cells denote accurate predictions, while orange cells denote inaccurate ones.

Appendix Figure C.18: Anticipated versus Actual Outcomes



*Notes:* This figure presents confusion matrices of participants' anticipated versus actual outcomes across the four experimental conditions (control vs. competition; gain vs. loss framing). Each cell shows the percentage of participants in that category. Green cells denote correct predictions (believing one would win and actually winning; believing one would lose and actually losing), while orange cells denote incorrect predictions (believing one would win but actually losing; believing one would lose but actually winning).

Overall, participants' predictive accuracy was relatively low, with accuracy rates in the competition groups even lower than in the control groups. As shown in the figure, approximately 65% of participants in the control conditions (both gain and loss framing) predicted correctly, whereas only about 50% of those in the competition conditions did so. This suggests that participants form more accurate expectations when evaluated against a fixed threshold than when competing against an uncertain opponent.

A noteworthy pattern is that around 60% of participants in the competition conditions (gain and loss framing) believed they would win, whereas roughly 80% of participants in the control groups believed they would exceed the threshold. In reality, however, only half



of participants in competition could win by design, and as Appendix Figure B.15 shows, nearly 40% of control participants failed to reach the threshold. These findings point to a general tendency toward overconfidence, consistent with prior literature (Niederle and Vesterlund, 2007; Deaves et al., 2010; Bhatt and Smith, 2025). Interestingly, competition appears to attenuate this bias: although participants remained overly optimistic, the gap between expected and actual outcomes was smaller under competition than under non-competition. This attenuation likely reflects the additional perceived uncertainty introduced by competing against an opponent rather than facing a fixed benchmark.

**Heterogeneity in Winning Expectations.** Participants with lower perceived ability experience greater enjoyment from competition than those who consider themselves higher ranked. As discussed in Section 5.2, competition influence utility through a belief channel by shaping individuals' expectations of winning. A natural question, therefore, is whether the observed heterogeneity across ability groups arises because competition differentially affects their winning expectations. Appendix Figure B.16 suggests this is not the case.

Panel A of Appendix Figure B.16 shows that competition significantly reduces winning expectations for both high- and low-ability participants ( $p < 0.001$  for both groups, pooled sample), and the difference between them is not statistically significant ( $p = 0.166$ , pooled sample). This suggests that the heterogeneity in utility is more likely to come through the intrinsic utility channel, whereby individuals with lower perceived relative ability derive greater enjoyment from competition than their higher-ranked counterparts. This inference is confirmed by the intrinsic-channel heterogeneity evidence presented later in this subsection. Panel B provides a more detailed view of the relationship between perceived rank and winning expectations across competition and control groups. Consistent with Panel A, participants in the competition condition hold systematically lower beliefs in winning, regardless of whether they view themselves as high or low ability.<sup>45</sup>

Appendix Figure B.11 explores heterogeneity in the effect of competition on winning expectations across demographic subgroups. By age, competition substantially lowers winning expectations among younger participants aged 18–39 ( $p < 0.001$ , pooled sample)<sup>46</sup> and among older participants aged 60–80 ( $p = 0.014$ ), while the effect for those aged 40–59 is smaller and statistically insignificant ( $p = 0.106$ ). By gender, both women ( $p = 0.018$ ) and men ( $p < 0.001$ ) show declines in winning expectations, with no statis-

<sup>45</sup>Panel B plots predicted probabilities of winning as a function of perceived rank for the pooled sample, based on estimates from a logit model, since belief in winning is a binary variable.

<sup>46</sup>All  $p$ -values reported in this paragraph are for the pooled sample.

tically significant difference between the two groups ( $p = 0.284$ ). In terms of education, the reduction is strongest among participants with a high school degree or less ( $p < 0.001$ ) and among those with graduate training ( $p = 0.008$ ), whereas the effect is smaller and only marginally significant among those with a bachelor's degree ( $p = 0.075$ ). Finally, the income gradient is clear: participants in the lower income groups, earning below \$35,000 ( $p < 0.001$ ) or between \$35,000 and \$75,000 ( $p = 0.007$ ), exhibit pronounced declines in winning expectations. In contrast, the effect is weaker and statistically insignificant among participants with higher incomes (\$75,000–\$150,000:  $p = 0.100$ ; above \$150,000:  $p = 0.245$ ).

**Heterogeneity in Intrinsic Enjoyment.** The preceding results indicate that heterogeneity in enjoyment across ability groups cannot be explained by differences in winning expectations (belief channel), pointing instead to the intrinsic utility channel as the underlying source. Appendix Table B.16 provides direct evidence supporting this interpretation. The regression specifications include winning beliefs as controls, effectively shutting down the belief channel and allowing the competition coefficient to be cleanly interpreted as capturing intrinsic enjoyment. The interaction terms are statistically significant, confirming that individuals who perceive themselves as lower ability derive greater intrinsic enjoyment from competition than those who view themselves as higher ability. This pattern holds in the pooled sample and under gain framing. Under loss framing, the interaction coefficient is not statistically significant, though its magnitude points in the same direction.

Appendix Table B.17 explores the heterogeneity in intrinsic enjoyment across demographic groups using the pooled sample. Results for the gain and loss framing samples are presented in Appendix Tables B.18 and B.19, which show broadly similar patterns. Among age groups, younger participants (under 40) experience positive enjoyment from competition, with no significant differences relative to those aged 40–59 or 60 and above. For gender, both men and women derive intrinsic enjoyment from competition, and the difference between them is not statistically significant. By education, individuals with a high school degree or less obtain substantial enjoyment, while those with a bachelor's or graduate degree experience lower levels. The reduction relative to bachelor's degree holders is not statistically significant, but the lower enjoyment among graduate degree holders is significant. In terms of income, participants with incomes below \$35,000 experience strong and significant enjoyment from competition, while higher-income groups gain somewhat lower enjoyment, though the differences are not statistically significant.

**Heterogeneity in Altruism.** Appendix Figure B.13 illustrates the heterogeneous effects of competition on altruism by participants' beliefs about winning and by their perceived ability. Panel A of Appendix Figure B.13 indicates that participants who believed they would win, or who rated their ability above the median, offered significantly more amounts in the competition condition than in the control group. In contrast, participants who expected to lose or who rated their ability below the median showed no significant differences. One interpretation for this pattern is that enjoyable competing experience motivates more altruistic behavior when individuals expect monetary payoff. Detailed regression results are shown in the Appendix Table B.10 and B.11. Panel B of Appendix Figure B.13 provides a more detailed view of altruism in the control and competition groups across the distribution of perceived ability. The same pattern emerges: among participants who rated their ability above the median, those in the competition group gave substantially more than those in the control group. The results are similar under both gain- and loss-framing, as shown in Appendix Figure B.14. By contrast, little heterogeneity is observed across age, gender, education, or income (Appendix Figure B.10).

## D Experimental Materials

### D.1 Recruitment

Participants were recruited via the Prolific platform for a study titled ‘A short quiz and a set of survey questions (15–25 minutes).’ The study description varied slightly depending on the contextual framing. Participants were randomly assigned to one of four experimental groups.

#### GAIN FRAMING: COMPETITION AND CONTROL GROUPS

In this study, you will complete a short quiz followed by a set of survey questions. The base payment for participating in the study is \$2.00. You can earn a bonus of up to \$4.00 in the first section, and additional bonuses in later sections, depending on your performance and decisions. The entire study will take approximately 15-25 minutes.

#### LOSS FRAMING: COMPETITION AND CONTROL GROUPS

In this study, you will complete a short quiz followed by a set of survey questions. The base payment for participating in the study is \$6.00. You may lose up to \$4.00 of this amount in the first section. Bonuses are available in later sections. Your final payment will depend on your performance and decisions. The entire study will take approximately 15-25 minutes.

To qualify for the study, participants had to be at least 18 years old, reside in the United States, have a task approval rate above 95%, and have completed at least 100 tasks on Prolific. Each participant could only take part in one experimental group.

### D.2 Welcome and Consent

Welcome! Thank you for participating in our study! We are academic researchers from Boston University. This study will take approximately 15–25 minutes to complete. Your participation contributes to valuable academic research. After completing the study, you will receive a completion code for payment.

**Important Guidelines.** (1) Engagement: Your careful attention is essential for the success of this research. Please read all instructions carefully, engage thoroughly in the tasks, and

answer questions truthfully. The study includes multiple checks to ensure participants are engaging properly. Responses that show signs of inattention may be flagged as low quality and excluded from our analysis. (2) Commitment: We kindly ask that you complete the entire study once you begin, as some sections involve interactions with other participants. Your participation is entirely voluntary, and you may withdraw at any point without penalty. However, please note that if you leave the study before finishing, your session will end, and you may not be able to resume later. (3) Anonymity: This study is completely anonymous. Your Prolific ID will be used only to process your payment. No one will know your decisions or performance, and you will never be personally identified.

#### GAIN FRAMING: COMPETITION AND CONTROL GROUPS

**Compensation.** The base payment for participating in the study is \$2.00. You can earn a bonus of up to \$4.00 in the first section, and additional bonuses in later sections, depending on your performance and decisions.

#### LOSS FRAMING: COMPETITION AND CONTROL GROUPS

**Compensation.** The base payment for participating in the study is \$6.00. You may lose up to \$4.00 of this amount in the first section. Bonuses are available in later sections. Your final payment will depend on your performance and decisions.

**Contact.** If you have any questions, you may contact us at [jiaruiw@bu.edu](mailto:jiaruiw@bu.edu). After the welcome page, participants are presented with an informed consent form. To proceed, they must agree to the following statement by checking a box: “I consent to participate in this study. By checking this box, I confirm that I have read and understood the information provided about the purpose, risks, and benefits of this study. I understand that my participation is voluntary and I agree to the use of my data as described in the study information.” Once consent is given, participants are asked to enter their Prolific ID and complete a reCAPTCHA verification to confirm they are not automated bots. They can then begin the study.

## D.3 Instructions

### A. *Gain Framing*

This study consists of Competition and Survey Sections. You will receive a base payment of \$2.00 for participating. You can earn a bonus of up to \$4 in the Competition Section (or Quiz Section for the control group), and additional bonuses in the Survey Section. The exact amount will depend on your performance and decisions. The entire study will take approximately 15 to 25 minutes to complete.

#### GAIN FRAMING: COMPETITION GROUP

**Competition section.** You will be randomly paired with another participant for an IQ Quiz Competition (Intelligence Quotient Quiz Competition). Only one of you will earn a \$4 reward. The participant with the higher score will be the top performer and receive the \$4. The one with the lower score will be the bottom performer and receive \$0. If both participants have the same score, one will be randomly selected to receive the \$4 reward. In the IQ Quiz Competition, you will earn 1 point for each correct answer, lose 0.25 points for each incorrect answer, and neither earn nor lose any points for skipped questions. Once you submit an answer or skip a question, you cannot go back to it. The quiz consists of fifty questions, and you will have 4 minutes to answer as many as possible. Throughout the competition, you will see your opponent's raw score in real time, which shows only the number of questions they have answered correctly (without any deductions for wrong answers). You will not see your own score during the competition. Your final score and competition results will be revealed at the end of the study. Before the competition begins, you will have the opportunity to answer a practice question to get familiar with the quiz format.

**GAIN FRAMING: CONTROL GROUP**

**Quiz section.** You will take an IQ Quiz (Intelligence Quotient Quiz). If you score 6 points or higher, you will receive a \$4 reward. In a previous session of the experiment, about half of the participants scored 6 points or higher. If you score below 6 points, you will receive \$0. In the IQ Quiz, you will earn 1 point for each correct answer, lose 0.25 points for each incorrect answer, and neither earn nor lose any points for skipped questions. Once you submit an answer or skip a question, you cannot go back to it. The quiz consists of fifty questions, and you will have 4 minutes to answer as many as possible. At the end of the study, you will see your score and the quiz results. Before the quiz begins, you will have the opportunity to answer a practice question to get familiar with the format.

**Survey section.** You will answer six survey questions and complete a decision-making task, followed by a few final survey questions. Some questions offer bonuses and will be clearly marked.

**Comprehension question.** Before we proceed, let's ensure you understand the instructions clearly. Please answer the following question correctly to continue.

**GAIN FRAMING: COMPETITION GROUP**

In the 4-minute IQ Quiz competition, Emily answers 20 questions (14 correct, 6 incorrect) while Lucy answers 12 questions (8 correct, 4 incorrect). Who receives the \$4 reward? A. Lucy B. Emily C. It's a tie. One of them will be randomly selected to receive the reward.

**GAIN FRAMING: CONTROL GROUP**

In the 4-minute IQ Quiz, Lucy answers 18 questions (14 correct, 4 incorrect). Does she receive the \$4 reward? A. No B. Yes C. Not enough information.

**B. Loss Framing**

This study consists of Competition and Survey Sections. You will receive a base payment of \$6.00 for participating. You may lose up to \$4 of this amount in the Competition Section (or Quiz Section for the control group). Bonuses are available in the Survey Section. Your final payment will depend on your performance and decisions. The entire study will take

approximately 15 to 25 minutes to complete.

#### LOSS FRAMING: CONTROL GROUP

**Quiz section.** You will take an IQ Quiz (Intelligence Quotient Quiz). If you score 6 points or higher, you will keep the initial payment of \$6. In a previous session of the experiment, about half of the participants scored 6 points or higher. If you score below 6 points, you will lose \$4 from the initial payment. In the IQ Quiz, you will earn 1 point for each correct answer, lose 0.25 points for each incorrect answer, and neither earn nor lose any points for skipped questions. Once you submit an answer or skip a question, you cannot go back to it. The quiz consists of fifty questions, and you will have 4 minutes to answer as many as possible. At the end of the study, you will see your score and the quiz results. Before the quiz begins, you will have the opportunity to answer a practice question to get familiar with the format.

#### LOSS FRAMING: COMPETITION GROUP

**Competition section.** You will be randomly paired with another participant for an IQ Quiz Competition (Intelligence Quotient Quiz Competition). Only one of you will keep the initial payment of \$6. The participant with the higher score will be the top performer and keep the \$6. The one with the lower score will be the bottom performer and lose \$4 from their initial payment. If both participants have the same score, one will be randomly selected to lose \$4. In the IQ Quiz Competition, you will earn 1 point for each correct answer, lose 0.25 points for each incorrect answer, and neither earn nor lose any points for skipped questions. Once you submit an answer or skip a question, you cannot go back to it. The quiz consists of fifty questions, and you will have 4 minutes to answer as many as possible. Throughout the competition, you will see your opponent's raw score in real time, which shows only the number of questions they have answered correctly (without any deductions for wrong answers). You will not see your own score during the competition. Your final score and competition results will be revealed at the end of the study. Before the competition begins, you will have the opportunity to answer a practice question to get familiar with the quiz format.

**Survey section.** You will answer six survey questions and complete a decision-making task, followed by a few final survey questions. Some questions offer bonuses and will be clearly marked.



**Comprehension question.** Before we proceed, let’s ensure you understand the instructions clearly. Please answer the following question correctly to continue.

**LOSS FRAMING: CONTROL GROUP**

In the 4-minute IQ Quiz, Emily answers 10 questions (6 correct, 4 incorrect). Does she lose \$4 from the initial payment of \$6? A. No B. Yes C. Not enough information.

**LOSS FRAMING: COMPETITION GROUP**

In the 4-minute IQ Quiz competition, Lucy answers 20 questions (14 correct, 6 incorrect) while Emily answers 12 questions (8 correct, 4 incorrect). Who loses \$4 from the initial payment of \$6? A. Lucy B. Emily C. It’s a tie. One of them will be randomly selected to lose \$4.

D.4 Practice Question

**Practice Question**

A pattern with one missing piece is displayed below. Your task is to **select the correct piece** from the four available options.

A

B

C

D

Which piece is the right complement?

☐ A

☐ B

☐ C

☐ D

Next

Appendix Figure D.1: Practice Question Interface

## D.5 IQ Quiz

### A. A Few Reminders

Before the IQ quiz begins, participants are presented with a reminder page that lists the following points: (1) This study is completely anonymous. Neither the organizers nor other participants will know your identity, and no one will know your performance or decisions. (2) To ensure high-quality research, please take the quiz in a quiet environment without distractions. Turn off notifications or other potential interruptions before starting.

#### **GAIN FRAMING: COMPETITION GROUP**

During the competition, you will see your opponent's real-time raw score (number of correct answers) and a 4-minute timer. You will not see your own score. Competition results will be revealed at the end of the study. Only the top performer receives the \$4 reward, while the bottom performer receives nothing. Once you submit an answer or skip a question, you cannot go back to it. When you are ready, click 'Next' to begin matching and start the competition.

#### **GAIN FRAMING: CONTROL GROUP**

During the quiz, you will see a 4-minute timer. Quiz results will be revealed at the end of the study. If you score 6 points or higher, you will receive a \$4 reward. If not, you will receive nothing. Once you submit an answer or skip a question, you cannot go back to it. When you are ready, click 'Next' to start the quiz.

#### **LOSS FRAMING: COMPETITION GROUP**

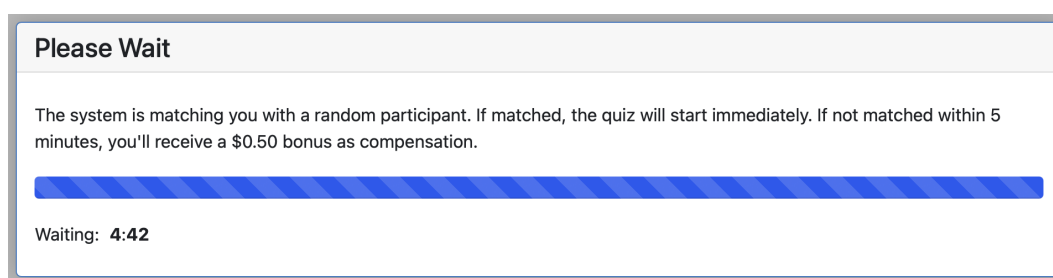
During the competition, you will see your opponent's real-time raw score (number of correct answers) and a 4-minute timer. You will not see your own score. Competition results will be revealed at the end of the study. Only the top performer keeps the full initial payment of \$6, while the bottom performer loses \$4. Once you submit an answer or skip a question, you cannot go back to it. When you are ready, click 'Next' to begin matching and start the competition.

### LOSS FRAMING: CONTROL GROUP

During the quiz, you will see a 4-minute timer. Quiz results will be revealed at the end of the study. If you score 6 points or higher, you will keep the full initial payment of \$6. If not, you will lose \$4. Once you submit an answer or skip a question, you cannot go back to it. When you are ready, click 'Next' to start the quiz.

### B. Matching

Following the reminder page, participants in the competition groups proceed to the matching page, where they wait to be paired in real time with another online participant.



Appendix Figure D.2: Matching Page Interface

### C. IQ Quiz Question

Below are screenshots of IQ quiz questions from each of the four experimental groups: Gain Framing–Competition, Gain Framing–Control, Loss Framing–Competition, and Loss Framing–Control.

Appendix Figure D.3: Sample IQ Quiz Question – Gain Framing (Competition Group)

Time Left: 2:25

IQ Quiz Competition - Question 1

Only one of you receives the \$4 bonus.

Top Performer: earn \$4 | Bottom Performer: earn \$0

Opponent's Raw Score: 

3

A

B

C

D

Which piece is the correct complement?

☐ A

☐ B

☐ C

☐ D

Next

55

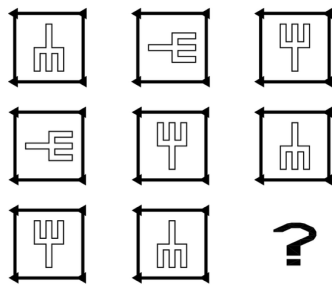
Appendix Figure D.4: Sample IQ Quiz Question – Gain Framing (Control Group)

Time Left: 3:32

### IQ Quiz - Question 1

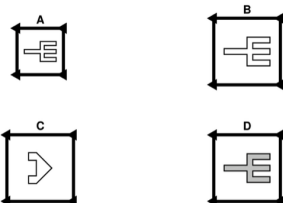
Reach the target score to receive the \$4 bonus.

Score 6 or above : earn \$4 | Score below 6 : earn \$0



Which piece is the correct complement?

- ☐ A
- ☐ B
- ☐ C
- ☐ D



Next

Appendix Figure D.5: Sample IQ Quiz Question – Loss Framing (Competition Group)

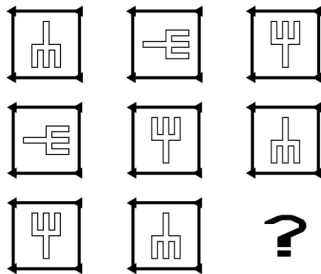
Time Left: 2:58

### IQ Quiz Competition - Question 1

Only one of you keeps the full initial payment of \$6.

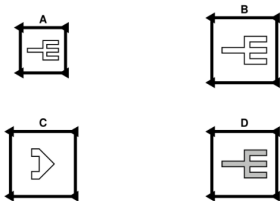
Top Performer: lose \$0 | Bottom Performer: lose \$4

Opponent's Raw Score: 5



Which piece is the correct complement?

- ☐ A
- ☐ B
- ☐ C
- ☐ D



Next

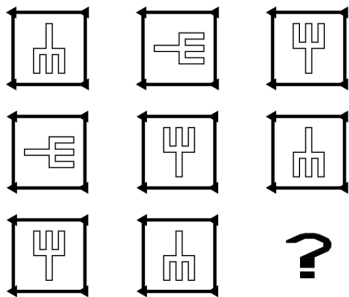
Appendix Figure D.6: Sample IQ Quiz Question – Loss Framing (Control Group)

### IQ Quiz - Question 1

Time Left: 3:22

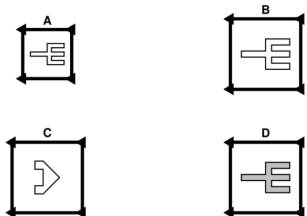
Reach the target score to keep the full initial payment of \$6.

Score 6 or above : lose \$0 | Score below 6 : lose \$4



Which piece is the correct complement?

- ☐ A
- ☐ B
- ☐ C
- ☐ D



Next

## D.6 Main Survey

### GAIN FRAMING: COMPETITION GROUP

**Survey Introduction.** The IQ Quiz Competition is complete. You will see the results at the end of the study. As a reminder, only the top performer will receive a \$4 bonus, while the bottom performer will receive nothing.

### GAIN FRAMING: CONTROL GROUP

**Survey Introduction.** The IQ Quiz is complete. You will see the results at the end of the study. As a reminder, if you score 6 points or higher, you will earn a \$4 bonus. Otherwise, you will receive nothing.

### LOSS FRAMING: COMPETITION GROUP

**Survey Introduction.** The IQ Quiz Competition is complete. You will see the results at the end of the study. As a reminder, only the top performer will keep the full initial payment of \$6, while the bottom performer will lose \$4.

### LOSS FRAMING: CONTROL GROUP

**Survey Introduction.** The IQ Quiz is complete. You will see the results at the end of the study. As a reminder, if you score 6 points or higher, you will keep the full initial payment of \$6. Otherwise, you will lose \$4.

Next, you will answer six survey questions and complete a decision-making task, followed by a few final survey questions. You can earn a bonus of up to \$1.30 in this section, depending on your performance and decisions. Bonus-eligible questions will be clearly marked.

Please note that your responses will remain anonymous. Your honest responses are greatly appreciated. To ensure high-quality research, attention checks are included throughout the survey. Responses indicating inattention may be excluded from the analysis.

### A. Utility

**Question1.** On a scale from 1 (not at all enjoyable) to 7 (extremely enjoyable), how enjoyable was the IQ Quiz Competition (or *IQ Quiz* for the control group)? [Options: 1 (not at all enjoyable), 2, 3, 4, 5, 6, 7 (extremely enjoyable)]



**Question2.** Thinking back to how you felt during the IQ Quiz Competition (or *IQ Quiz* for the control group), please indicate to what extent you were experiencing the following, on a scale from 1 (not at all) to 7 (extremely) [Stress, Excitement, Satisfaction, Embarrassment, Anxiety. Options: 1 (not at all), 2, 3, 4, 5, 6, 7 (extremely)]

*B. Task Preference*

**Question1.** On a scale from 1 (not at all likely) to 7 (extremely likely), how likely would you be to voluntarily take another 10 IQ questions, similar to the ones you just saw in the IQ quiz, with no monetary rewards? [Options: 1 (not at all likely), 2, 3, 4, 5, 6, 7 (extremely likely)]

**Question2.** Imagine you have another IQ quiz in front of you right now, similar to the one you just completed, but with a different set of fifty questions. You again have 4 minutes to answer as many as possible. If you get at least 6 questions correct within the 4 minutes, you will receive a payment. Below is a list of payment amounts. For each amount, please indicate 'Yes' if you would be willing to take the quiz for that payment, or 'No' if you would not. [\$0.20, \$0.40, \$0.60, \$0.80, \$1.00, \$1.20, \$1.40, \$1.60, \$1.80, \$2.00, \$2.50, \$3.00, \$5.00. Options: Yes, I would take; No, I would not take.]

*C. Zero-sum Thinking*

**Question1.** Consider a local job market where both immigrants and locals work as day laborers. On Day 1, immigrant workers found 20 jobs, while local workers found 40 jobs. On Day 2, immigrant workers found 40 jobs. Which of the following estimates is more likely to be accurate regarding the number of jobs local workers found on Day 2? [Options: 20 jobs, 80 jobs]

|                                                                                                                                                 |                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Day 1</p>  <p><i>Immigrant workers: 20 jobs found</i></p> | <p>Day 2</p>  <p><i>Immigrant workers: 40 jobs found</i></p> |
| <p>Day 1</p>  <p><i>Local workers: 40 jobs found</i></p>     | <p>Day 2</p>  <p><i>Local workers: ____ jobs found</i></p>   |

**Question2.** Below are two statements. Where do your views fall on a scale from 1 to 10? 1 means you agree completely with the statement 1; 10 means you agree completely with the statement 2. *Statement 1:* People can only become wealthy at the expense of others. *Statement 2:* Wealth can grow so there's enough for everyone. If your views fall somewhere in between, you can choose any number in between. [Options: 1 (statement1), 2, 3, 4, 5, 6, 7, 8, 9, 10 (statement2)]

#### D. Public Goods Game

Appendix Figure D.7: Public Goods Game Instructions

\$ All payoffs are real and will be added as a bonus to your payment.

### Decision-Making Task

You are now paired with another random participant (not the person you interacted with earlier).

Each of you is given **\$0.50** and will independently decide whether to contribute it into a **shared pool** or to **keep it** for yourself. Your final payment will depend on both of your decisions, as shown below:

| Other Participant's Decision | Your Decision  | Other Participant's Payoff | Your Payoff |
|------------------------------|----------------|----------------------------|-------------|
| Put into pool                | Put into pool  | \$0.80                     | \$0.80      |
| Keep the money               | Keep the money | \$0.50                     | \$0.50      |
| Keep the money               | Put into pool  | \$0.90                     | \$0.40      |
| Put into pool                | Keep the money | \$0.40                     | \$0.90      |

*Please answer the following question to ensure you understand the rules:*

If the other participant chooses to keep the money, and you choose to put the money into the pool, how much will you receive?

Your Answer:

☐ \$0.80  
☐ \$0.90  
☐ \$0.50  
☐ \$0.40

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## Appendix Figure D.8: Public Goods Game-Decision Page

 All payoffs are real and will be added as a bonus to your payment.

**Please make your decision:**



Drag your \$0.50 to one of the options below.

**Put into pool**

**Keep the money**

Reminder:

| Other Participant's Decision | Your Decision  | Other Participant's Payoff | Your Payoff |
|------------------------------|----------------|----------------------------|-------------|
| Put into pool                | Put into pool  | \$0.80                     | \$0.80      |
| Keep the money               | Keep the money | \$0.50                     | \$0.50      |
| Keep the money               | Put into pool  | \$0.90                     | \$0.40      |
| Put into pool                | Keep the money | \$0.40                     | \$0.90      |

Your final bonus from this task depends on **both** decisions. Take your time to decide. If dragging does not work, please click directly on a box to select. Once you are satisfied with your choice, press "Continue."

Continue

**Question.** Approximately 800 participants are taking part in this study today. What percentage of them do you think choose to keep the money? If your answer is correct, you will receive a \$0.20 bonus. [Options: 0–10%, 11–20%, 21–30%, 31–40%, 41–50%, 51–60%, 61–70%, 71–80%, 81–90%, 91–100%. ]

### E. Dictator Game

Appendix Figure D.9: Interface of the Dictator Game

**How would you split \$100 between yourself and a randomly-selected participant in this study?**

The closer you drag the slider to one individual, the more money you allocate to that individual. The randomly-selected individual would never find out that it was you who sent them the money. Please drag the slider to the point where you feel satisfied with the way the money is split.



You  
\$50

How would I split the money?

\$100





A random participant  
\$50

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### F. Attention Check

**Question.** This is not a question that needs to be answered. Instead, the goal of this question is to check whether you are reading all instructions carefully. To indicate this, please click the next button without selecting any of the options below. You must click next without making any selections in order to have your response approved. [Options: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

## D.7 Supplemental Survey

### A. *Belief in Winning*

| COMPETITION GROUPS                                                                       |
|------------------------------------------------------------------------------------------|
| Do you think you were the top performer in the IQ Quiz Competition? [Options: Yes, No. ] |

| CONTROL GROUPS                                                           |
|--------------------------------------------------------------------------|
| Do you think you scored 6 or higher on the IQ Quiz? [Options: Yes, No. ] |

### B. *Perceived Percentile Rank*

**Question.** Based on your experience in the IQ Quiz, how well do you think you performed compared to all other participants in this study, on a scale from 0 to 10? The lowest value (0) indicates that you believe you performed worse than almost everyone else, and the highest value (10) indicates that you believe you performed better than nearly all other participants. For example, if you think you performed better than approximately 70% of all other participants, you should select 7. Please make your choice. [Options: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

### C. *Competition Preferences*

**Question1.** On a scale from 1 (not at all like me) to 7 (exactly like me), please indicate the extent to which the following statement describes you: "Competition brings the best out of me." [Options: 1 (not at all like me), 2, 3, 4, 5, 6, 7 (exactly like me)]

**Question2.** On a scale from 1 (not at all like me) to 7 (exactly like me), please indicate the extent to which the following statement describes you: "I enjoy competing against others." [Options: 1 (not at all like me), 2, 3, 4, 5, 6, 7 (exactly like me)]

### D. *Understanding of Instructions*

**Question.** Did you understand the instructions throughout the study? [Options: I fully understood them, I understood them almost fully, I only partly understood them, I did not understand them]

### *E. Demographics*

**Question1.** Which year were you born? [Options: 1920 to 2010]

**Question2.** What's your gender? [Options: Male, Female, Other]

**Question3.** How would you describe your race or ethnicity? [Options: White, African-American, Hispanic, American Indian or Alaska Native, Asian, Other]

**Question4.** How would you describe your educational attainment? [Options: No high school graduation, High school graduate, Bachelor's degree, Graduate or professional degree]

**Question5.** What's your approximate annual household income before taxes? [Options: Below \$10,000, \$10,000–\$14,999, \$15,000–\$24,999, \$25,000–\$34,999, \$35,000–\$49,999, \$50,000–\$74,999, \$75,000–\$99,999, \$100,000–\$149,999, \$150,000–\$199,999, \$200,000 or more]

**Question6.** Are you employed full-time? [Options: Yes, No]

**Question7.** On economic policy matters, where do you see yourself on the left–right spectrum? [Options: Extreme Left, Leaning Left, Center, Leaning Right, Extreme Right]

### *F. Comments*

Thank you for participating in our study! We would appreciate any comments or feedback about your experience. If any part of the instructions was unclear or if you ran into any issues during the study, please let us know. Your valuable feedback will help us improve our research.” After this page, participants are shown their IQ Quiz results and informed whether they were the top performer in the competition (or scored 6 or higher).